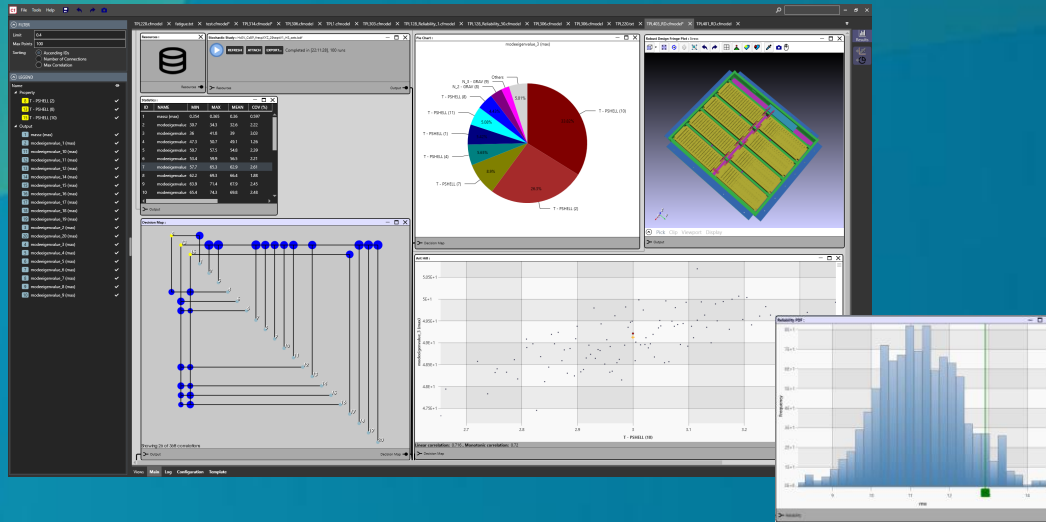


MSC Nastran Advanced Training CAEfatigue and Robust Design



Stochastic analysis With CAEfatigue Robust Design

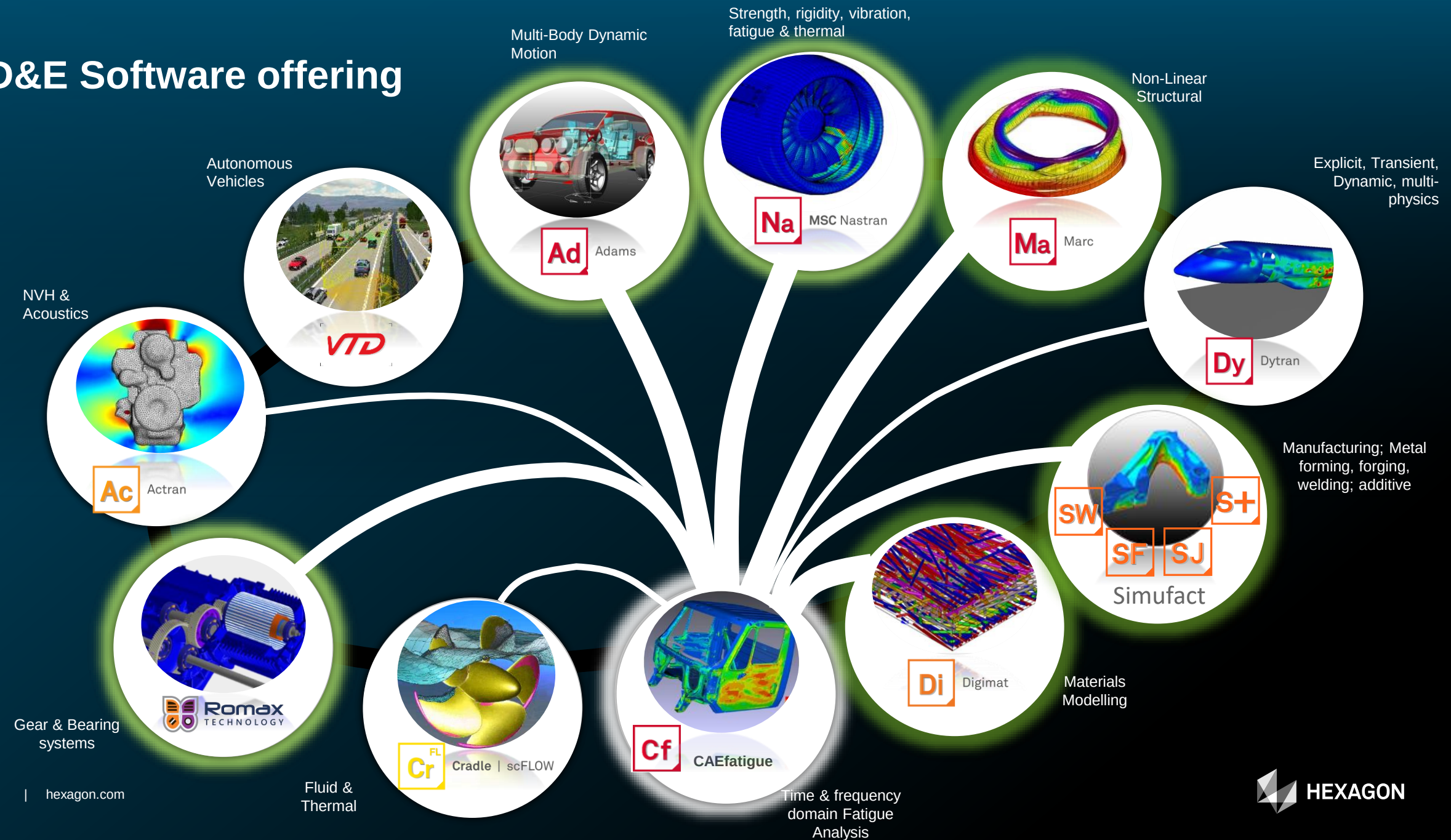


Marco Veltri, PhD

Business Enablement Manager, Multiphysics
Product Management & Business Development

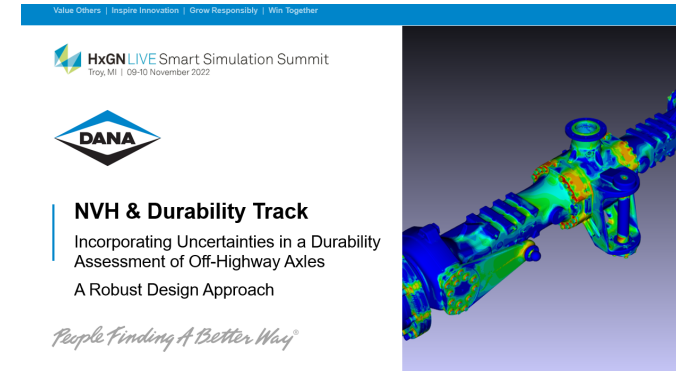
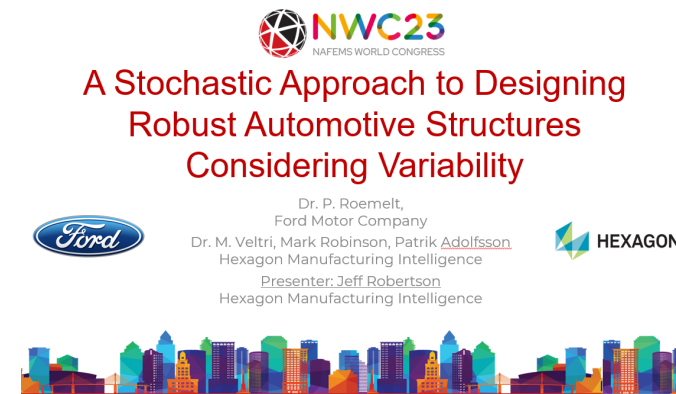
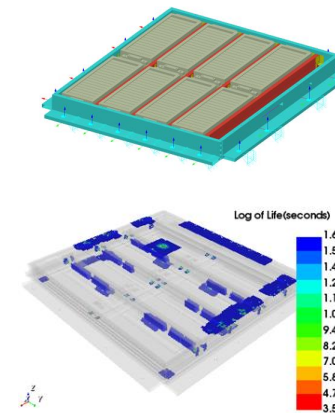
marco.veltri@hexagon.com

D&E Software offering



agenda

- Intro to CAEf fatigue and Robust Design
 - CAEf fatigue process Flow and FE embedded operation
- Intro to stochastic simulation workflow with Robust Design
 - Input randomization and function study
 - Building actionable knowledge
 - Stochastic Design Improvement
- Use case review:
 - HV battery pack certification
 - Automotive testing (Ford NAFEMS paper)
 - Heavy vehicle axle testing (DANA at HxGN Smart simulation Summit)



Fatigue, Loads Management Robust Design : Where are your challenges?

Better
Products
Quicker

Time, Budget,
Quality &
Repeatability
important for a
project delivery
organisation

new platforms
– More
complex
engineering,
more levels of
integration

Digital
handshake
between
engineering
toolsets

Digital
handshake
between
engineering and
operations

Product quality
and robustness

Insights into
complex
systems
limited

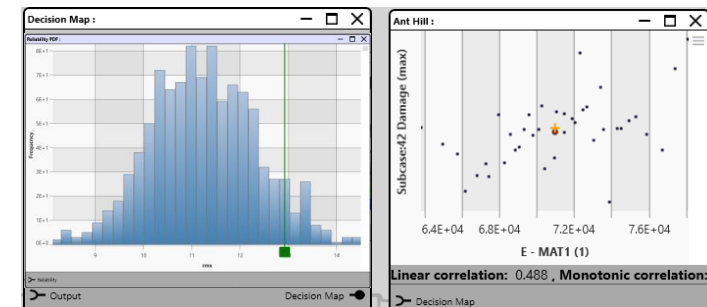
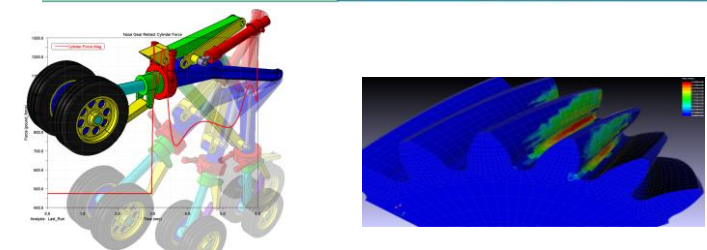
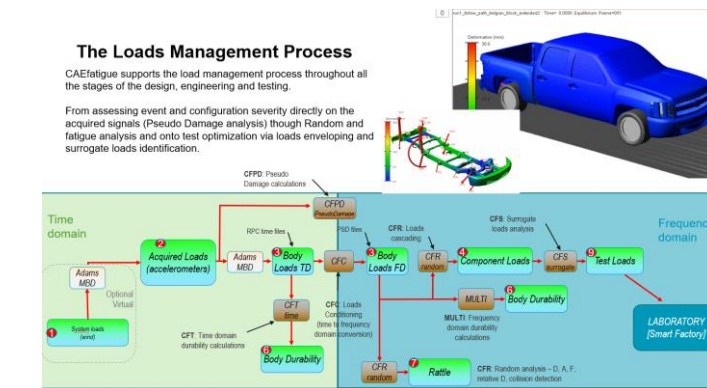
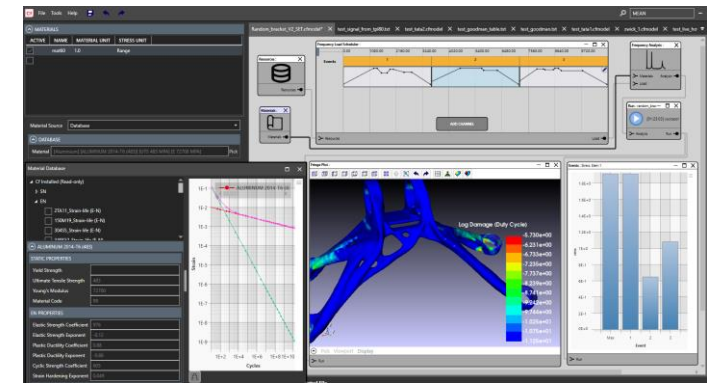
... what we have heard or experienced based from Customer challenges

- Time and Quantity: Solution Speed, Repeatability, Ease of Use, Support (from self-help tutorials to multidomain and Multiphysics solution experts)
- Complexity: Skill-set requirements in challenging domain (dynamics, fatigue, random signals, stochastics...), embedding / accessing company and experts knowhow, integration between CAE, testing, manufacturing processes
- Digital Handshake: operations loads management (e.g. proving ground or MBD), handling certification standards in simulations or generating surrogate loads for physical tests
- Design Quality/Robustness : handling uncertainties and variability, reduce liability, improve correlation with physical world, define cost effective quality checks and tolerances

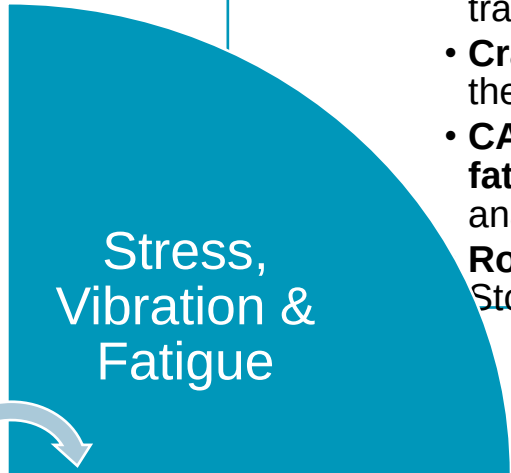
...From challenges to solutions

A quick glance at CAEfatigue & Robust Design value *what makes it unique*

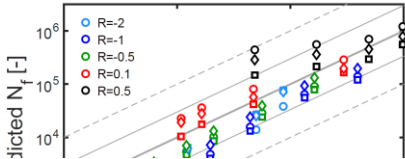
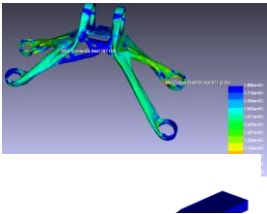
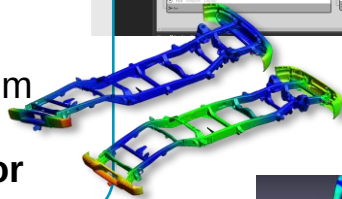
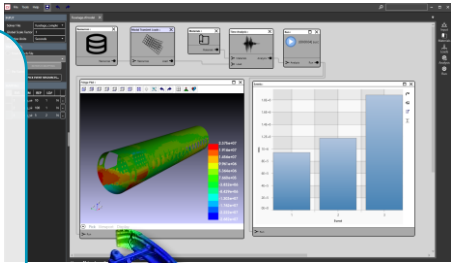
- CAEfatigue architecture
 - Fast and intuitive Process Flow templates
 - Solver neutral (Nastran, Abaqus, Ansys...)
 - batch runs and easy integration in enterprise portals (e.g. SimManager)
 - Visual Random and Vibration response tools
- Load Management Process
 - Digital Signal Processing and automatic conditioning for time to PSD loading conversion
 - Using CAEfatigue for test optimization (envelopes and surrogates)
 - Virtual Strain gauge analysis
- Fatigue analysis integration in CAE processes
 - Embedded fatigue solutions, Manufacturing effects, mechanism and transmission durability, advanced materials
- Robust Design for stochastic simulation and stochastic design improvements



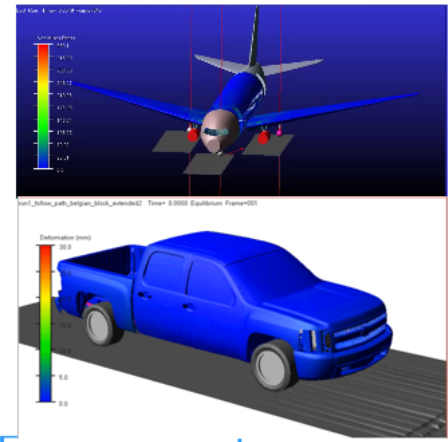
CAEfatigue and Robust Design for Loads Management, Durability Solutions Integration (digital handshake)



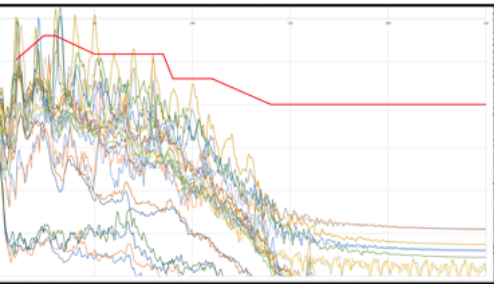
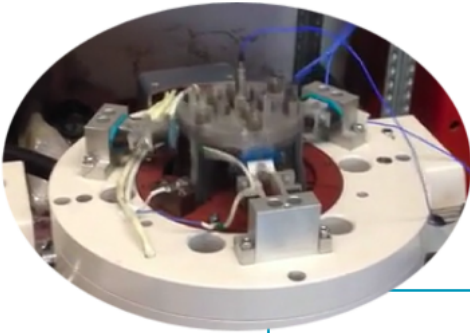
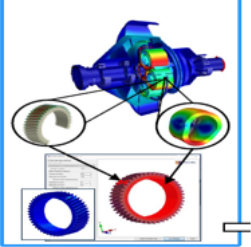
- **Apex/ Nastran** for modal reduction and transfer function
 - **Cradle/Marc** for thermal-structural
 - **CAEfatigue** for **fatigue** and random analysis
- Robust Design for Stochastic Analysis**



Loads Management & Durability Solutions Integration and digital handshake



- **Adams/Apex** for loads in Dynamic transient
- **Romax** for engine, transmission loads, gear/bearing fatigue
- **CAEfatigue** Pseudo damage, load conditioning



- **CAEfatigue** for loads specification and test optimization
- **Robust Design** for test correlation

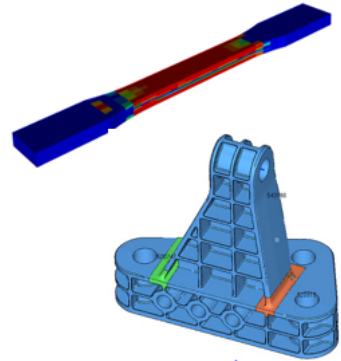
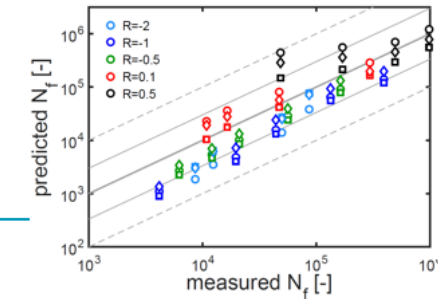
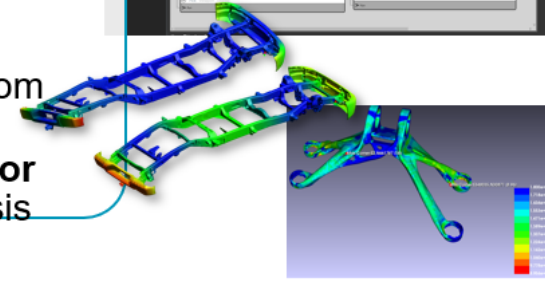
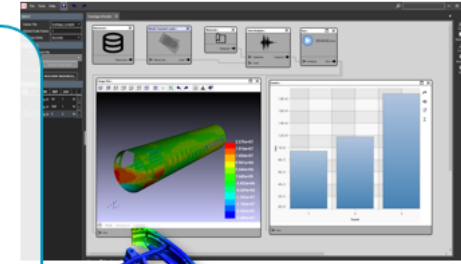
Vehicle
Dynamics &
Durability

Stress,
Vibration &
Fatigue

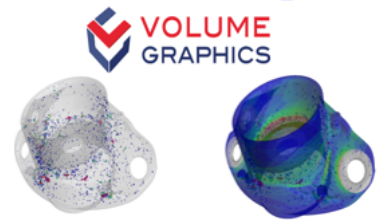
Testing &
Certification

Materials &
Manufacturing
effects

- **Apex/ Nastran** for modal reduction and transfer function
- **Cradle/Marc** for thermal-structural
- **CAEfatigue** for fatigue and random analysis
- **Robust Design for Stochastic Analysis**



- **Digimat** for material engineering
- **Simufact** for seam welds, cold forming processes, AM
- **Volume Graphics CT scan analysis** for defect identification



CAEfatigue technologies and operation modes

*Includes Robust Design
(New in 2022)*

**Cf
TIME**

**Cf
RANDOM**

**Cf
FREQUENCY**

**Cf
PREMIUM**

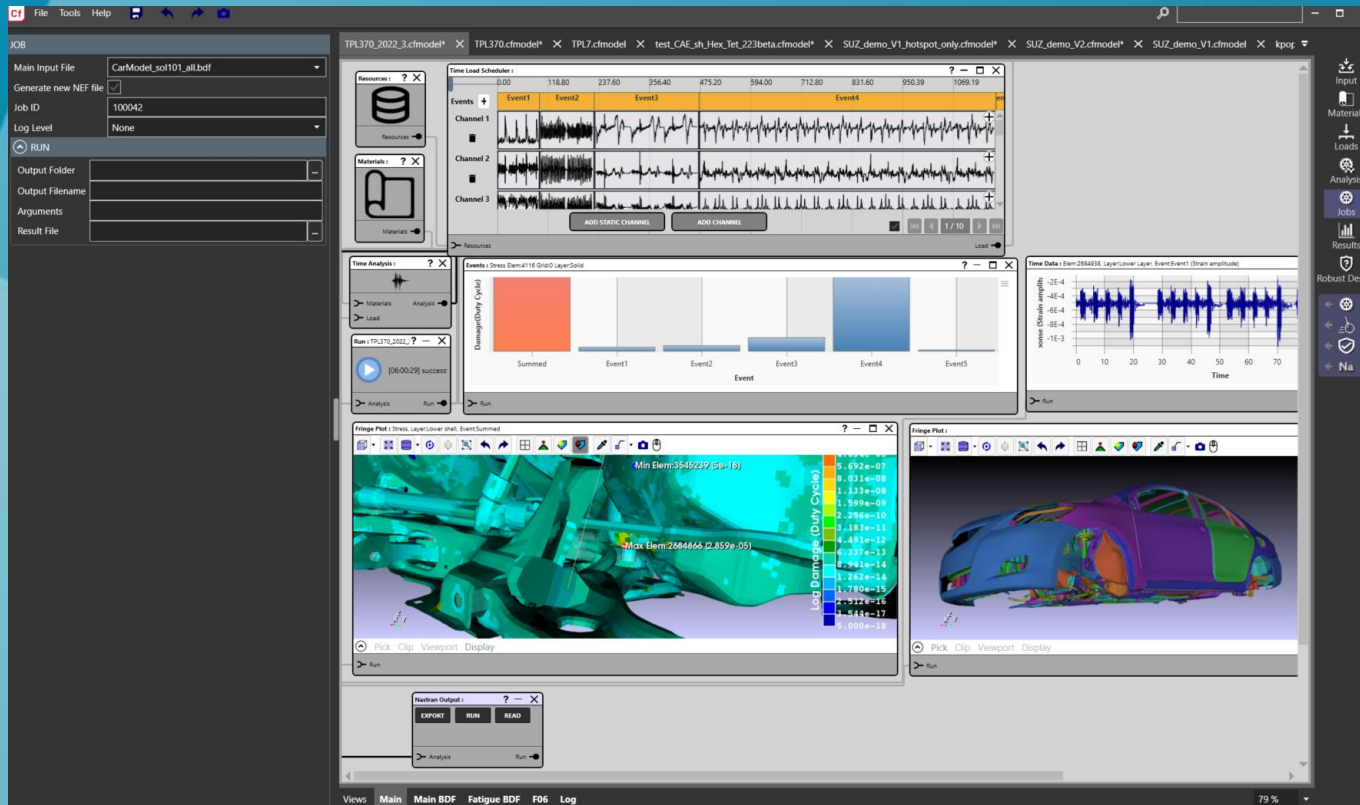
**FE Solver
Embedded
(Nastran/Marc)**

Control File

**Stand Alone
(APEX Aligned)**

Support ANSYS, Nastran,
Optistruct, Abaqus, Marc...
LSDyna (New in 2022)

One unified Durability, Random Response and Loads Management solution



CAEfatigue (Cf) Process Flow and FE embedded

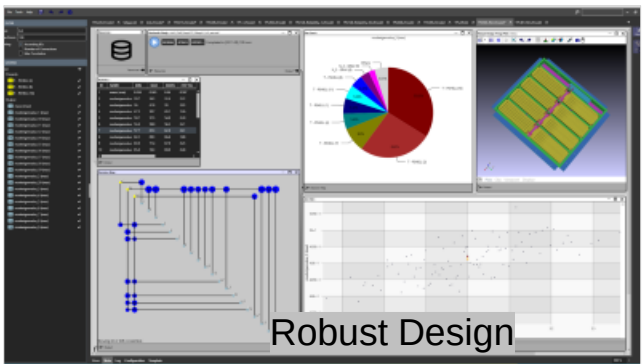
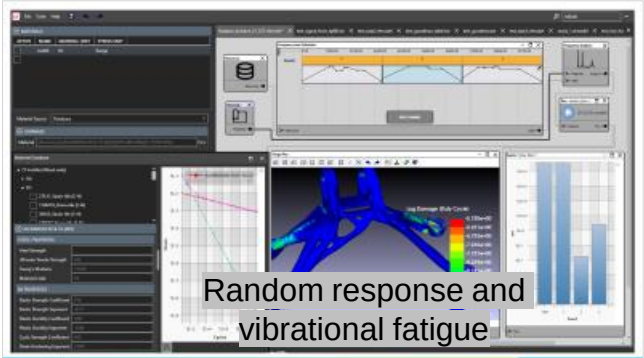
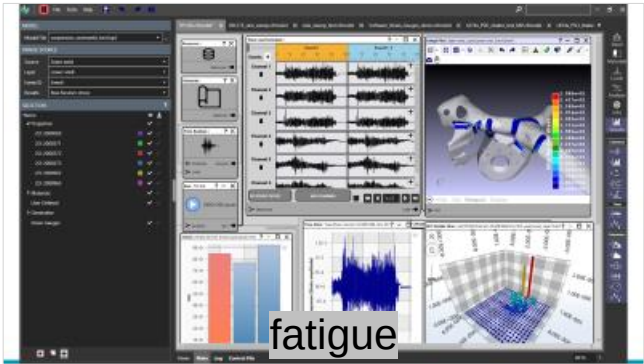
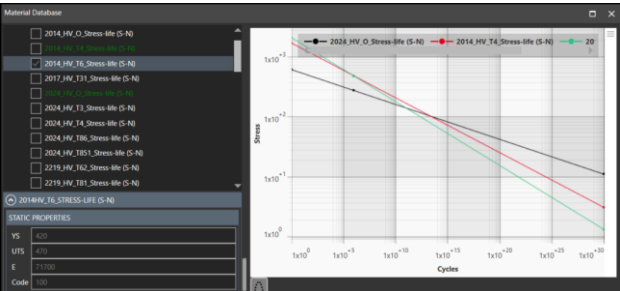
From interactive process flow to Nastran embedded fatigue

CAEfatigue & Robust Design: intuitive, repeatable, transferable process flows

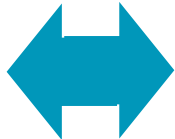
- Process orientated
- Pre-set templates, customizable
- Drag and drop
- Indexed help
- Multi tab / multi output / Multi run

Multiple material sources

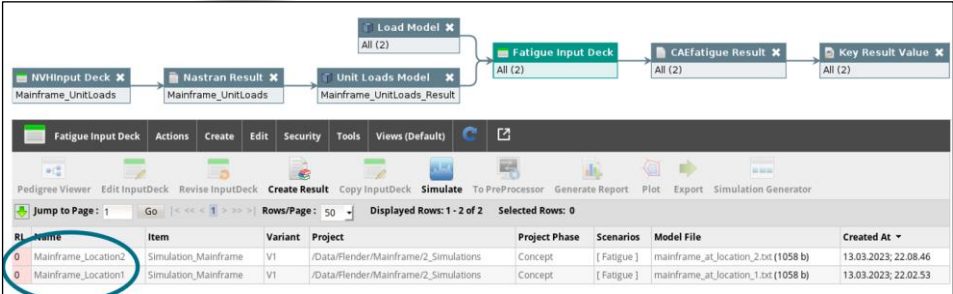
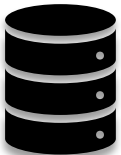
- own or custom DB
- MaterialCenter
- Digimat



Bidirectional flow with batch operatable control file

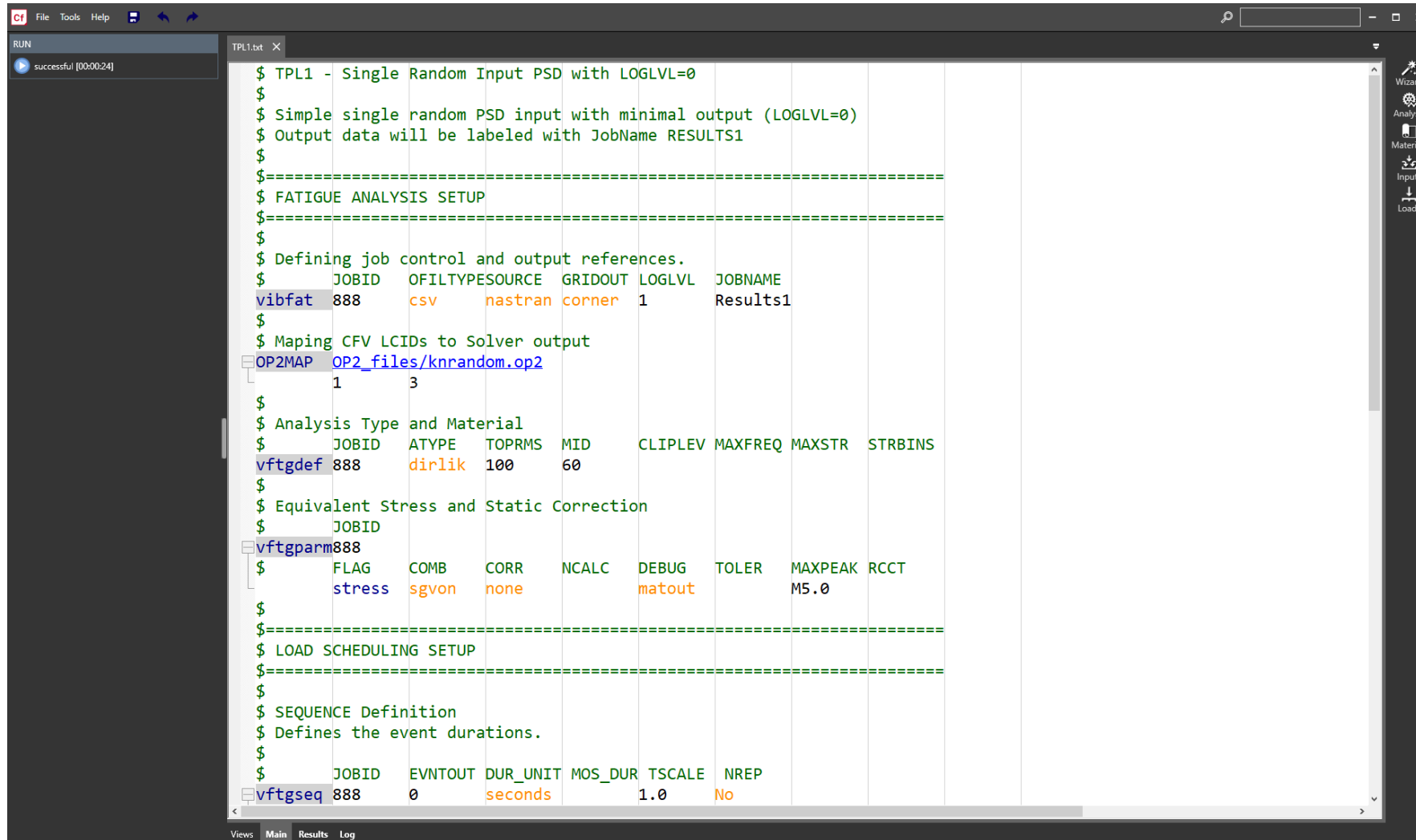


```
$ Generated by CAEfatigue (2023.1 7114)
$
vibfat 777 csv nash5 corner 1 Digi_Marc_H5_test
$ @matid[60]="mat1"
vmatftg 60 MPa range
file 60 sn ID=Stress-life (S-N) ParentID=2014_HV_T6
vudname 60
C:\MSC.Software\CAEfatigue_Beta\Windows\Resources\Materials\cfv_material_db.xml
fe_units 777 MPA 1 B N mm
vftgdef 777
timeout S
histrcc elset
fset3 10001 ELEM 10001
13247
vftgparm 777
stress sgvon none naver none 1E-05
dsp filter peakvaly
vftgseq 777 1 seconds 1 yes
101 10 102 1
vftgevt 101 402
name flight_1
vftgevt 102 403
name flight_2
casemap C:\A_presale\Industry_based\Digimat\CF\QW22001_Demonstrator_CAEfatigue_DigimatCoupled.hs
1 subcase 1, step 1
vftgload 402 nqstatic 0.0005
vftgload 403 nqstatic 0.0001
C:\A_presale\Industry_based\Digimat\CF\QW22001_Demonstrator_CAEfatigue_DigimatCoupled.hs
C:\A_presale\Industry_based\Digimat\CF\QW22001_Demonstrator_CAEfatigue_DigimatCoupled.hs
```



Powerful Control File Editor Still Available for Advanced Users

Work in advanced mode by using just the Control File editor



The screenshot shows the Hexagon Control File Editor interface. On the left, a 'RUN' panel indicates a successful execution at 00:00:24. The main window displays a control file named 'TPL1.txt' with the following content:

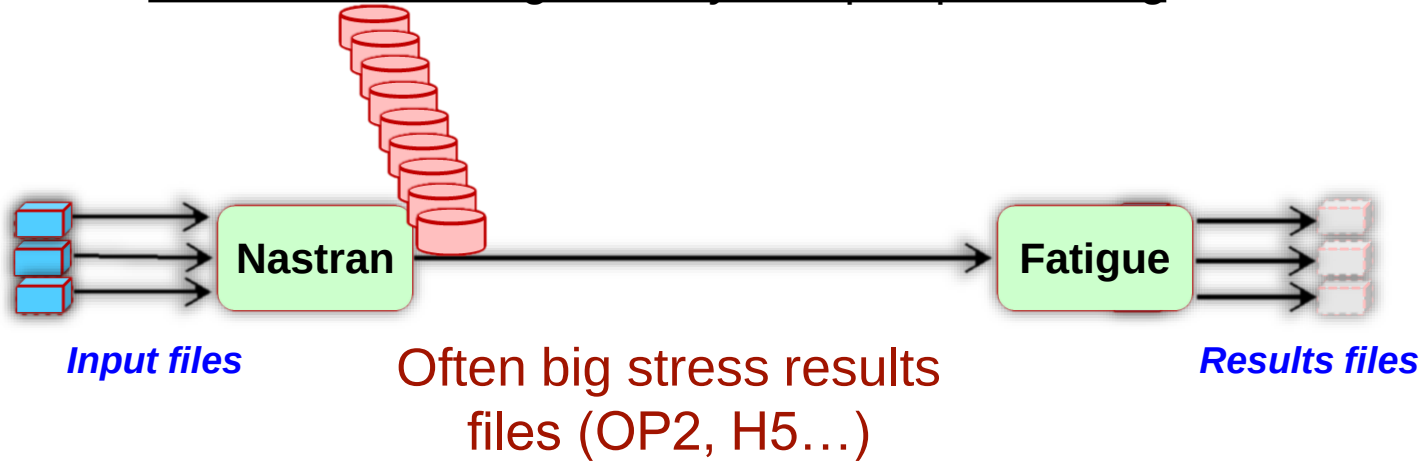
```
$ TPL1 - Single Random Input PSD with LOGLVL=0
$
$ Simple single random PSD input with minimal output (LOGLVL=0)
$ Output data will be labeled with JobName RESULTS1
$
$=====
$ FATIGUE ANALYSIS SETUP
$=====
$
$ Defining job control and output references.
$      JOBID  OFILTYPE  SOURCE  GRIDOUT  LOGLVL  JOBNAME
vibfat  888      csv       nastran  corner  1       Results1
$
$ Mapping CFV LCIDs to Solver output
OP2MAP  OP2_files/knrandom.op2
1       3
$
$ Analysis Type and Material
$      JOBID  ATYPE  TOPRMS  MID  CLIPLEV  MAXFREQ  MAXSTR  STRBINS
vftgdef  888      dirlik  100   60
$
$ Equivalent Stress and Static Correction
$      JOBID
vftgparm 888
$      FLAG  COMB  CORR  NCALC  DEBUG  TOLER  MAXPEAK  RCCT
stress    sgvon  none  NCALC  matout  M5.0
$
$=====
$ LOAD SCHEDULING SETUP
$=====
$
$ SEQUENCE Definition
$ Defines the event durations.
$
$      JOBID  EVNTOUT  DUR_UNIT  MOS_DUR  TSCALE  NREP
vftgseq  888      0       seconds    1.0     No
```

The interface includes a menu bar (File, Tools, Help), a toolbar, and a sidebar with icons for Wizard, Analysis, Materials, Inputs, and Loads. The bottom status bar shows 'Views: Main Results Log'.

Stand alone CAEfatigue or embedded NEF(time) and NEVF (frequency)

Dual path for efficient fatigue analysis

Conventional Fatigue Analysis a postprocessing

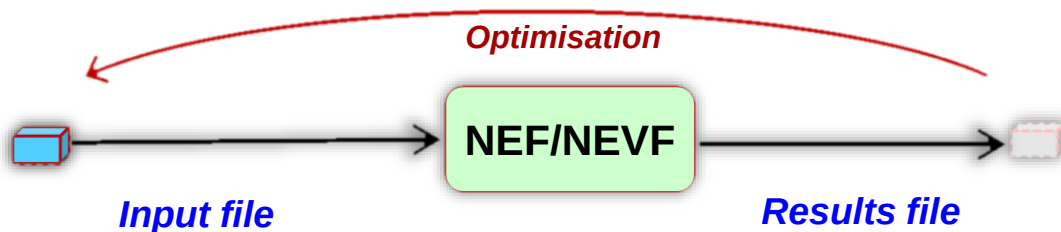


CAEfatigue Process Flow and control files

INTERACTIVE or BATCH: GUI driven
INTUITIVE: easy and repeatable workflows
WHAT IF: easy interrogate and modify
REPORTING: nested postprocessing
ASSISTED: linked training and documentation

Embedded Fatigue Analysis

(no need for intermediate stress results)



Nastran Embedded Fatigue NEF(time) / NEVF(frequency)

FASTER: Drastic reduction of fatigue analysis time
SMALLER: Large intermediate files eliminated
SIMPLER: combines FE and Fatigue
PORTABLE: Process embedded in the solver file
TRANSPARENT: Process is transparent (text lines)
DESIGNS Optimization: coupling fatigue analysis and FE optimization and **Robust Design**

How to setup a fatigue analysis with NEF?

Case Control (defines fatigue life output request):

FATIGUE[([SORT1, SORT2] , [PRINT, PLOT] , [FORMAT=CODE] , [BULK])] = **n**

Bulk Data:

	1	2	3	4	5	6	7	8	9	10
FIGPARAM	ID	TYPE	FACTOR	NTHIRD	LOGVAL					
	"VIRESS"	COMB	CORR	PLAST	LOG					
	"POS"	DURATION								
	"VEFTG"	ATYPE	MAXSTR	CLIPVAL	MAXPEAK	STRENGTH	MAXFREQ	NCALC		
FIGDEF	ID	IOPEMS								
	"ELSET"	ELSID1		ELSID2		ELSID3				
	"XELSET"	XELSID1	XELSID2	XELSID3	XELSID4	XELSID5				
FIGSEQ	ID	EVENIOU1	METHOD							
	FID1	N1	FID2	N2	FID3	N3				
FIGEVEN1	ID	FLOAD1	FLOAD2	FLOAD3	FLOAD4	FLOAD5	FLOAD6			
FIGLOAD	ID	TID	LCID	LDM	SCALE	OFFSE1	TYPE			
RANDPS	SID	J	K	X	Y	TID				
TABRND1	TID	XAXIS	YAXIS	Z	E	Z3				
	x1	y1	x2	y2	z3					
UDNAME	UID									
	NAME									
TIMEPSD	ID	STATE								
	"EVENT"	EVNTID1								
	"MAP"	LCID1	CHAN1	LCID2	CHAN2					
MAIFIG	MID	CONV1	MAIOU1							
	"STAIN"	YES	UIS							
	"SN"	SEN	N1	N2	N3					
	"TABLE"	VALUE1	TID1							
	"MMPDS"			A3	A4					
	"EN"			Ef	n	K	GAMMA			
				SEc	Ne	FSN	TOLER			
TABLEM1	TID	XAXIS	YAXIS							
	x1	y1	x2	y2	x3	y3				

What to Calculate

Where to Calculate

Random Loading

Materials

Points to SET1, SET3, SET4 entries to define elements

Points to SUBCASE or TF stress/strain extraction depending on TYPE

Points to UDNAME for FFT conversion or STATIC offset depending on TYPE

Points to FTGLOAD of TYPE = PSD for FFT conversion

Associated to PSHELL, PSOLID, etc and corresponding MAT1 entries.

Points to TABLEM1 for defining discrete S-N curves.

- Required case control cards:
 - FATIGUE (select what to run)
- Required case control cards:
 - FTGDEF (where to calculate)
 - FTGSEQ (sequence of loading)
 - FTGEVENT (simultaneous loads)
 - FTGLOAD (define loading)
 - MATFTG (define S-N or e-N data)
 - DTI,UNITS (set units needed)

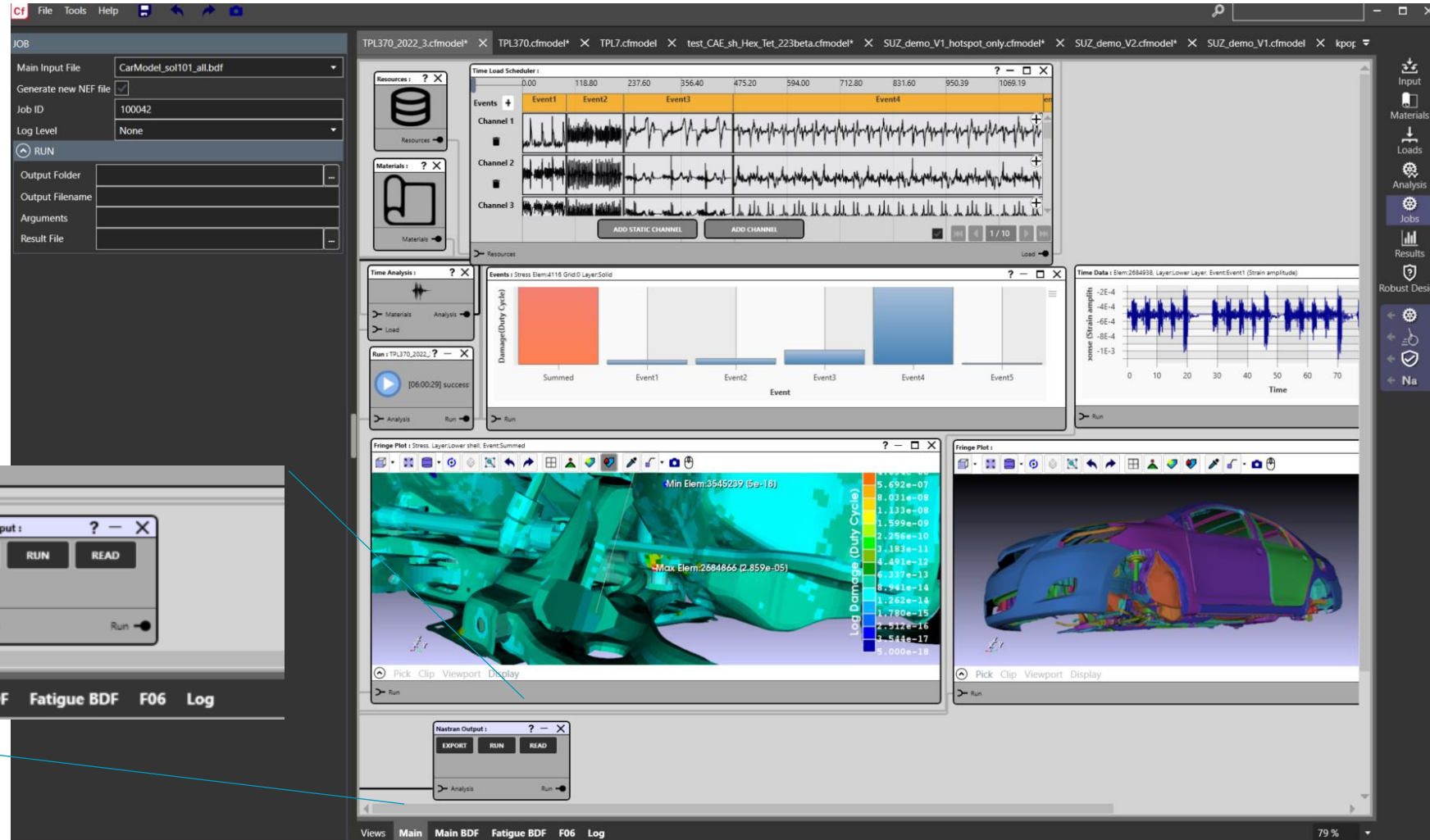
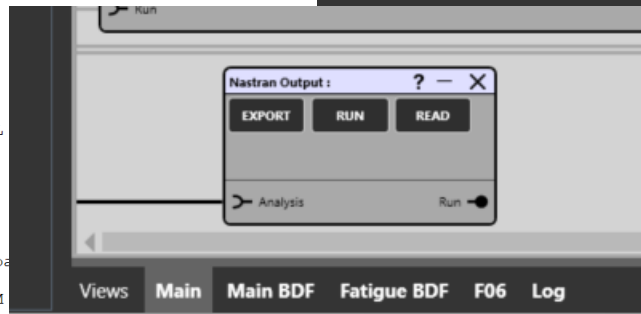
The CAEfatigue GUI takes care of this for you!

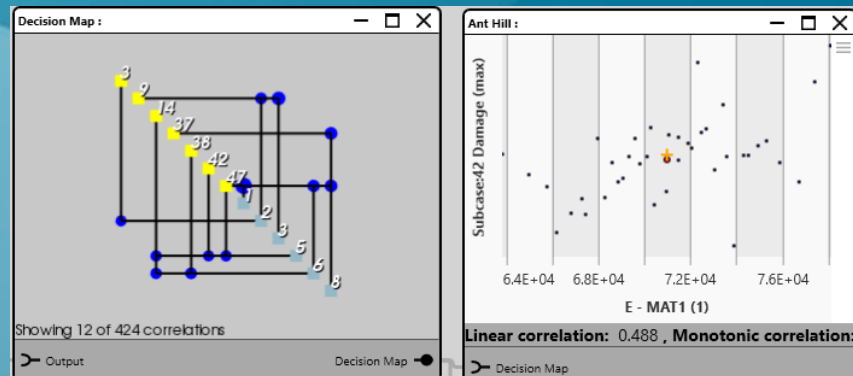
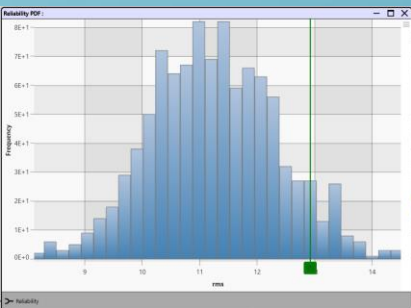
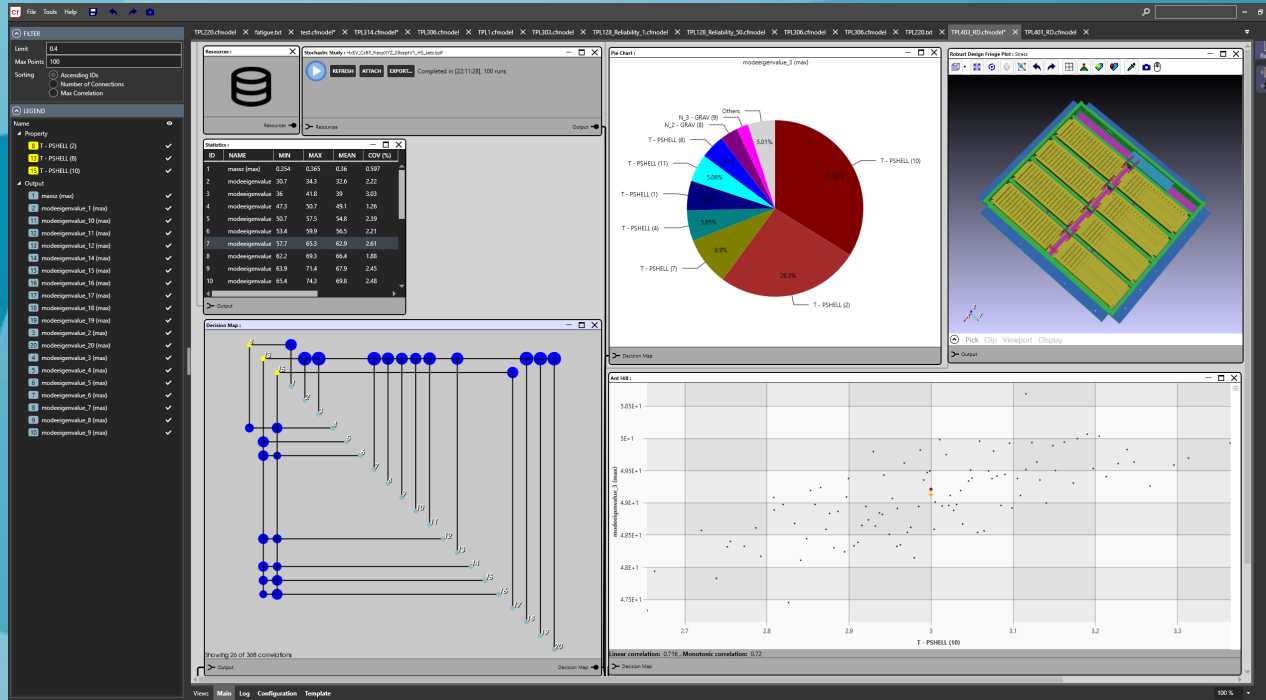
Support for Nastran Embedded Fatigue “NEF” pre-post processing

Control File input can be directly added to an existing Nastran time domain job

- CAEfatigue Process Flow increments a Nastran input file with the additional cards needed to run NEF
- NEF can be run Interactively or in batch
- Use CAEfatigue for Postprocessing or your preferred tool (via Nastran Op2 or H5)

```
SOL 112
CEND
TITLE = CANTILEVER BEAM NEF MODEL
ECHO = NONE
RESVEC = NO
LOADSET = 5
FATIGUE (SORT1, FORMAT=64, BULK)=1
SUBCASE 1
  SUBTITLE=ALL
  METHOD = 1
  TSTEP = 1
  SPC = 2
  DLOAD = 6
  STRESS (PLOT, SORT2, REAL, VONMISES, BILIN) =ALL
$ Direct Text Input for this Subcase
BEGIN BULK
PARAM POST 1
PARAM GRDPNT 0
PARAM PRTMAXIM YES
DTI UNITS 1
$ SN Analysis, Load Sequence: Simultaneous_Load
FTGPARM 1 SN 1. 1 0
          STRESS ABSMAXPR GOODMAN ELEM
          CERTNTY50.
FTGSEQ 1 0 0
        1 1.
        UNITS 1. Repeats
TSTEP 1 108 1. 1
EIGRL 1 10 0
$ Elements and Element Properties for region : Shells
PSHELL 1 1 .001 1 1
PFTG 1 1 NONE 1. 1. 0.
        1
```





Robust Design

Robust Design Applications:
A stochastic approach incorporating
uncertainties in FE and Durability

What is Robust Design?

- pioneered by Dr. Genichi Taguchi, Robust Design methodologies aim to develop products and processes that are robust or insensitive to variation of their environment
- **ro·bust** [roh-buhst, *adjective*, from latin *robustus*]
(of a material object) sturdily constructed, solidly build, resilient, not easily damaged
(of a system/process) able to withstand or overcome adverse conditions, capable of performing without failure under a wide range of conditions
- in engineering design, the object is product robustness towards a function and purpose without compromising other qualities such weight, production or material costs

Challenges in Stochastic Simulation: measuring uncertainty

$$CV = \frac{\sigma}{\mu}$$

Relevant variation can be found in parameters such as

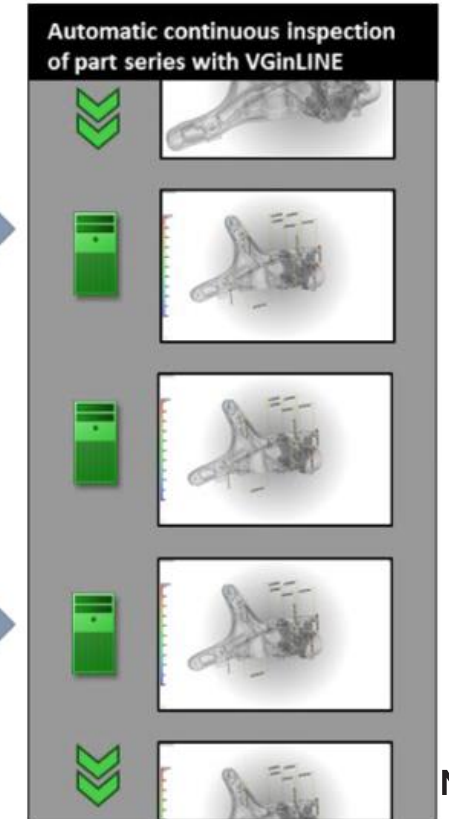
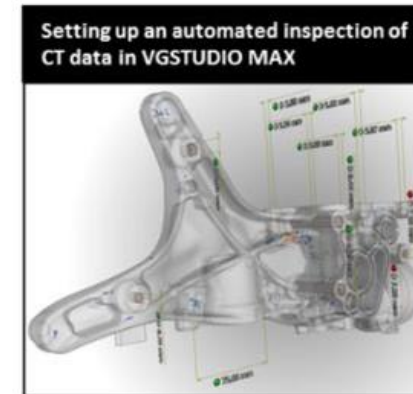
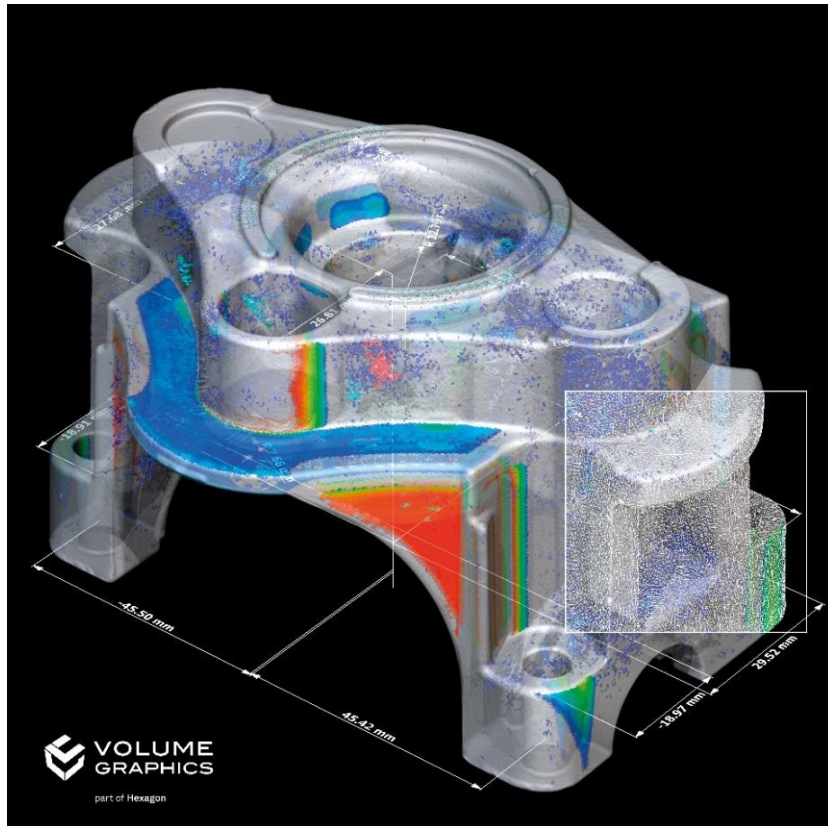
- Material properties
- Component dimensions due to tolerances
- Variation in positional tolerances for assemblies
- Geometry imperfections (flatness, offset, tissue fiber orientation & direction)
- Manufacturing process failures
 - Rivets not riveted properly
 - Bonds not bonded totally
 - Welds not welded completely
- Variation in environmental conditions
 - Loading
 - Boundary and initial conditions
 - Temperature

Quantifying variability is challenging, but a precise probability density function may not be needed for many variables where natural distributions occur and coefficients of variation may be assumed

Material	Characteristic	C of V (%)
Metallic	Rupture Buckling	8-15 14
Carbon fiber	Rupture	10-17
Screw, rivet, welding	Rupture	8
Bonding	Adhesive strength Metal/metal	12-16 8-13
Honeycomb	Tension Compression Face wrinkling	16 10 8
Inserts	Axial loading	12
Thermal protection (AQ60)	On-plane tension In-plane compression	12-24 15-20
Load Type	Source	C of V (%)
Launch vehicle thrust	STS, Ariane	5
Launch Vehicle, quasi-static loads	STS, Ariane, Delta	30
Transient	Ariane 4	60
Thermal	Thermal tests	8-20
Deployment shocks	Aerospatiale	10
Vibration	Satellite tests	20

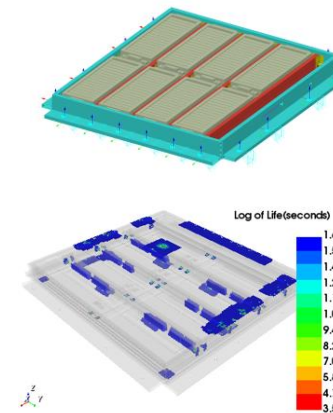
measuring uncertainty with non-destructive evaluation

- Using physical components, Industrial CT and Optical Scanning hardware and analysis software provide unprecedented insight into coefficient of **variation** for **key measurements**
- Based on the robustness investigation, **appropriate quality controls** can be implemented in manufacturing plants, even integrated in production lines



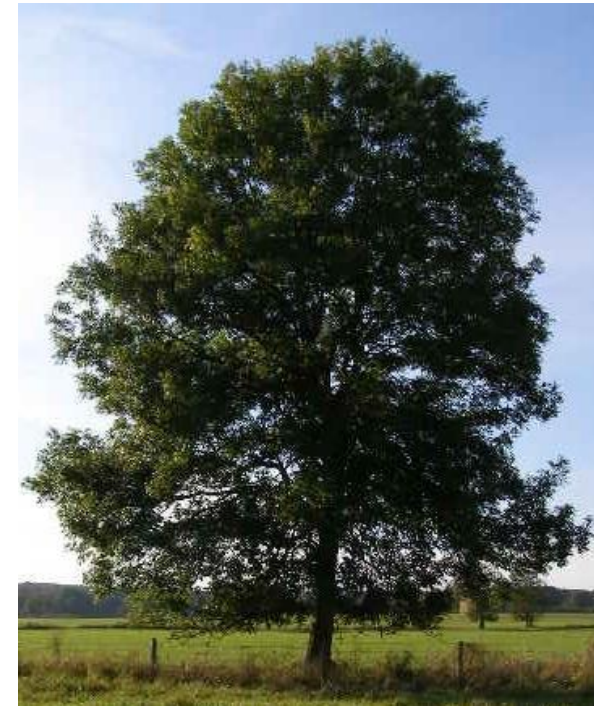
agenda

1. Intro to stochastic simulation workflow with Robust Design
 - Input randomization and function study
 - Building actionable knowledge
 - Stochastic Design Improvement
2. Use case: HV battery pack certification and ground vehicle testing (Ford and Dana example)
 - Physical test
 - Deterministic durability analysis
 - Stochastic Design and improvement cycles



Incorporating uncertainty... by design

- **OEMs strive to create improved and more robust designs** by identifying potential weak points, susceptible to variability
- **Uncertainty** exists in everything, from dimensions to material properties, loads, and boundary conditions.
- Incorporating these uncertainties leads to models of unprecedented levels of **realism, credibility and resilience**
 - **Reduces warranty and product liability** costs by identifying, correcting unwanted behaviors
 - **Reduces** weight penalty from **overdesign**



A perfect tree

Stochastic Simulation Workflow: randomization and function study

- Nominal FE: a given FE model corresponding to an assumed nominal reference point
- Input Randomization: specify the tolerances and variable scatter distributions on all FE inputs subjected to variability, including customer usage, manufacturing, load factors, material and geometrical
- Stochastic function study: a Monte Carlo simulation (MCS) where the model is executed multiple times (typically 100), randomly changing each of the input variables within the assigned limits

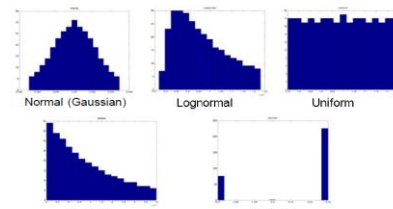
Nominal FE + Fatigue input/output model

VARIABLES	
Name	
▶ Material	
▶ Property	
▶ Connector	
▶ Temperature	
▶ Load	
▶ Output	
▶ SOL 101	
▶ Subcase 3	
▶ Subcase 4	
▶ Fatigue 100042	
▶ Global	

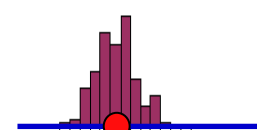
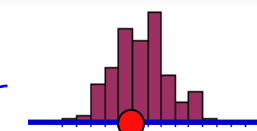
Name	
▶ Materials	
1 Static E - VMATFTG (60)	✓
2 EN b - VMATFTG (60)	✓
3 EN C - VMATFTG (60)	✓
4 EN Epsilon' - VMATFTG (60)	✓
5 EN K' - VMATFTG (60)	✓
6 EN N' - VMATFTG (60)	✓
7 EN Sigma' - VMATFTG (60)	✓
8 Static UTS - VMATFTG (60)	✓
9 Static Ys - VMATFTG (60)	✓
▶ Output	
1 Damage (max)	✓
3 Max Stress (max)	✓
2 RMS (max)	✓

Input randomization

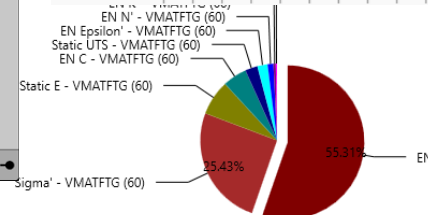
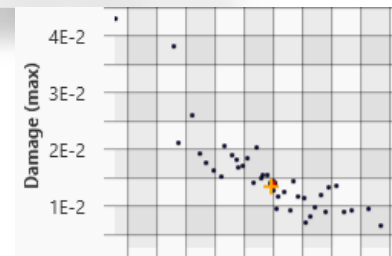
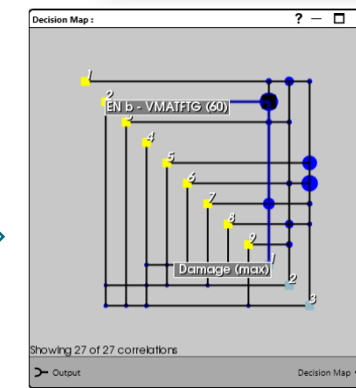
PSHELL (121)	✓	
PSHELL (122)	✓	
PSHELL (123)	✓	
Name	Value	Randomize
T	1.1	☒
Distribution	Gauss Normal	
Cutoff (#std.dev)	3	
Coef. of Variation (%)	5	



(model properties pdfs)

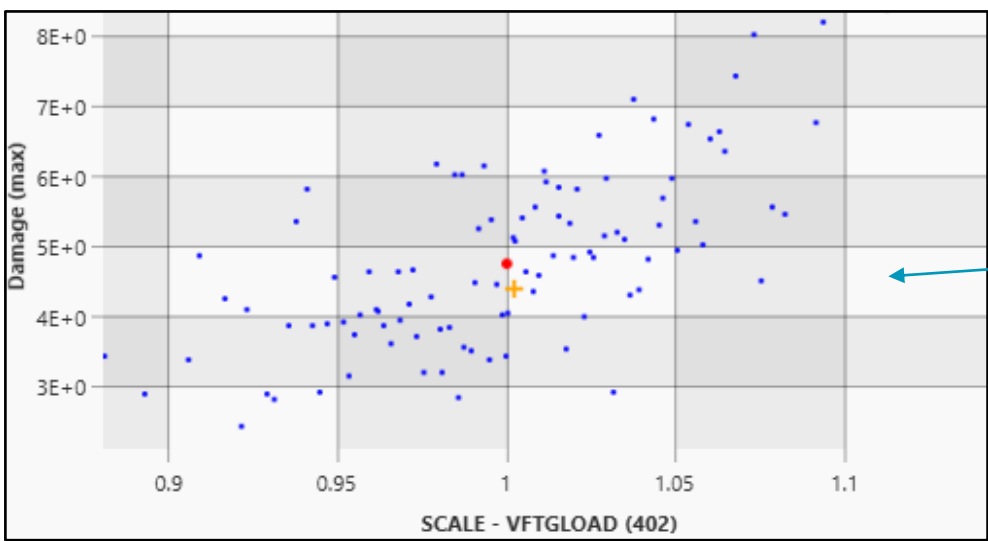


Function Study

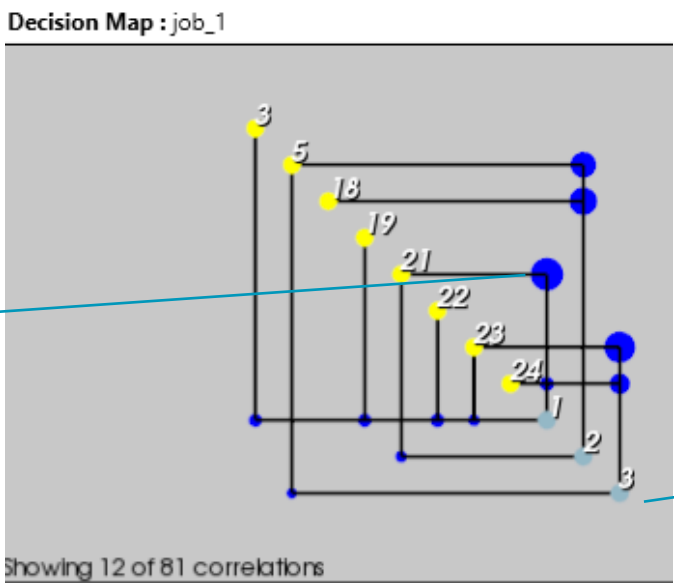


Stochastic Simulation Workflow: building actionable knowledge

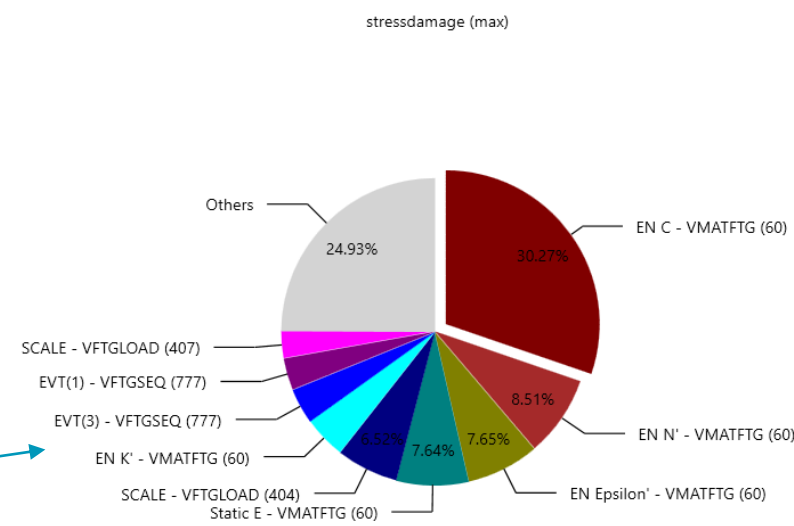
Accessible Data analysis tools enabling informed decision making about model modifications and enforcing or recommending limits for specific maneuvers and weight



Ant hill plots : distribution of input-output pairs, relationships, and correlation shape (thresholds, outliers, probability of exceedance)



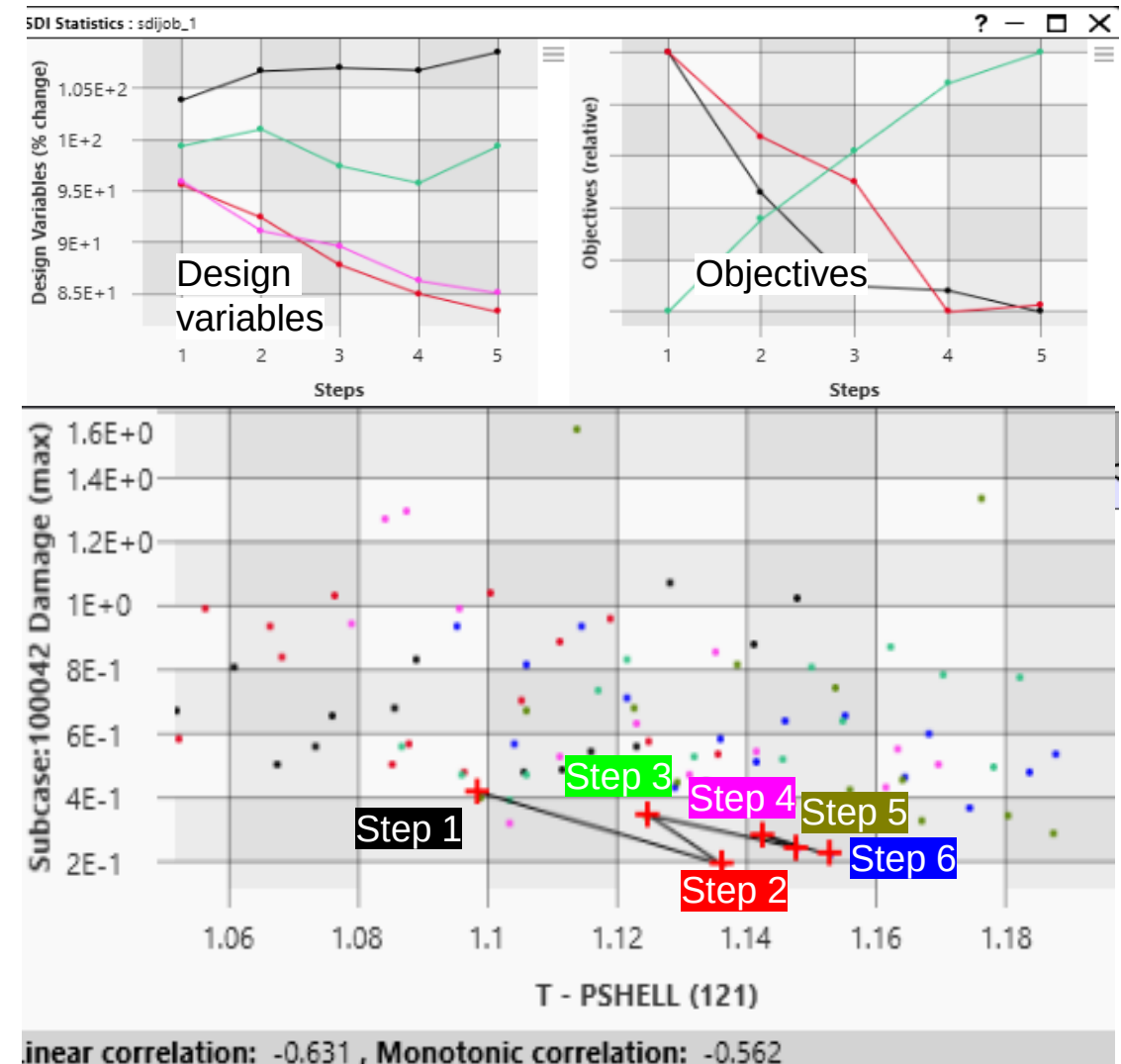
Decision maps: visual input-output correlation relationship influencing the functioning of a product



Pie charts: based on correlations ranking, reflects the relative weight and influence of any parameter variation on key outputs

Stochastic Simulation Workflow: Stochastic Design Improvement

- As a follow-on to a stochastic function study, Stochastic design improvement cycles (SDI) support gradual and robust design changes
- A gradual variation of selected design variables pushes the design towards one or more targets, achieving improved performance over the nominal analysis
 - A subset of Inputs become *Design Variables*
 - A subset of Outputs become *Objectives*
- Each stochastic simulation represents a single step. The SDI process terminates when the target *distance* is within a stated *tolerance*



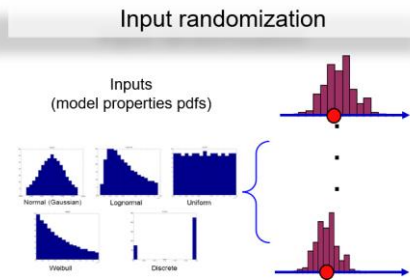
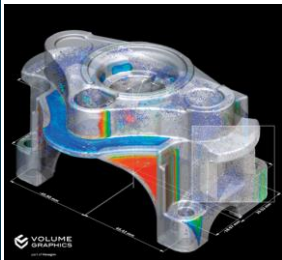


(CAEfatigue) Robust Design

Incorporating uncertainty by design

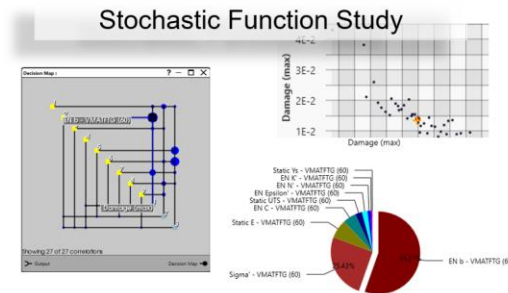
DEFINITION

- **Uncertainty** exists in everything from dimensions to material properties, loads, and boundary conditions. The incorporation of these uncertainties leads to models of unprecedented levels of realism and credibility.
- **OEMs strive to create improved and more robust designs** by identifying potential weak points, susceptible to variability
- Defining **appropriate quality controls** to be implemented in manufacturing plants



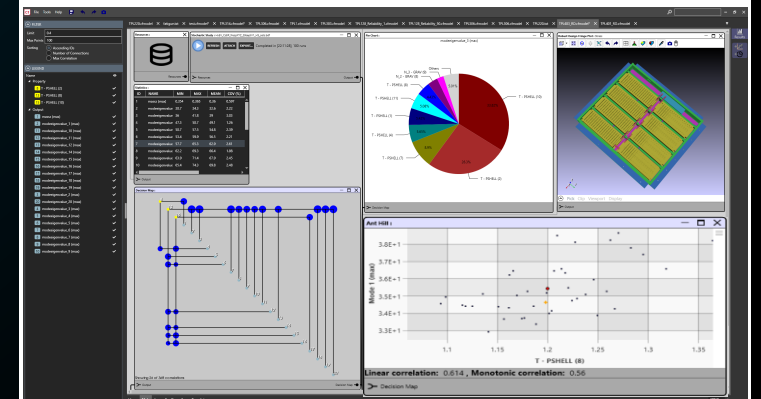
RECIPE

- **MSC Nastran + CAEfatigue technologies**
 - Monte Carlo DoE with intuitive randomization
 - Decision Maps and pie charts describing linear or non-linear correlation between inputs
 - Ant hill plots for determination of probability of exceedance
- **Industrial CT and optical scanning** for Non-Destructive uncertainty measurement
- **Stochastic Design improvement:** Reach design targets through robust improvement cycles



PURPOSE & BENEFITS

- **Simulation Health checks:** assess model numerical quality
- **Design and engineering beyond margin of safety and certification**
- **Reduce warranty and product liability costs** by identifying, correcting unwanted behaviors
- **Improving Correlation** with physical test or operational events/conditions



Robust Design to Alter Nastran Parameters and Rerun Jobs

CAEfatigue has **significantly enhanced** the feature ROBUST DESIGN, which allows you to run rerun Nastran many times while altering the input parameters by a User specified amount. Outputs show you the sensitivity of the output variable to the changing input parameters.

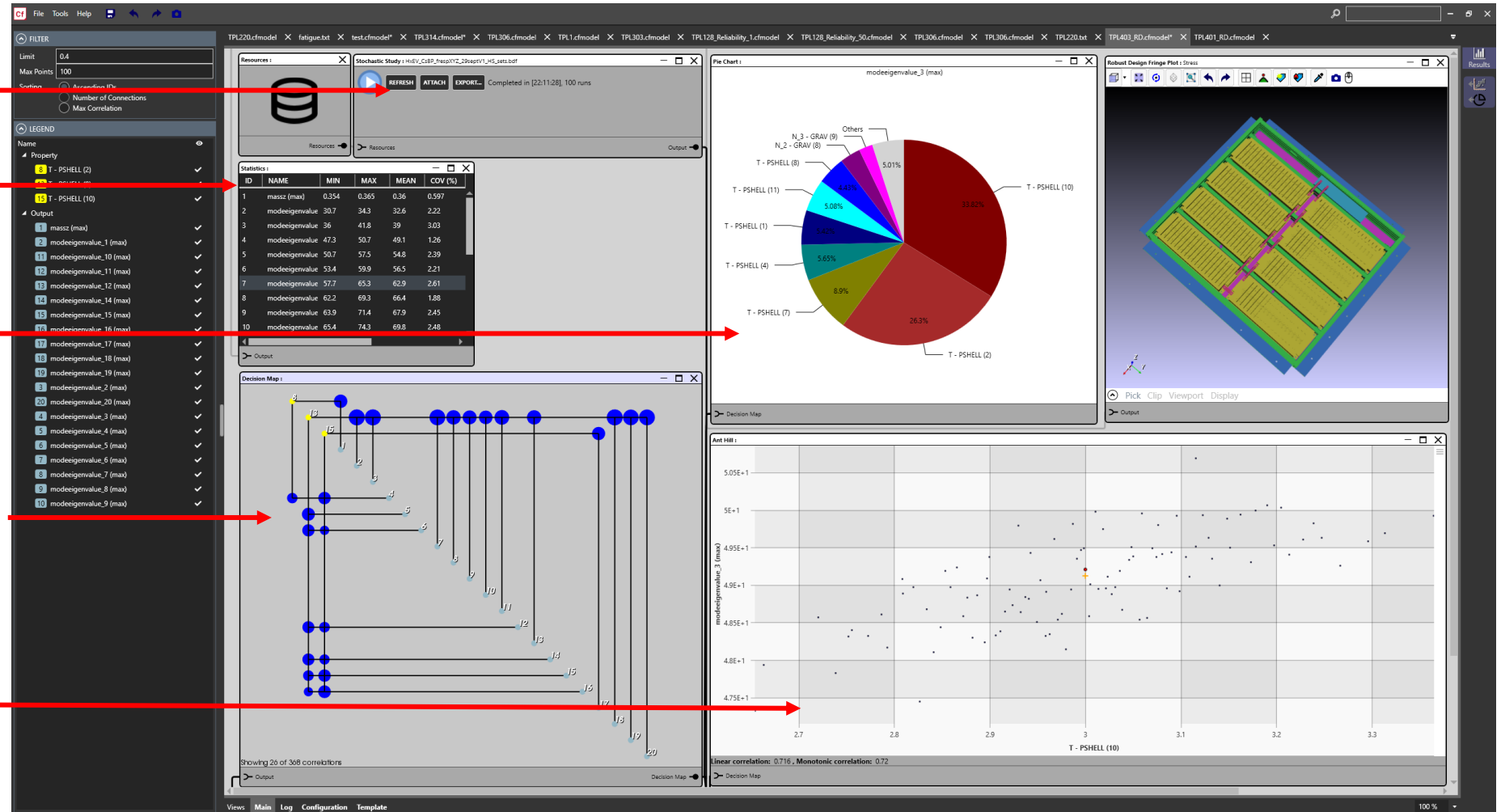
The **Stochastic Study** permits the selection of the inputs and output and controls the job operation.

The **Statistics** plot shows summary data for the Output variables

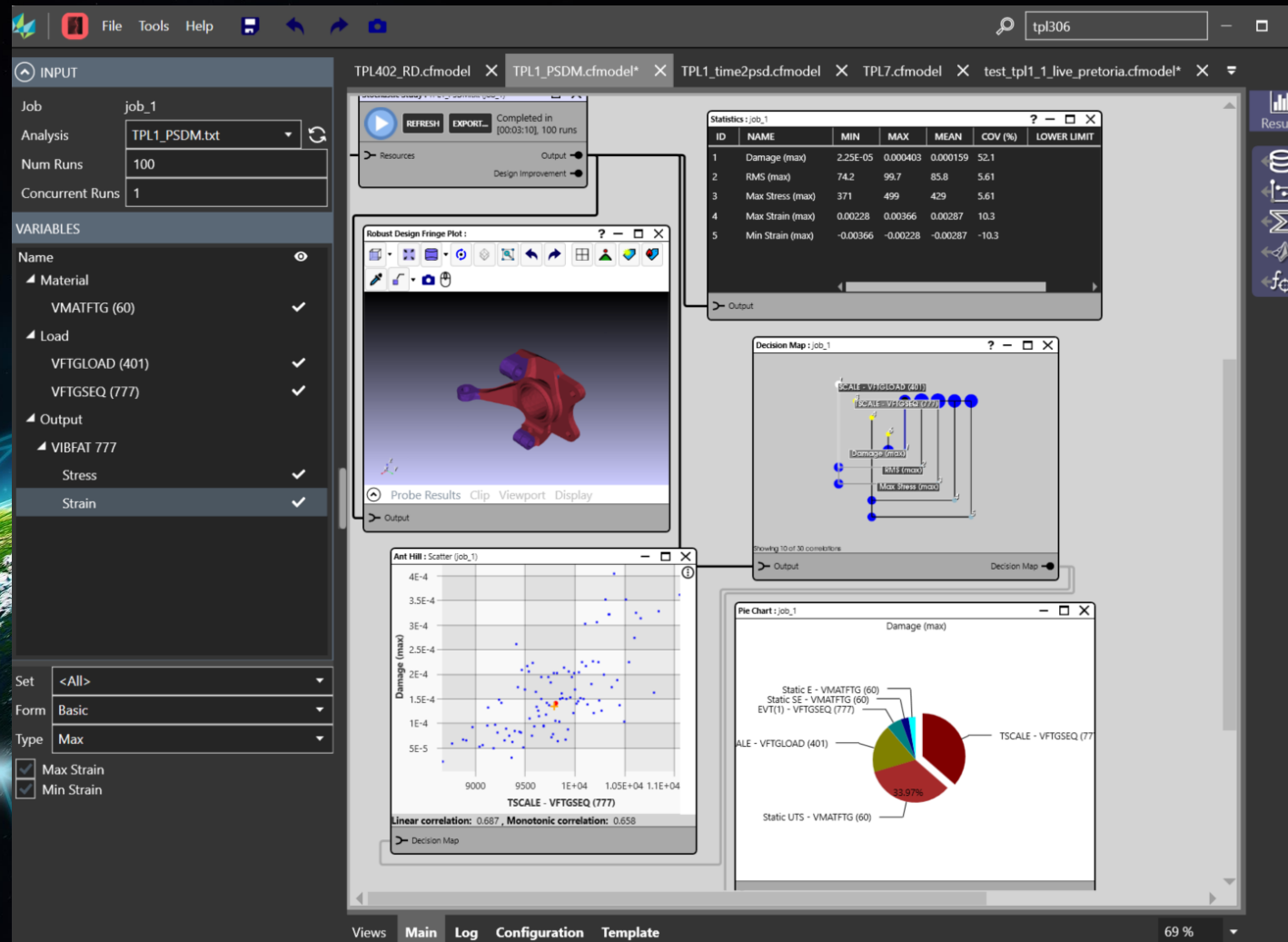
Pie charts: relative influence of the input variables on any observable output

The **Decision Map** shows the input to output correlation results and how different variables influence the functioning of a product

The **ANT HILL** plot is visual distribution of input output pairs, relationships and correlation shape



Workshop TPL 1 with PSD matrix and Robust Design



Virtual testing for HV EV battery pack

2021

1

© 2021 carhs.training gmbh

automotive
CAE
GRAND
CHALLENGE

Virtual vibration fatigue testing of an automotive EV battery pack

Presenter Marco Veltri, PhD
Hexagon | MSC Software

automotive CAE Grand Challenge 2021
October 19 - 20, 2021
Hanau, Germany & Online

HEXAGON

MSC Software

carhs
Empowering Engineers

BIW
Torsional Stiffness 22 kNm/°

+45%

BIW + Battery
Torsional Stiffness 32 kNm/°

Log of Life(seconds)

1.646e+01
1.529e+01
1.411e+01
1.294e+01
1.177e+01
1.059e+01
9.419e+00
8.246e+00
7.072e+00
5.899e+00
4.726e+00
3.552e+00

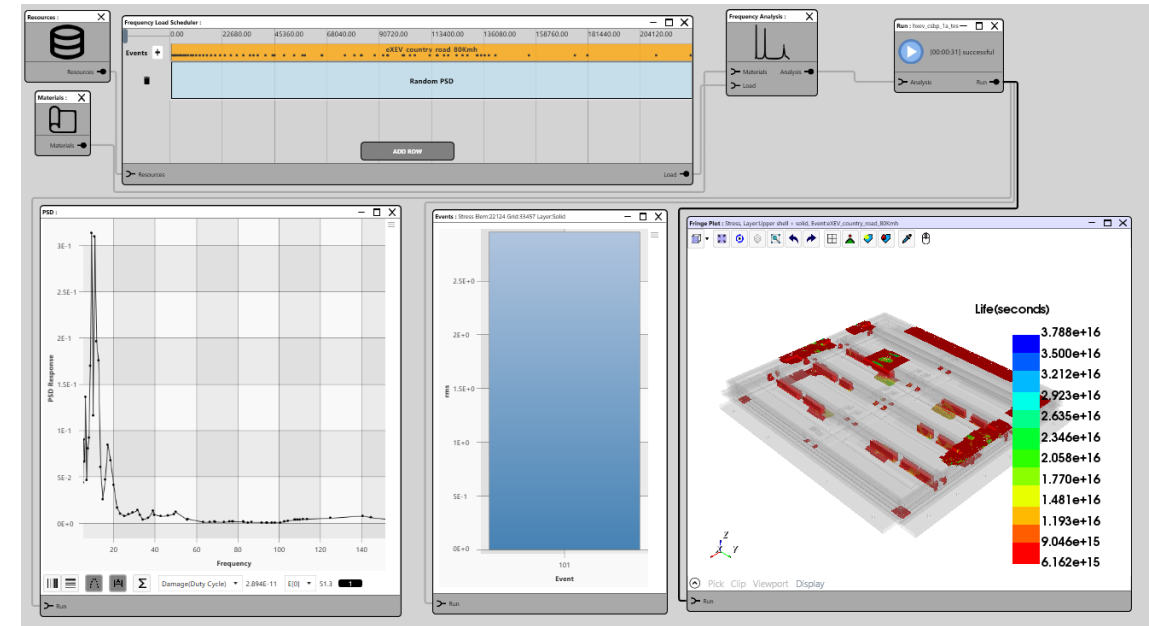
43 | hexagonmi.cc

Application example Robust Design to HV battery pack design

Incorporating Design Uncertainties in a Durability assessment – a Robust Design Approach

Dr. M. Veltri (presenter), M. Robinson, P. Adolfsson, Dr N.W. Bishop

Hexagon Manufacturing Intelligence

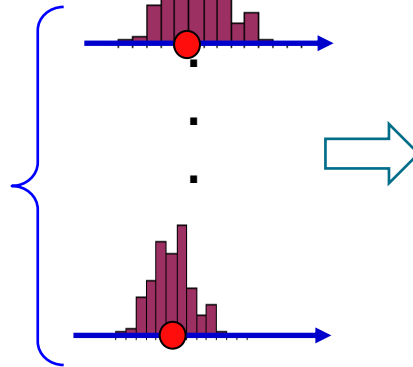
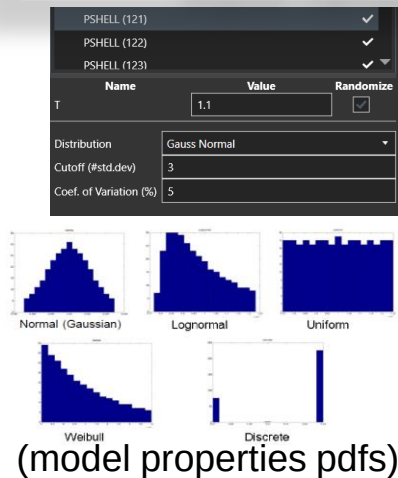


Nominal FE + Fatigue input/output model

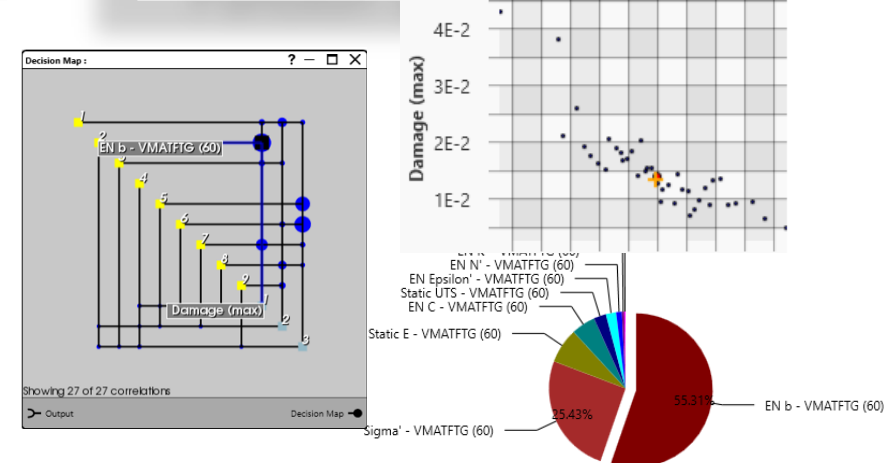
VARIABLES	Name
Material	1 Static E - VMATFTG (60)
Property	2 EN b - VMATFTG (60)
Connector	3 EN C - VMATFTG (60)
Temperature	4 EN Epsilon' - VMATFTG (60)
Load	5 EN K' - VMATFTG (60)
Output	6 EN N' - VMATFTG (60)
SOL 101	7 EN Sigma' - VMATFTG (60)
Subcase 3	8 Static UTS - VMATFTG (60)
Subcase 4	9 Static Ys - VMATFTG (60)
Fatigue 100042	1 Damage (max)
Global	2 Max Stress (max)
	3 RMS (max)



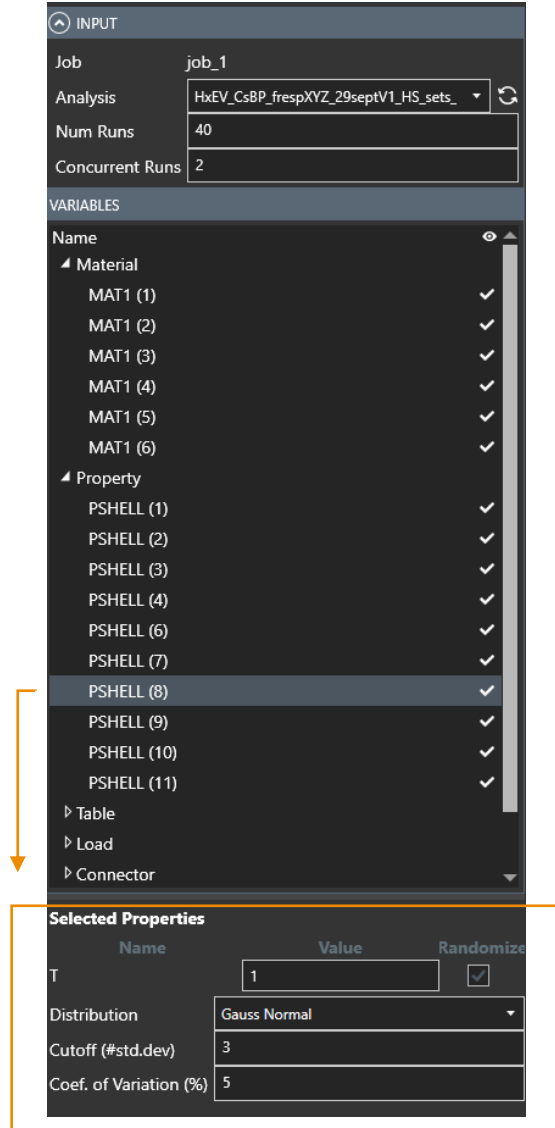
Input randomization



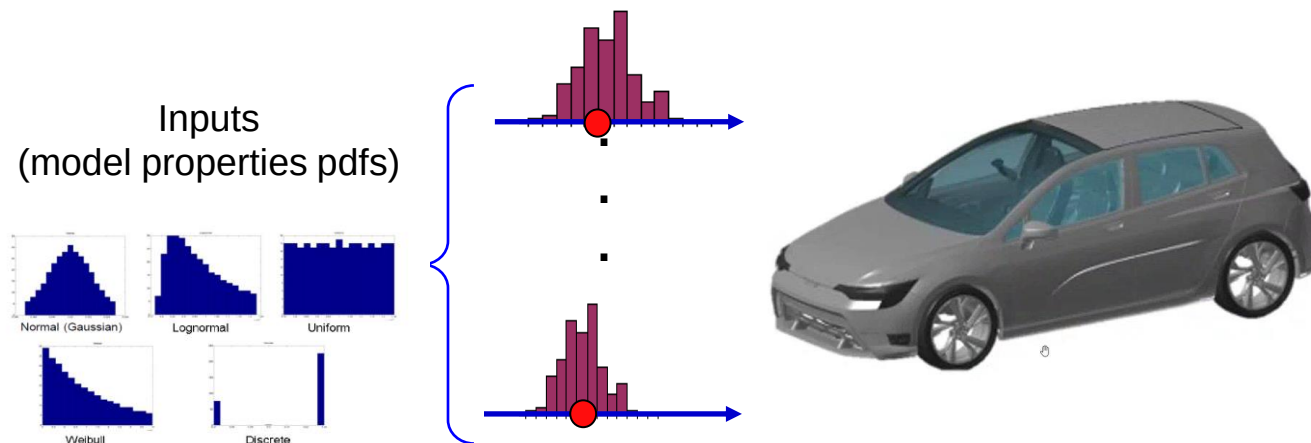
Function Study



Randomization of input variables

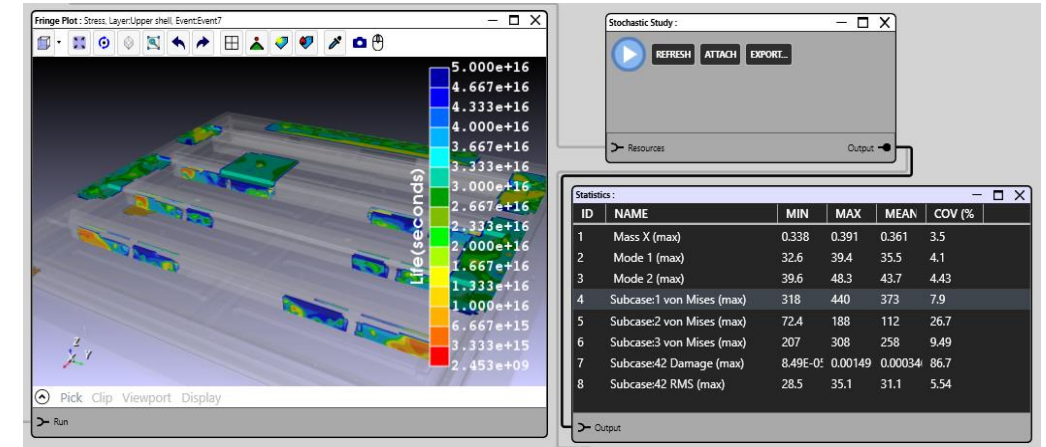


- For input properties that are not deterministic we use 3 pieces of information
 - The mean (nominal) value
 - The range (tolerances)
 - The Probability Density Function (PDF)
- Include every tolerance, everywhere, in a mouse “click”



Simulation and generating input-output statistics meta-model

- Concurrent runs are submitted and monitored (maybe on a server)
 - The model can be any size with any number of stochastic variables
- Typically, around 100 analysis runs are sufficient
 - Each time, different inputs values are selected randomly from the bounded PDFs you defined
 - Special sampling logic ensures similar values are not selected and that the full range of the PDFs are covered during randomization selection
- Only input-output statistic “meta-model” data is collected (typically a few KB)
 - Using Nastran embedded fatigue allows tackling effect of design changes both stress and fatigue in a single run



AutoSave: Off | From_CAEfatigue2Odyssey_metamodel.csv | Search (Alt+Q) | VELTRI Marco

File Home Insert Page Layout Formulas Data Review View Help Adams Drill

Clipboard | Font: Calibri 11 | Alignment | Number | Styles | Cells | Editing | Analysis

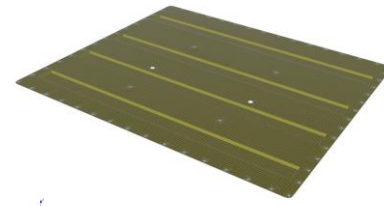
A1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	E - MAT1 (1)	NU - MAT1 (1)	RHO - MAT1 (1)	T - PSHELL	T - PSHELL	T - PSHELL	T - PSHELL	T - PSHELL	T - PSHELL	T - PSHELL	T - PSHELL	T - PSHELL	G(1) - TAB	F(2) - TAB	G(2) - TAB	X(1) - TAB
2	71000	0.3	2.70E-09	3	2	5	3	1.5	1.2	2.7	3	2	1.5	0.08	1.00E+20	0.08
3	69358.48	0.3033702	2.74E-09	3.096177	1.90445	4.918653	2.642749	1.432129	1.174048	2.574454	3.094832	2.225486	1.544489	0.077642	9.21E+19	0.079392
4	71983.77	0.2977175	2.65E-09	3.33165	2.037135	4.781589	2.934176	1.420954	1.084636	2.711683	3.173373	1.969501	1.426158	0.081122	1.01E+20	0.080832
5	69583.95	0.294003	2.91E-09	2.94269	1.94941	5.368108	2.825169	1.519727	1.166161	2.5561	3.023136	1.939997	1.474864	0.080888	9.60E+19	0.074191
6	66279.52	0.3158797	2.78E-09	3.172369	2.08999	4.98346	2.870163	1.510479	1.05943	2.483402	2.996739	2.185566	1.481194	0.084167	1.05E+20	0.081747
7	71237.16	0.3023751	3.10E-09	2.891588	1.958299	5.059357	2.970934	1.59948	1.186518	2.714929	3.138191	2.123425	1.520473	0.079322	1.09E+20	0.07525
8	75906.77	0.2701943	2.63E-09	2.977673	2.153155	4.891282	3.02703	1.533264	1.112925	2.791723	2.985675	1.77022	1.549322	0.088101	9.97E+19	0.077614
9	69070.98	0.2771473	2.54E-09	2.882452	1.886077	5.139475	3.076386	1.456328	1.139815	2.700911	2.882312	2.054195	1.455644	0.075076	9.29E+19	0.083657
10	67520.31	0.314597	2.61E-09	2.858714	2.061579	5.173768	3.267903	1.440229	1.261732	2.843232	3.067936	1.890725	1.364702	0.08311	1.00E+20	0.080407
11	68408.38	0.3057182	2.59E-09	2.816522	2.165951	5.108294	3.034629	1.502744	1.22723	2.796839	3.00386	1.971618	1.596454	0.082867	1.07E+20	0.078779
12	71697.02	0.3037985	2.73E-09	3.071218	2.051469	4.857668	3.214802	1.47209	1.228227	2.856809	3.121292	2.080666	1.503617	0.073096	9.65E+19	0.079071
13	69875.21	0.2806539	2.79E-09	2.785708	2.065908	4.801069	2.74837	1.536568	1.26919	2.360835	3.149377	1.8136	1.479181	0.082251	8.94E+19	0.084443
14	63720.43	0.2987474	2.80E-09	2.995339	1.914841	5.156013	3.085918	1.514043	1.302743	2.751605	2.811997	2.013282	1.683296	0.077038	1.02E+20	0.083541
15	74575.46	0.2688427	2.77E-09	3.03408	1.989176	5.480575	2.9928	1.49615	1.210949	2.592572	3.13051	1.90472	1.41365	0.079672	1.02E+20	0.084028
16	64986.33	0.3095066	2.68E-09	3.229442	2.033421	5.195103	3.141644	1.499647	1.244846	2.770534	2.765157	1.828021	1.568093	0.085049	1.02E+20	0.089007

From_CAEfatigue2Odyssey_metamodel

We observe MODE 1, well correlated with thickness of cover plate

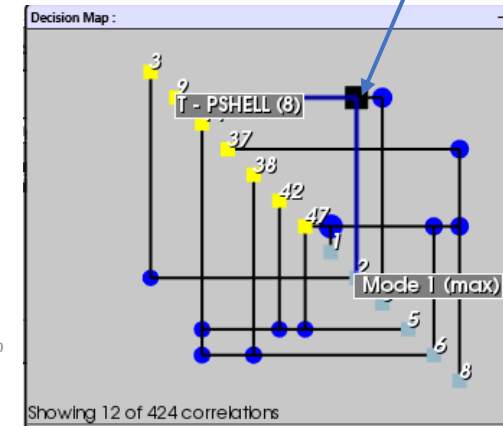
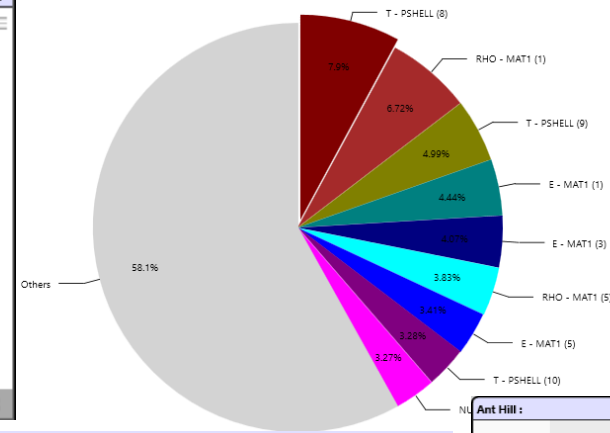
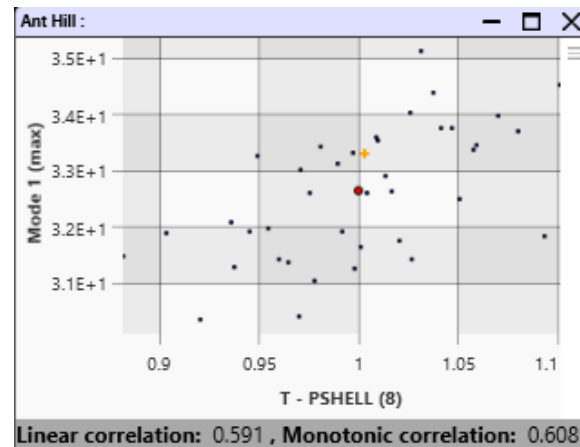
After our first stochastic study (50 runs), fatigue was fine but...

- Mode 1 showed unexpected high variation (30.4Hz to 35.1Hz)
- Mode 1 mean value is below recommended 35Hz
- The decision map identified a good correlation between mode 1 and cover plate “PSHELL 8”, nominally 1 mm thick.



Cover plate
“PSHELL 8”

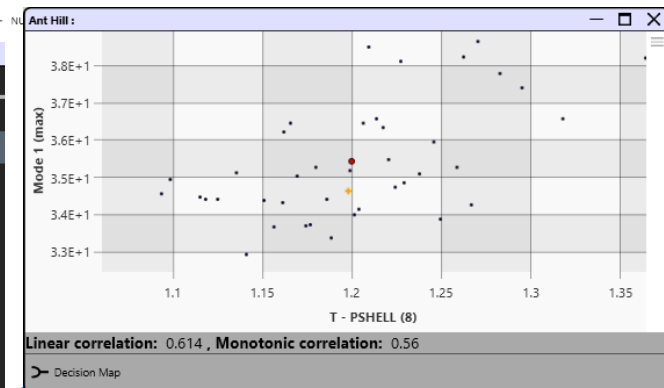
ID	NAME	MIN	MAX	MEAN	COV (%)
1	Mass X (max)	0.325	0.401	0.36	3.88
2	Mode 1 (max)	30.4	35.1	32.6	3.53
3	Mode 2 (max)	35.8	42.2	39	4.08
4	Subcase:1 von Mises (max)	291	446	374	9.12
5	Subcase:2 von Mises (max)	154	252	193	12.7
6	Subcase:3 von Mises (max)	196	359	267	14.5
7	Subcase:42 Damage (max)	7.87E-12	9.08E-12	8.4E-12	2.93
8	Subcase:42 RMS (max)	2.31	3.32	2.73	8.76



A new stochastic function study increasing the cover plate's thickness to 1.2 mm and no fear of affecting other parameters resulted in

- Mode 1 mean value >35Hz,
- fatigue damage, mass etc. near unaffected

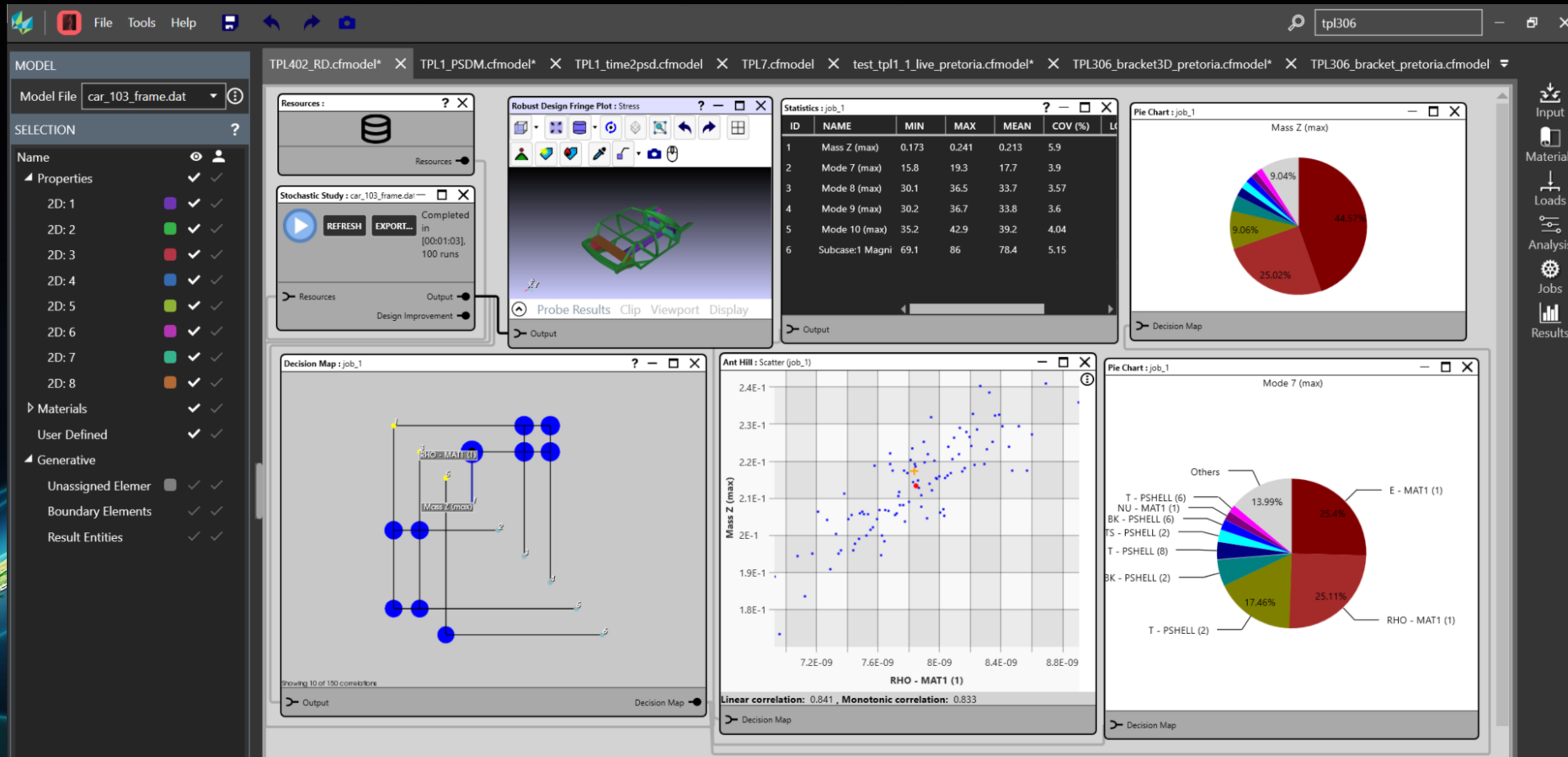
ID	NAME	MIN	MAX	MEAN	COV (%)
1	Mass X (max)	0.339	0.39	0.362	3.36
2	Mode 1 (max)	32.8	38.6	35.4	4.43
3	Mode 2 (max)	39.6	47.9	43.6	4.95
4	Subcase:1 von Mises (max)	320	472	376	10.9
5	Subcase:2 von Mises (max)	160	230	192	9.98
6	Subcase:3 von Mises (max)	182	352	258	12
7	Subcase:42 Damage (max)	8.11E-12	1.06E-11	8.9E-12	6.93
8	Subcase:42 RMS (max)	2.35	3.25	2.73	7.25



HEXAGON

carhs
Empowering Engineers

Workshop TPL 402RD



Application example Robust Design to automotive durability assessment



A Stochastic Approach to Designing Robust Automotive Structures Considering Variability



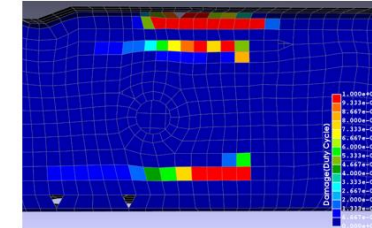
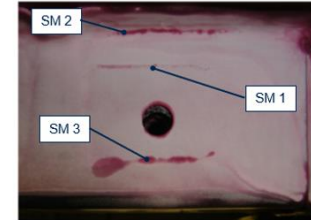
Dr. P. Roemelt,
Ford Motor Company
Dr. M. Veltri, Mark Robinson, Patrik Adolfsson
Hexagon Manufacturing Intelligence
Presenter: Jeff Robertson
Hexagon Manufacturing Intelligence



Comparison with physical test



- the fatigue analysis using nominal parameters shows good alignment with the critical areas identified in the physical test

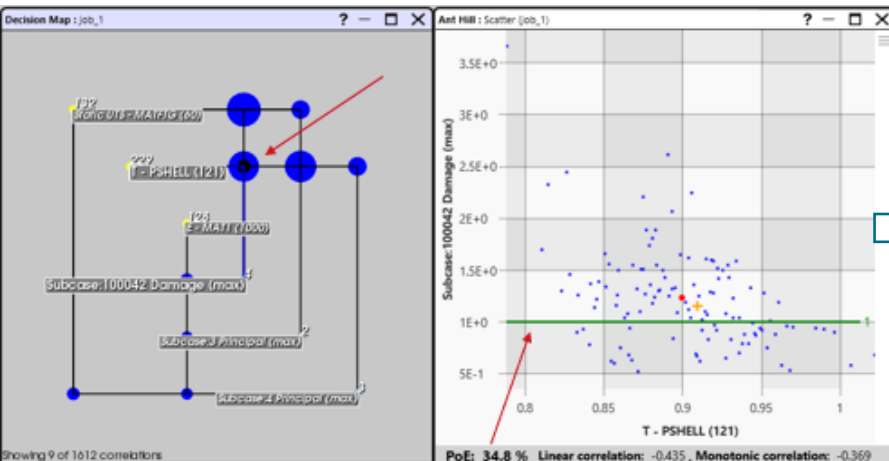


- An ensuing stochastic function study automatically incorporates uncertainties with the aim of building knowledge about model robustness

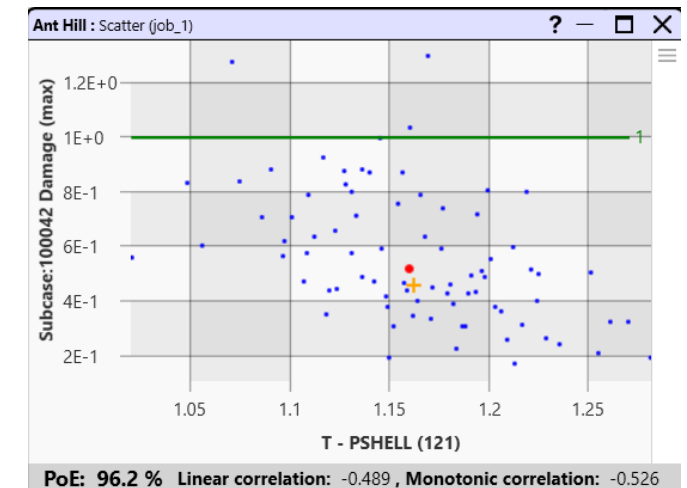
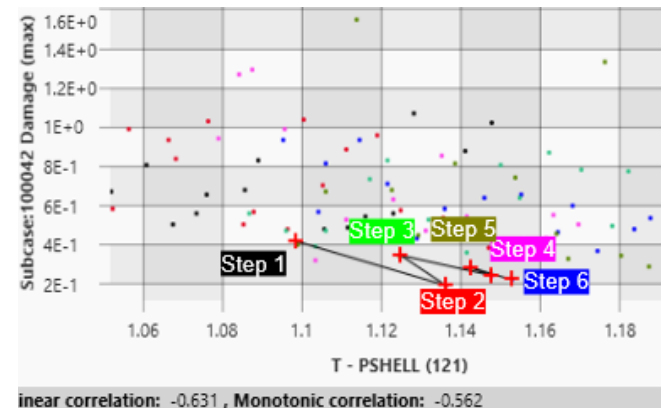


NAFEMS World Congress 2023 | Tampa, Florida, USA | May 15-18 2023

From stochastic Design function study



To stochastic Design Improvement cycles

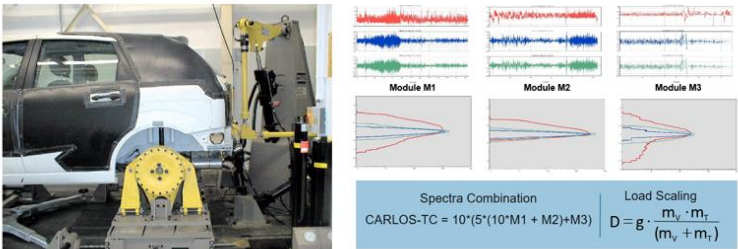


Test and simulation comparison

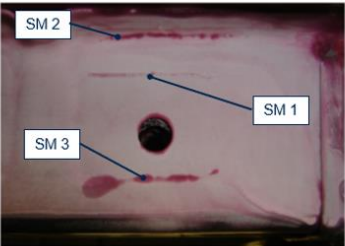
Trailer towing use case

- An “ideal” nominal structure is evaluated for durability under “CARLOS” standard spectrum fatigue test (ECE Regulation 94/20)

CARLOS test setup and load spectrum, with loading is applied in three directions at the tow ball



During test, discrete fatigue cracks developed in the sheet metal where towing device is bolted to the vehicle side rails

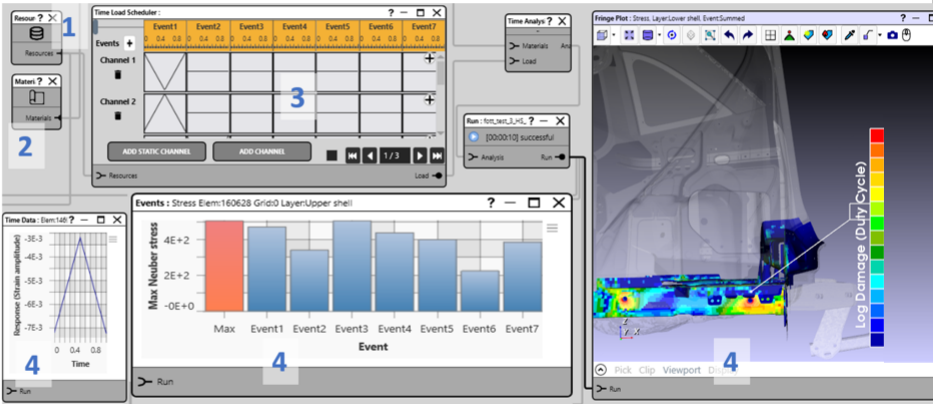


- A Stochastic Design Function Study measures the effects of different parameters tolerance against the nominal performance
- A subsequent Stochastic Design Improvement demonstrates an approach to ensure robust designs under assumed tolerances and material uncertainties



Strain-Life duty cycle process flow

1. Resources: input FE model and results
2. Materials: library for Strain life modelling
3. Time-Load scheduler: multiple events through linear superposition of 14 channels/subcases
4. Outputs: per event as well as per collective “duty cycle” schedule



critical sheet metal area

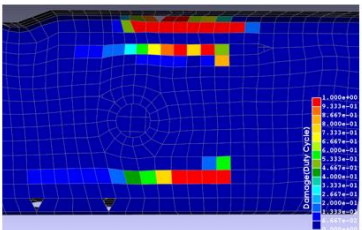
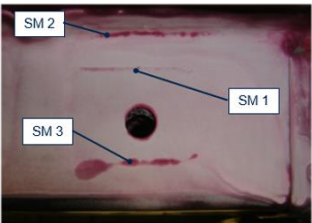
The interactive process generated in CAEfatigue is transparently nested into the FE solver (MSC Nastran)



Comparison with physical test

the fatigue analysis using nominal parameters shows good alignment with the critical areas identified in the physical test

- the fatigue analysis using nominal parameters shows good alignment with the critical areas identified in the physical test



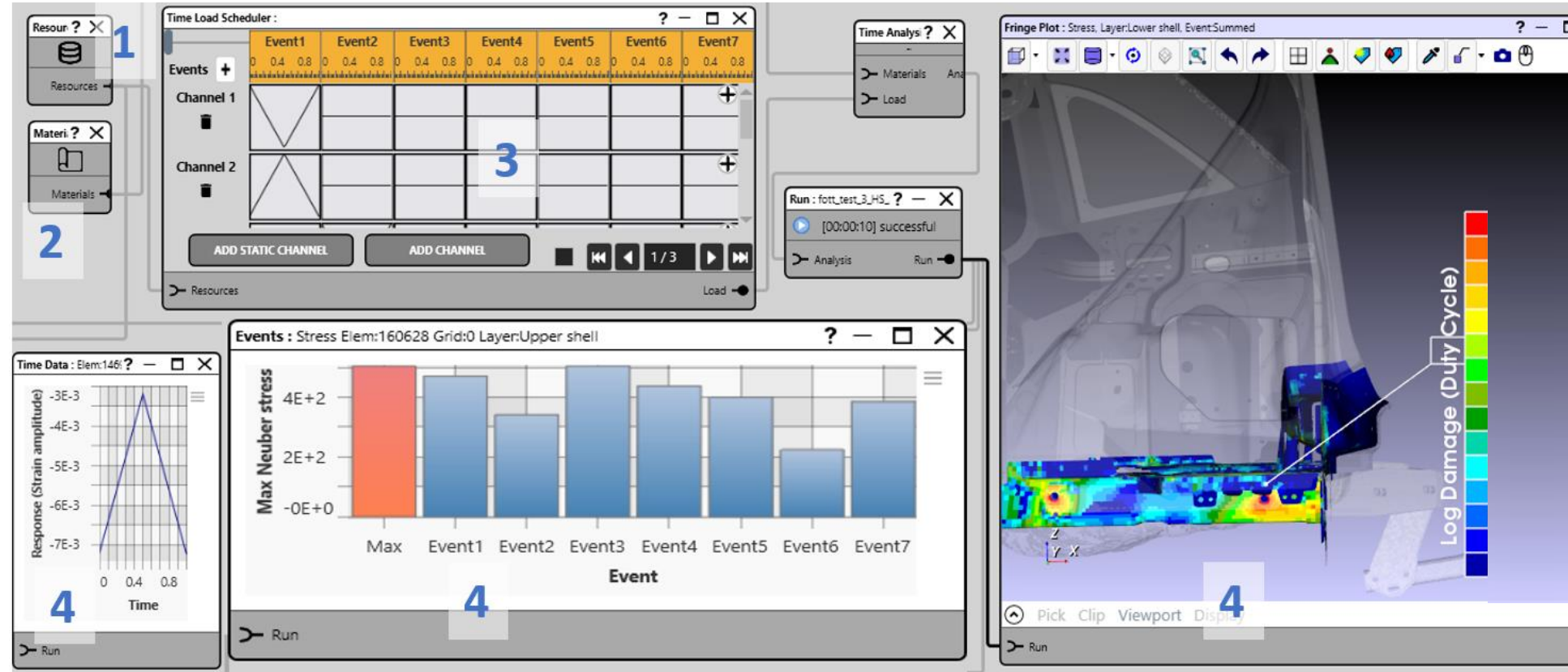
An ensuing stochastic function study automatically incorporates uncertainties with the aim of building knowledge about model robustness





Strain-Life duty cycle process flow

1. Resources: input FE model and results
2. Materials: library for Strain life modelling
3. Time-Load scheduler: multiple events through linear superposition of 14 channels/subcases
4. Outputs: per event as well as per collective “duty cycle” schedule



critical sheet metal area

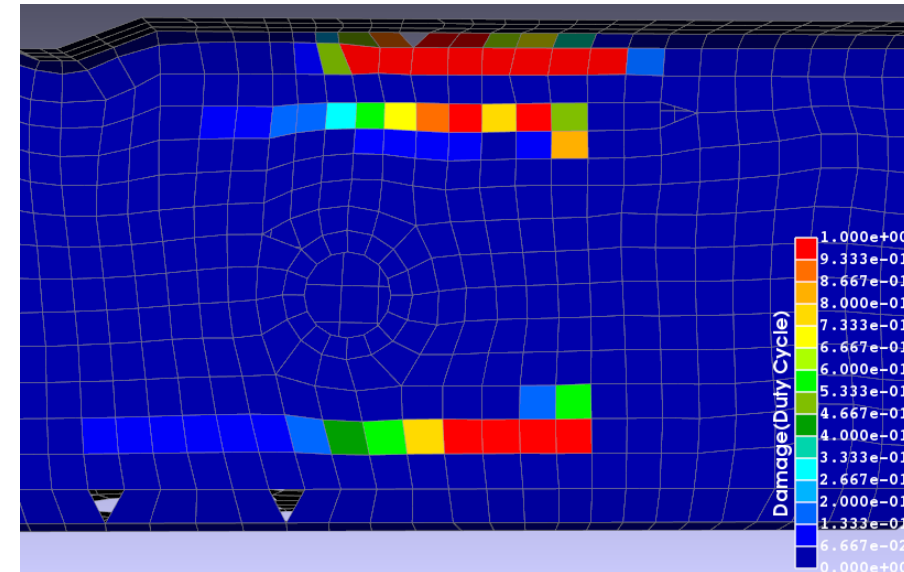
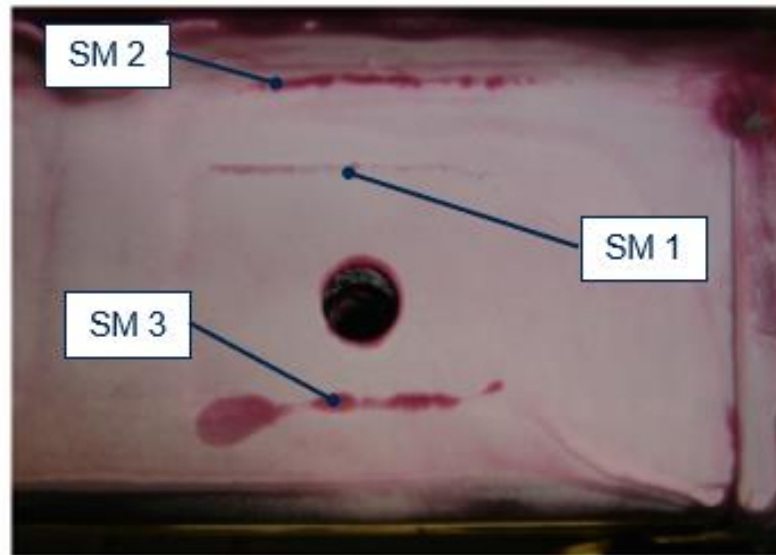
The interactive process generated in CAEfatigue is transparently nested into the FE solver (MSC Nastran)





Comparison with physical test

- the fatigue analysis using nominal parameters shows good alignment with the critical areas identified in the physical test

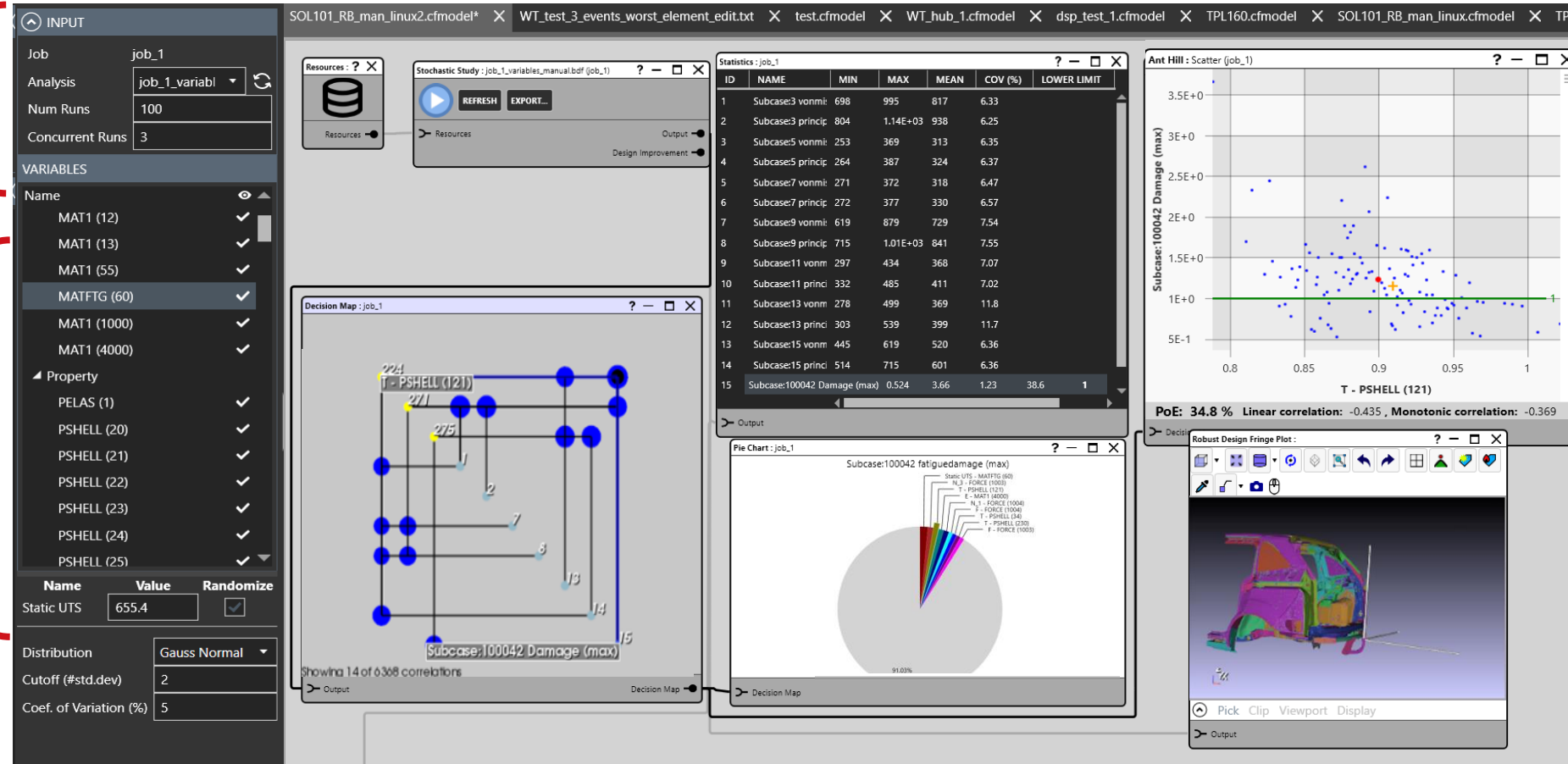


- An ensuing stochastic function study automatically incorporates uncertainties with the aim of building knowledge about model robustness

Stochastic function study process flow



- Managing the FE solver MC queue (e.g. 100 runs, 3 concurrent jobs)
- Allows automatic input randomization of large number of parameters (FE and fatigue input file simultaneously)
- Run in batch on a HPC server while interactively postprocessing locally





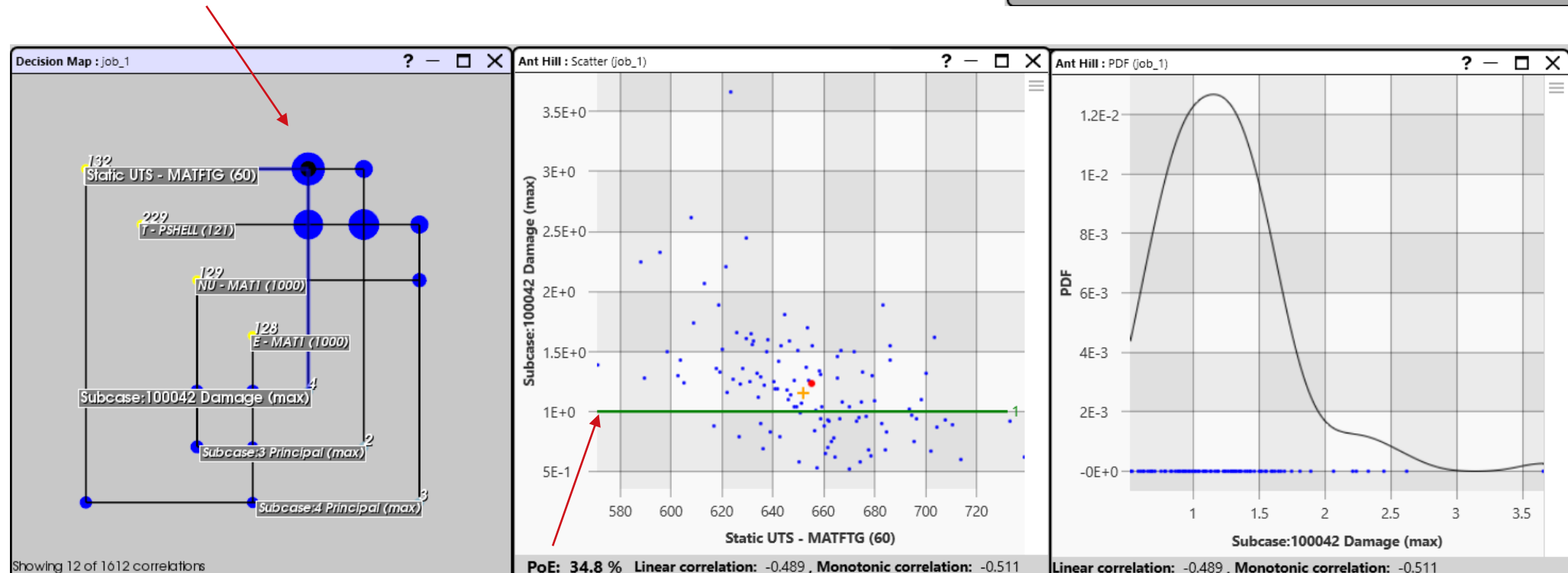
Trailer towing use case

120 runs stochastic function study review

- probability of survival ($\text{Damage} < 1$) is 34.8%
- Greatest correlation is found with fatigue e-N curve (“static UTS”) and sheet metal thickness in the area of failure (“T-PSHELL 121”)

ID	NAME	MIN	MAX	MEAN	COV (%)	LOWER LIMIT
1	Mass Z (max)	0.000844	0.00109	0.000962	4.84	
2	Subcase:3 Principal (max)	807	1.14E+03	934	5.75	
3	Subcase:4 Principal (max)	98.2	483	311	22	
4	Subcase:100042 Damage (max)	0.524	3.66	1.23	38.6	1

Output



DAMAGE scatter plot around nominal value of material UTS



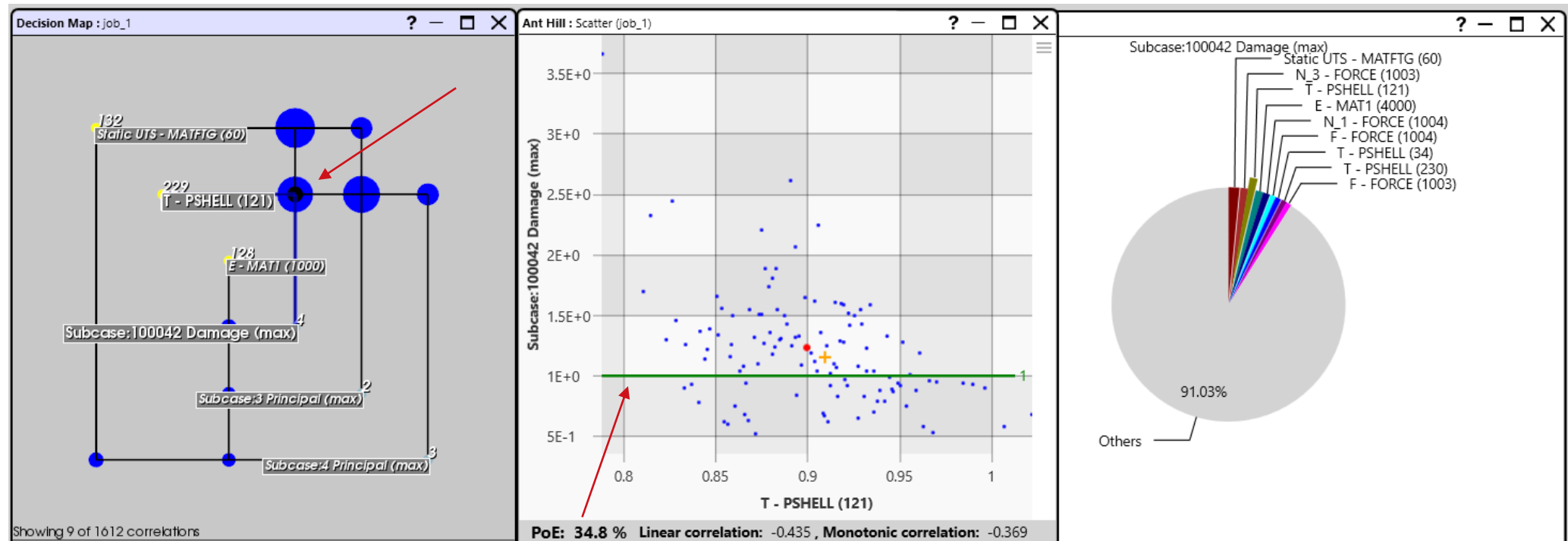


Trailer towing use case

120 runs stochastic function study review

- probability of survival ($\text{Damage} < 1$) is 34.8%
- Greatest correlation is found with fatigue e-N curve (“static UTS”) and sheet metal thickness in the area of failure (“T-PSHELL 121”)

ID	NAME	MIN	MAX	MEAN	COV (%)	LOWER LIMIT
1	Mass Z (max)	0.000844	0.00109	0.000962	4.84	
2	Subcase:3 Principal (max)	807	1.14E+03	934	5.75	
3	Subcase:4 Principal (max)	98.2	483	311	22	
4	Subcase:100042 Damage (max)	0.524	3.66	1.23	38.6	1



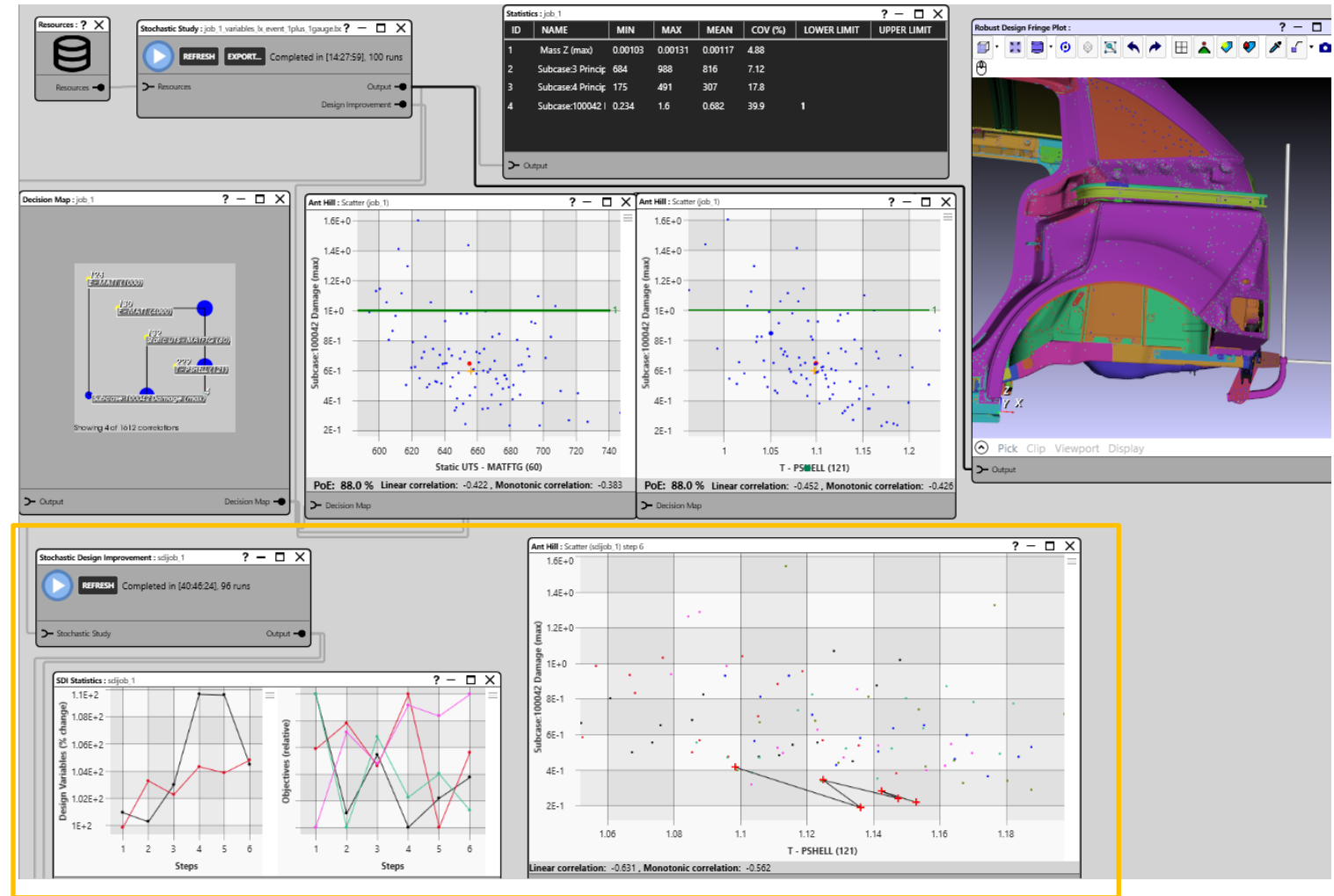
DAMAGE scatter plot around nominal thickness value of PSHELL 121



Stochastic Design Improvement



- As a follow-on to a stochastic function study, Stochastic design improvement cycles (SDI) support gradual and robust design changes
- The local sheet thickness and the adjacent solid panel are selected as design variables
- A damage of 0.05 with a 10% tolerance is selected as “upper limit” hard target
- Each stochastic simulation represent a single step. The process terminates when the target reaches the target distance within a stated tolerance



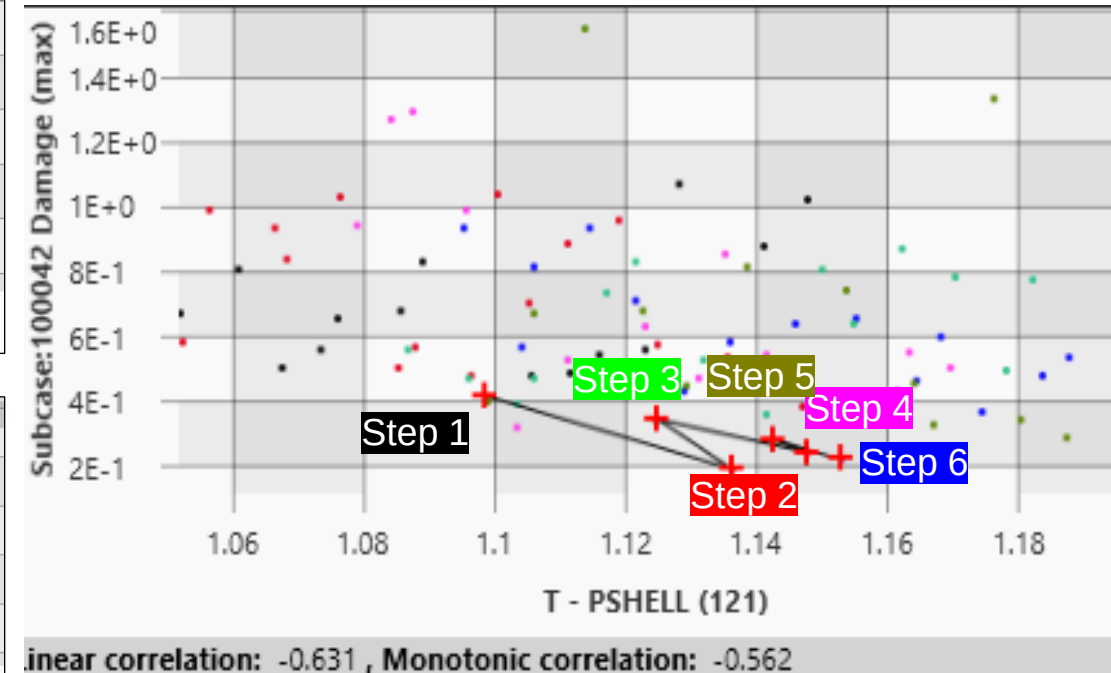
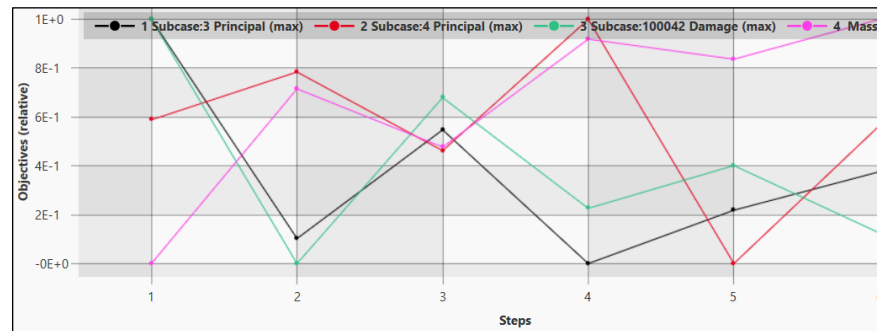
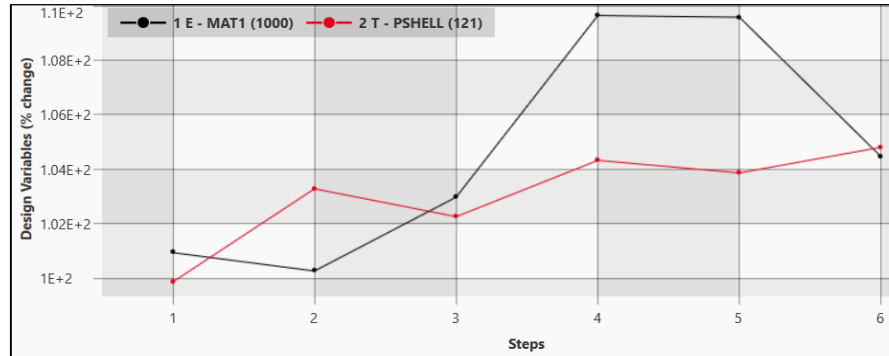
Stochastic Design Improvement



Settings:

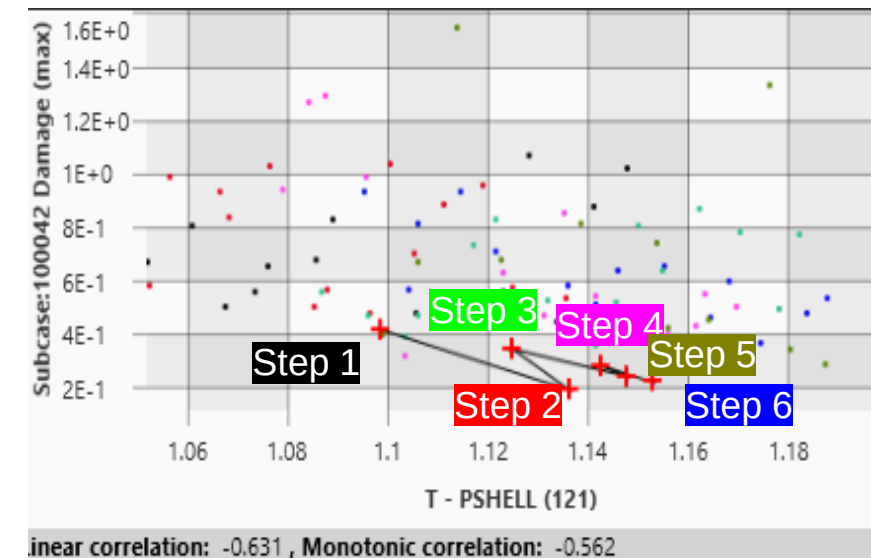
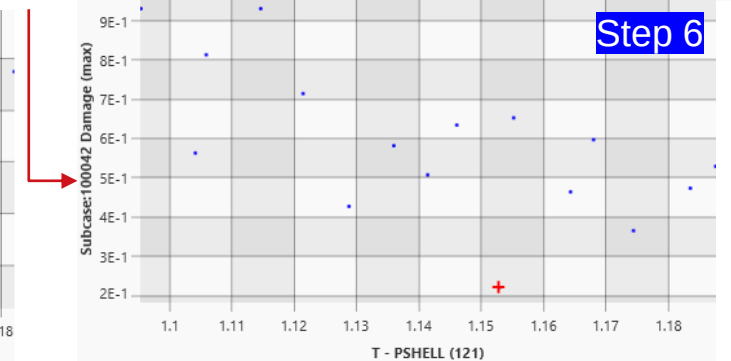
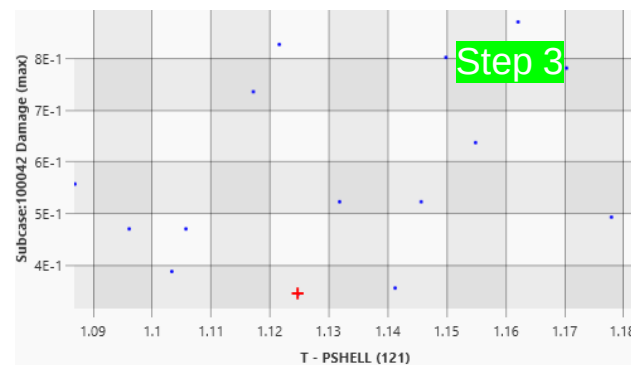
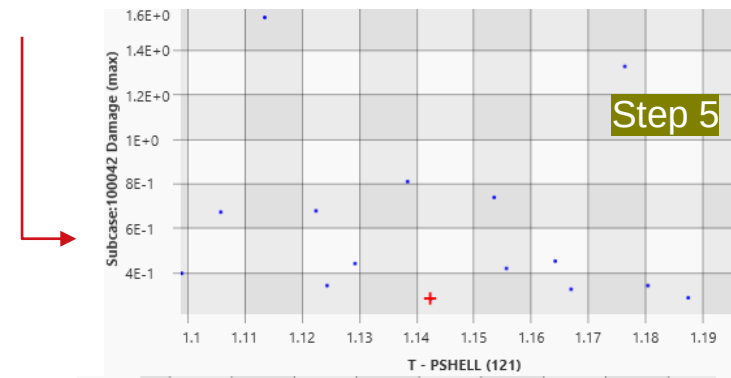
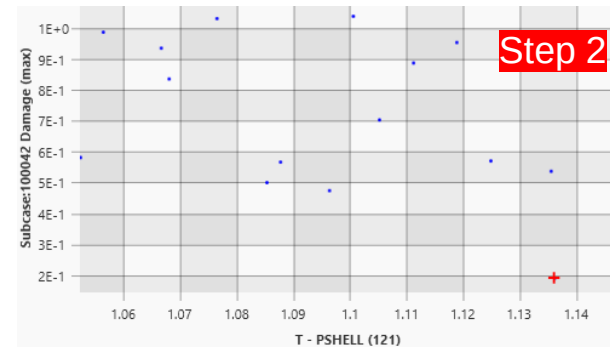
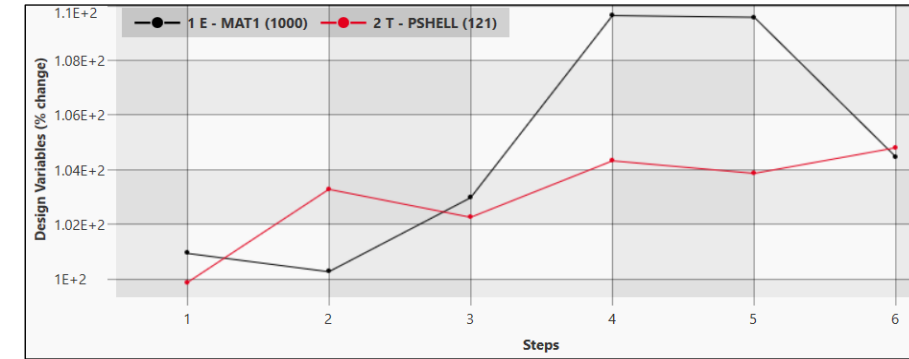
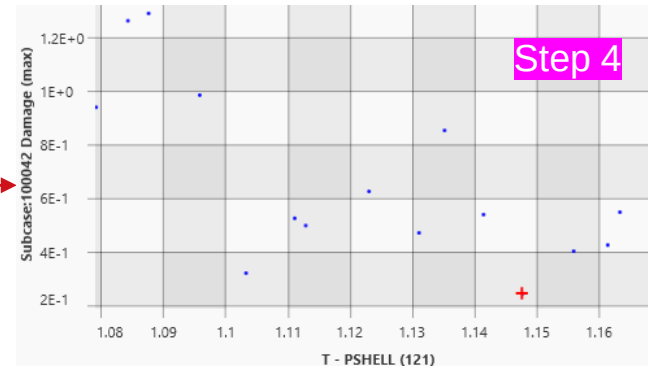
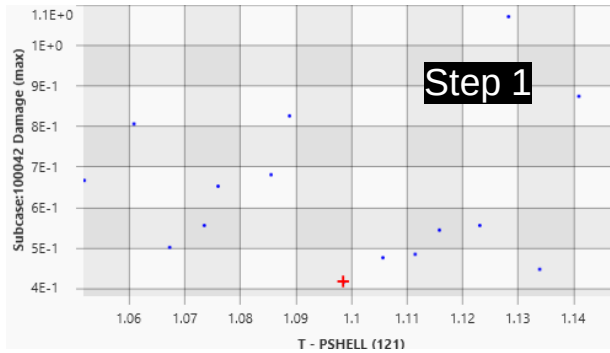
- 2 parallel jobs
- 6 SDI steps (i.e. 6 stochastic studies) each of 16 samples
- 10% tolerance
- 2 design variables representing relevant gauges (MAT 1000 and PSHELL 121)
- Damage target 0.05 (tolerance distance 10%)

PARAMETERS	
Job	sdijob_1
Num Concurrent Jobs	2
Num SDI Steps	6
Num Samples per Step	16
Threshold (%)	10
Design Variables	
NAME	OBJECTIVES
PSHELL (113) - T	✓ 1.2 - 1.6
PSHELL (114) - T	✓ 1.2 - 1.6
PSHELL (115) - T	✓ 0.76 - 1
PSHELL (116) - T	✓ 0.76 - 1
PSHELL (117) - T	✓ 0.6 - 0.8
PSHELL (118) - T	✓ 0.6 - 0.8
PSHELL (119) - T	✓ 1.2 - 1.6
PSHELL (120) - T	✓ 1.2 - 1.6
PSHELL (121) - T	✓ 0.94 - 1.3
PSHELL (122) - T	✓ 0.94 - 1.3
PSHELL (123) - T	✓ 1.3 - 1.7
PSHELL (124) - T	✓ 1.3 - 1.7
PSHELL (125) - T	✓ 1.3 - 1.7
PSHELL (126) - T	✓ 1.3 - 1.7
PSHELL (127) - T	✓ 1.3 - 1.7
PSHELL (128) - T	✓ 1.3 - 1.7
PSHELL (129) - T	✓ 1.3 - 1.7
PSHELL (130) - T	✓ 1.3 - 1.7
DESIGN VARIABLE RANGE	
Range	0.935 - 1.265
Mean/Nominal	1.1
Step Size	0.1





SDI – 6 steps of 16 runs

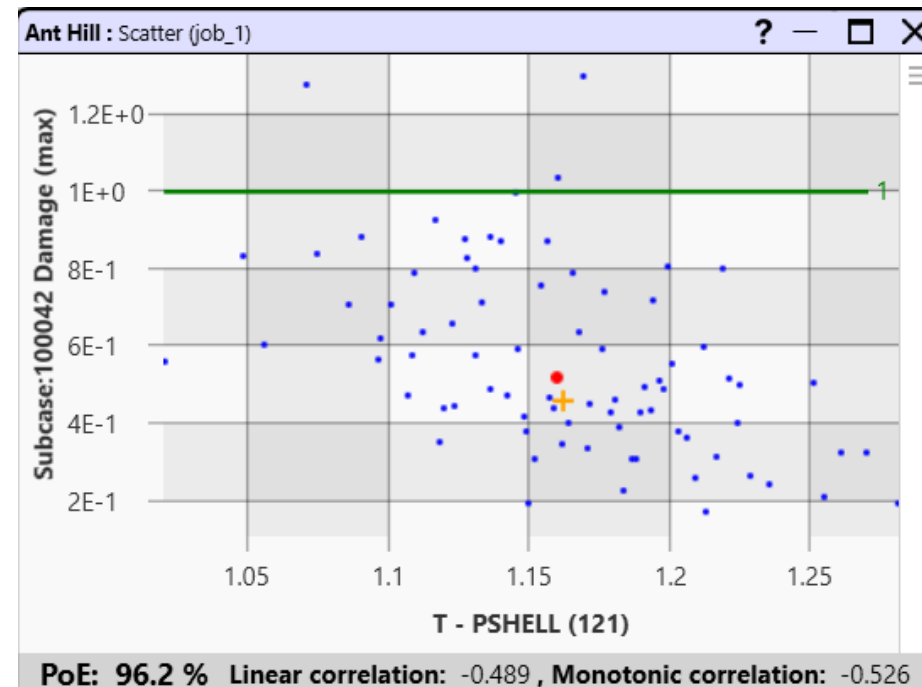


Checking Robustness of modified model



- The new position reached through the SDI is tested again through a stochastic function study
- The probability of survival is now within a desirable boundary (96%)

Statistics : job_1						
ID	NAME	MIN	MAX	MEAN	COV (%)	LOWER LIMIT
1	Mass Z (max)	0.00109	0.00136	0.00124	4.43	
2	Subcase:3 Princip	626	914	781	7.37	
3	Subcase:4 Princip	183	491	302	19.2	
4	Subcase:100042 I	0.169	1.3	0.56	43.2	1



Application example Robust Design to automotive durability assessment



A Stochastic Approach to Designing Robust Automotive Structures Considering Variability



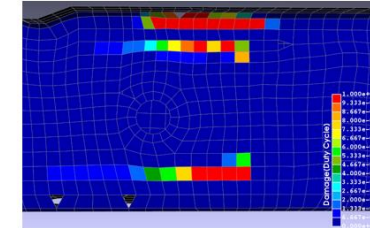
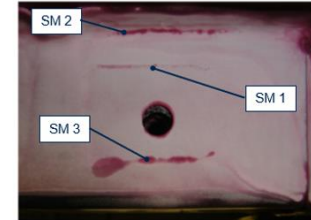
Dr. P. Roemelt,
Ford Motor Company
Dr. M. Veltri, Mark Robinson, Patrik Adolfsson
Hexagon Manufacturing Intelligence
Presenter: Jeff Robertson
Hexagon Manufacturing Intelligence



Comparison with physical test



- the fatigue analysis using nominal parameters shows good alignment with the critical areas identified in the physical test

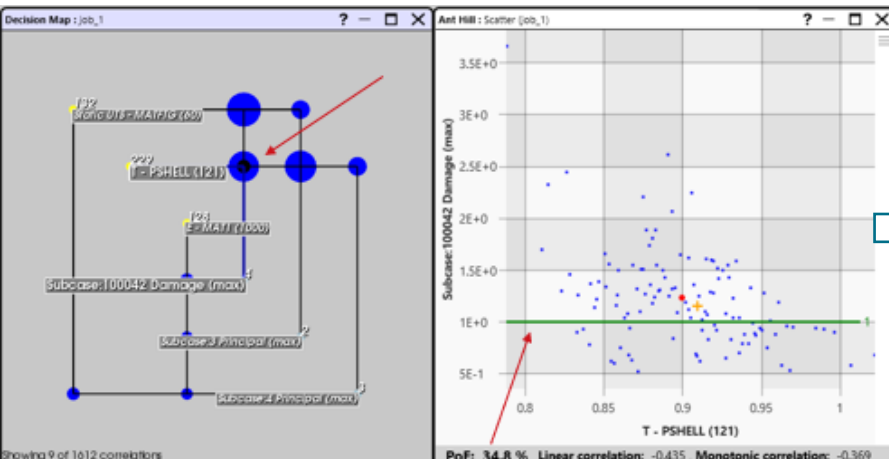


- An ensuing stochastic function study automatically incorporates uncertainties with the aim of building knowledge about model robustness

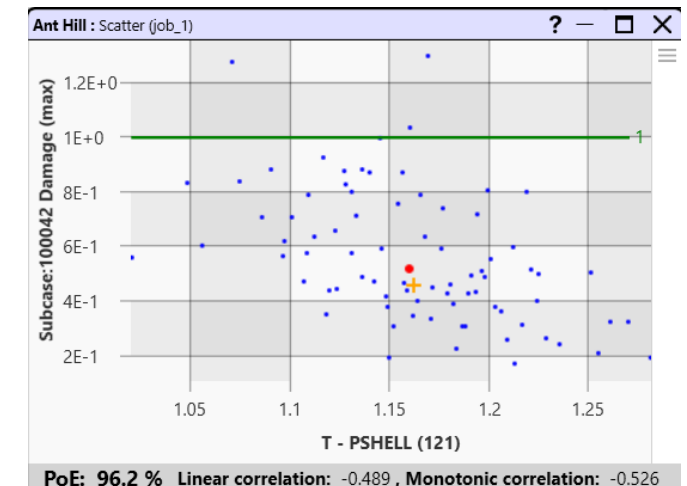
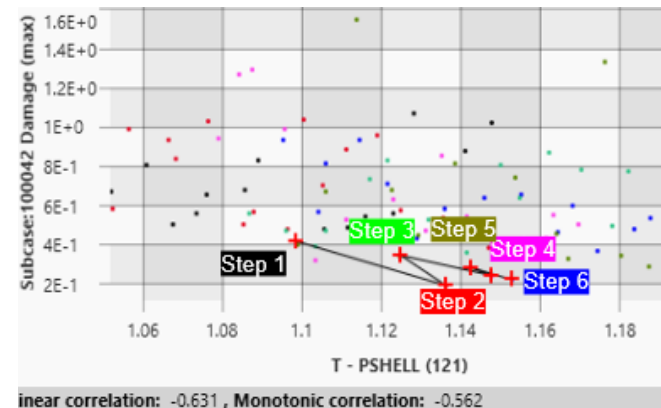


NAFEMS World Congress 2023 | Tampa, Florida, USA | May 15-18 2023

From stochastic Design function study



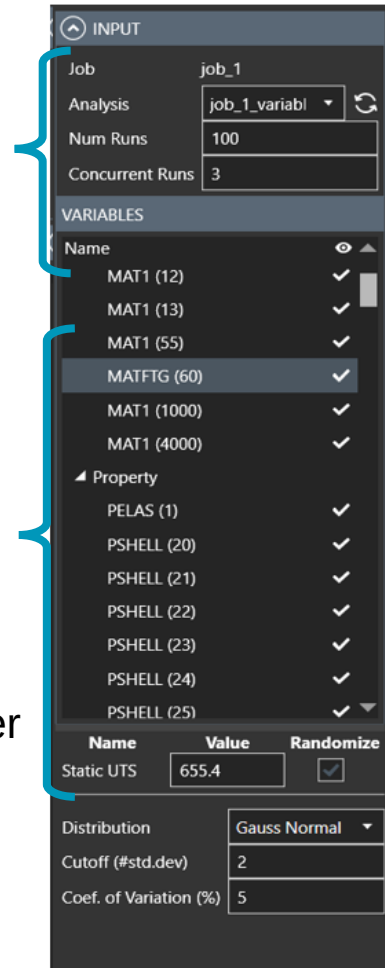
To stochastic Design Improvement cycles



Running stochastic function study on nominal model

Probability of failure (Damage>1) around 65%

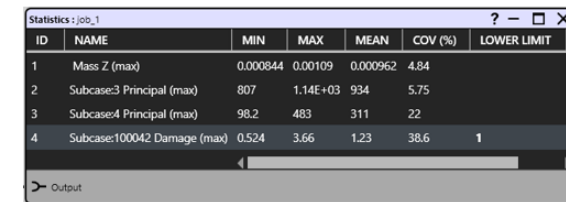
- Managing the FE solver MC queue (e.g.100 runs, 3 concurrent jobs)
- Allows automatic input randomization of large number of parameters (FE and fatigue input file simultaneously)
- Run in batch on a HPC server while interactively postprocessing locally



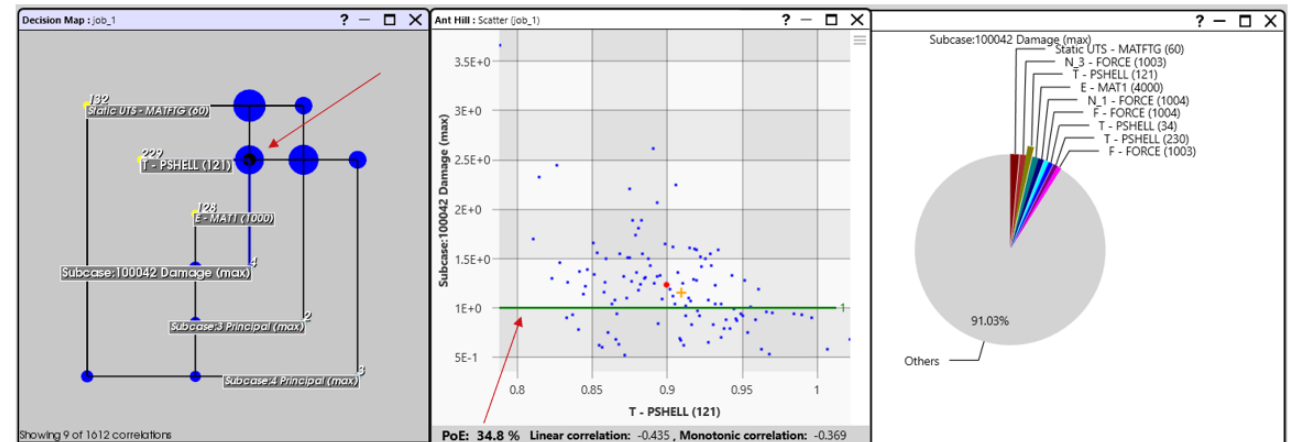
Trailer towing use case

120 runs stochastic function study review

- probability of survival (Damage<1) is 34.8%
- Greatest correlation is found with fatigue e-N curve ("static UTS") and sheet metal thickness in the area of failure ("T-PSHELL 121")



ID	NAME	MIN	MAX	MEAN	COV (%)	LOWER LIMIT
1	Mass Z (max)	0.000844	0.00109	0.000962	4.84	
2	Subcase3 Principal (max)	807	1.14E+03	934	5.75	
3	Subcase4 Principal (max)	98.2	483	311	22	
4	Subcase100042 Damage (max)	0.524	3.66	1.23	38.6	1



DAMAGE scatter plot around nominal thickness value of PSHELL 121

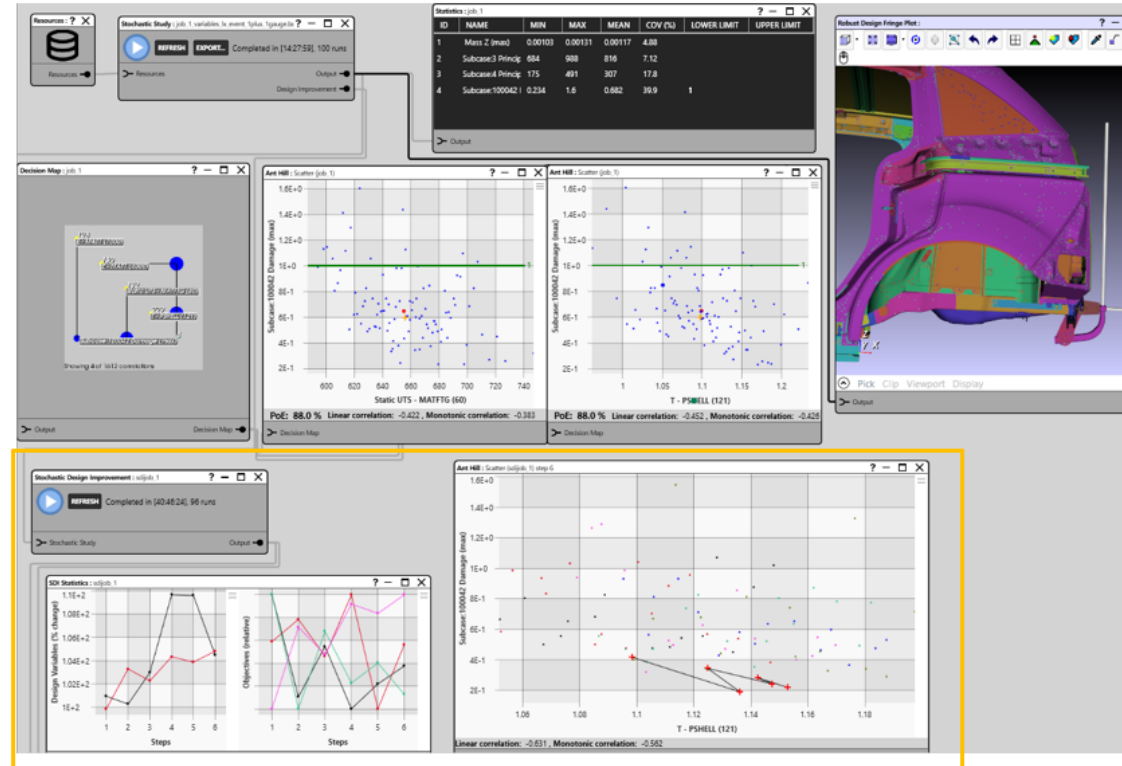


Running stochastic Design Improvement

Stochastic Design Improvement



- As a follow-on to a stochastic function study, Stochastic design improvement cycles (SDI) support gradual and robust design changes
- The local sheet thickness and the adjacent solid panel are selected as design variables
- A damage of 0.2 with a 10% tolerance is selected as “upper limit” hard target
- Each stochastic simulation represent a single step. The process terminates when the target reaches the target distance within a stated tolerance



Application example Robust Design to large axles durability assessment

Value Others | Inspire Innovation | Grow Responsibly | Win Together

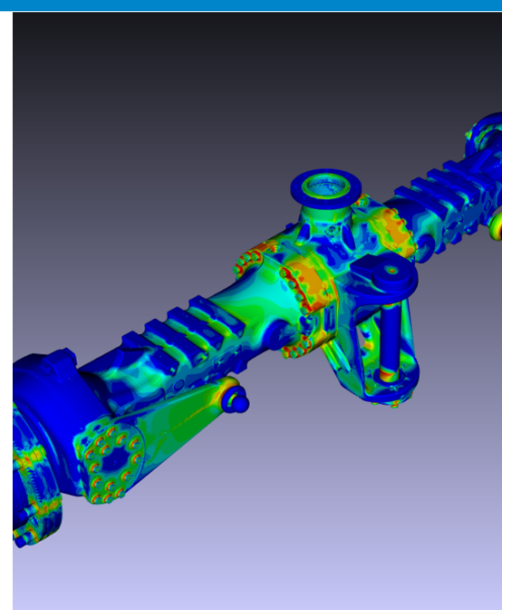
HxGN LIVE Smart Simulation Summit
Troy, MI | 09-10 November 2022



NVH & Durability Track

Incorporating Uncertainties in a Durability Assessment of Off-Highway Axles
A Robust Design Approach

People Finding A Better Way®



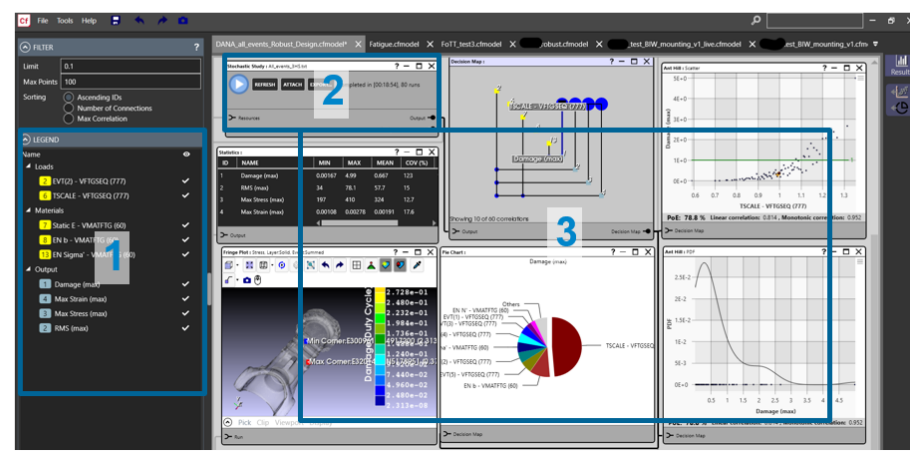
Stochastic design process with Robust Design



Robust Design stochastic function study to build knowledge about model robustness

Process flow driven in CAEfatigue

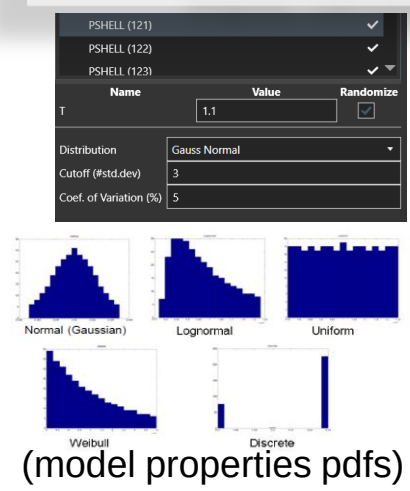
1. FE and Fatigue input randomization and output section
2. DOE queue set-up and monitoring
3. Output section



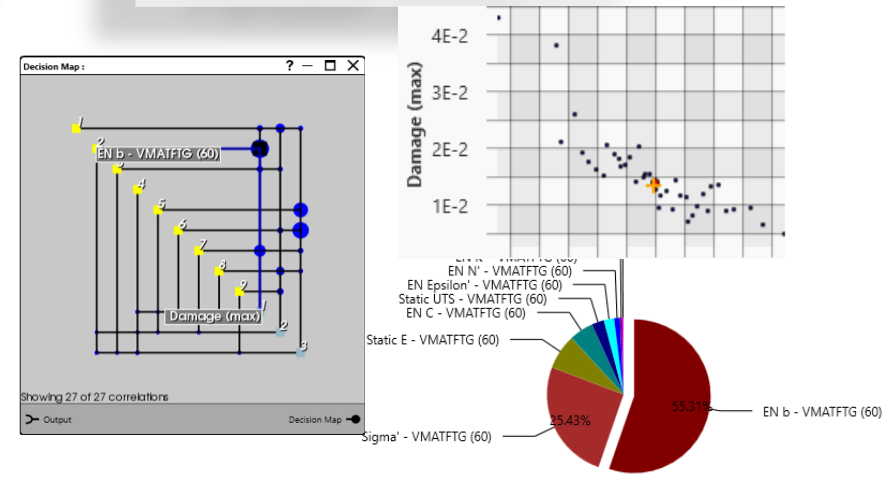
Nominal FE + Fatigue input/output model

VARIABLES	Name
Material	1 Static E - VMATFTG (60)
Property	2 EN b - VMATFTG (60)
Connector	3 EN C - VMATFTG (60)
Temperature	4 EN Epsilon' - VMATFTG (60)
Load	5 EN K' - VMATFTG (60)
Output	6 EN N' - VMATFTG (60)
SOL 101	7 EN Sigma' - VMATFTG (60)
Subcase 3	8 Static UTS - VMATFTG (60)
Subcase 4	9 Static Ys - VMATFTG (60)
Global	1 Damage (max)
	2 Fatigue 100042
	3 RMS (max)

Input randomization



Function Study



Application example Robust Design to large axles durability assessment

Value Others | Inspire Innovation | Grow Responsibly | Win Together

HxGN LIVE Smart Simulation Summit
Troy, MI | 09-10 November 2022

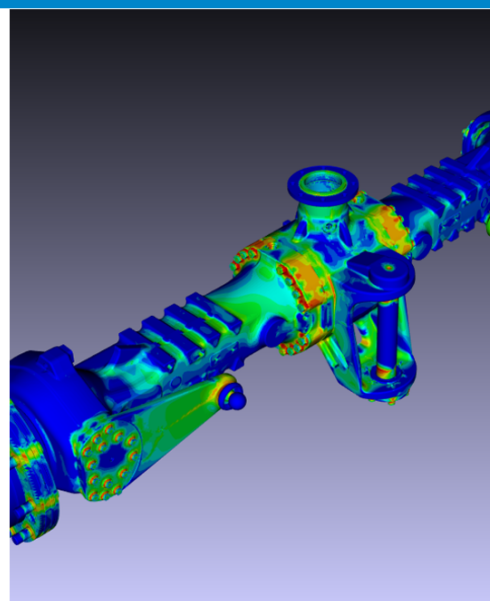


NVH & Durability Track

Incorporating Uncertainties in a Durability Assessment of Off-Highway Axles

A Robust Design Approach

People Finding A Better Way®



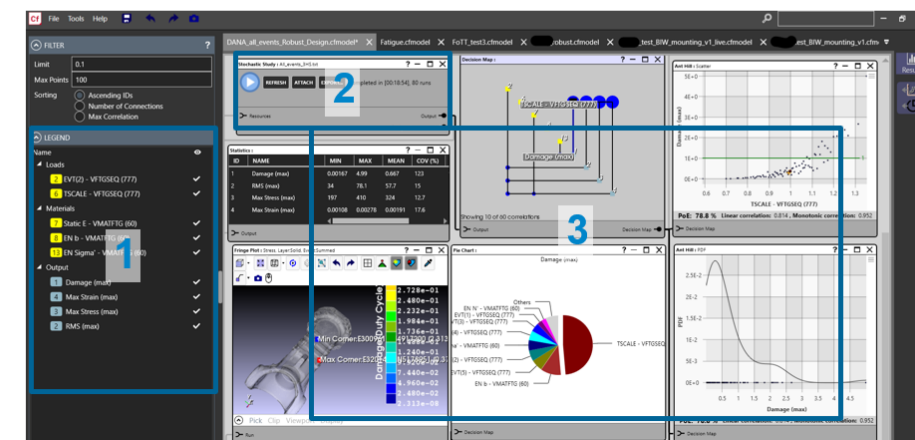
Stochastic design process with Robust Design



Robust Design stochastic function study to build knowledge about model robustness

Process flow driven in CAEfatigue

1. FE and Fatigue input randomization and output section
2. DOE queue set-up and monitoring
3. Output section



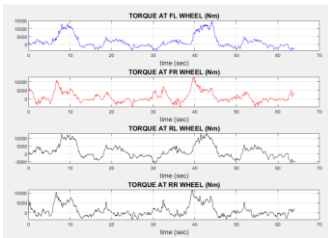
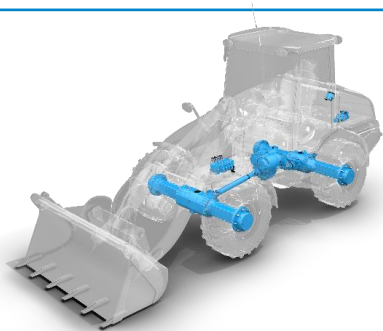
- ✓ Durability workflow for operation loads (Abaqus ODB results)
- ✓ Stochastic Design with Robust Design for assessing tolerances and uncertainties
- ✓ Stochastic design improvement to identify a surrogate simplified test responding to operation use

Project scope



Wheel Loader field measurements postprocess :

- To identify and put under control axle design characteristics and
- To identify axle structural requirements and test parameters



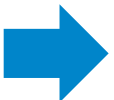
Analysis



Structure (casting)

Input:

- Real loads history from field measurements
- Variation of cycle duration
- Variation of cycle amplitude
- Variation of material's prop
- ...



Output:

- Damage evaluation
- Sensitivity analysis (influence evaluation of inputs on outputs)
- Identification of critical parameters

DOE: possible design configurations

Driveline (gears)

Input:

- Real loads history from field measurements
- Variation of cycle duration
- Variation of cycle amplitude
- Variation of material's mechanical characteristics
- Microgeometry variation for the gears

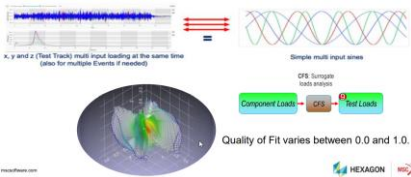


Output:

- Pressure distribution on the tooth
- Sensitivity analysis (influence evaluation of inputs on outputs)
- Identification of critical parameters

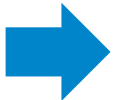
DOE: possible design configurations

Synthesis



Input:

- Real loads history from field measurements
- Bench boundary conditions



Output:

- Condensed loads as structural requirement for sizing and optimization
- Test bench parameters definition

Surrogate Loads evaluation

Input:

- Real loads history from field measurements
- Bench boundary conditions



Output:

- Condensed loads as structural requirement for sizing and optimization
- Test bench parameters definition

Surrogate Loads evaluation

Fatigue loads and duty cycle

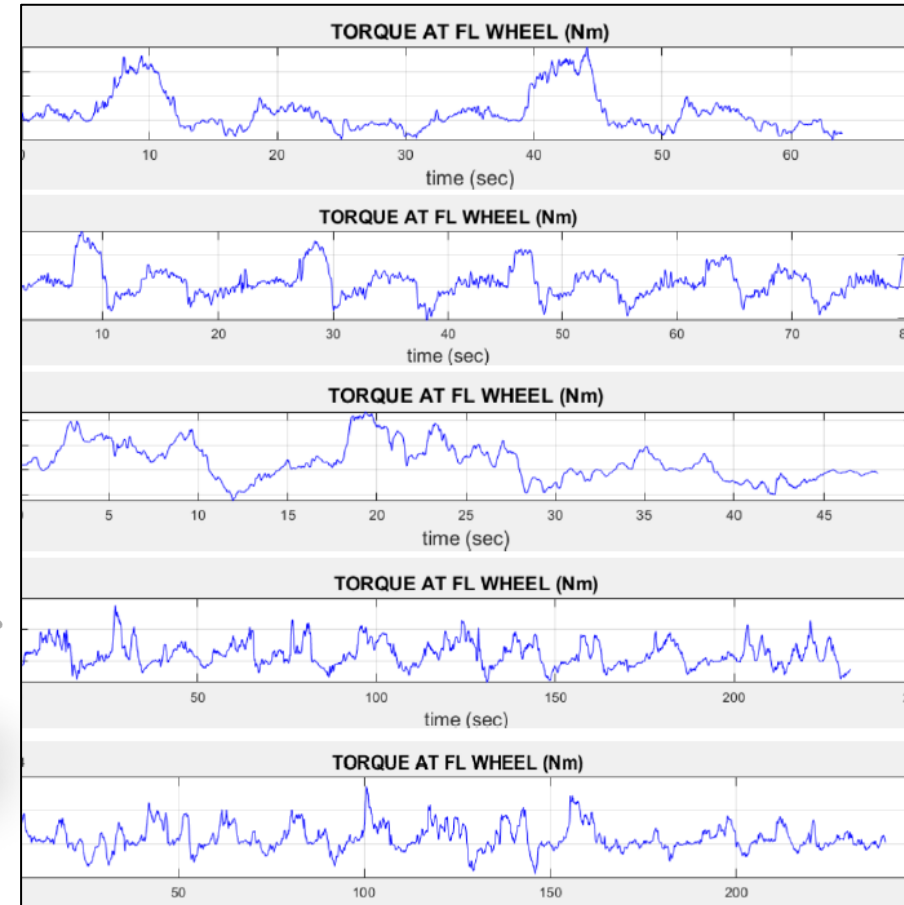
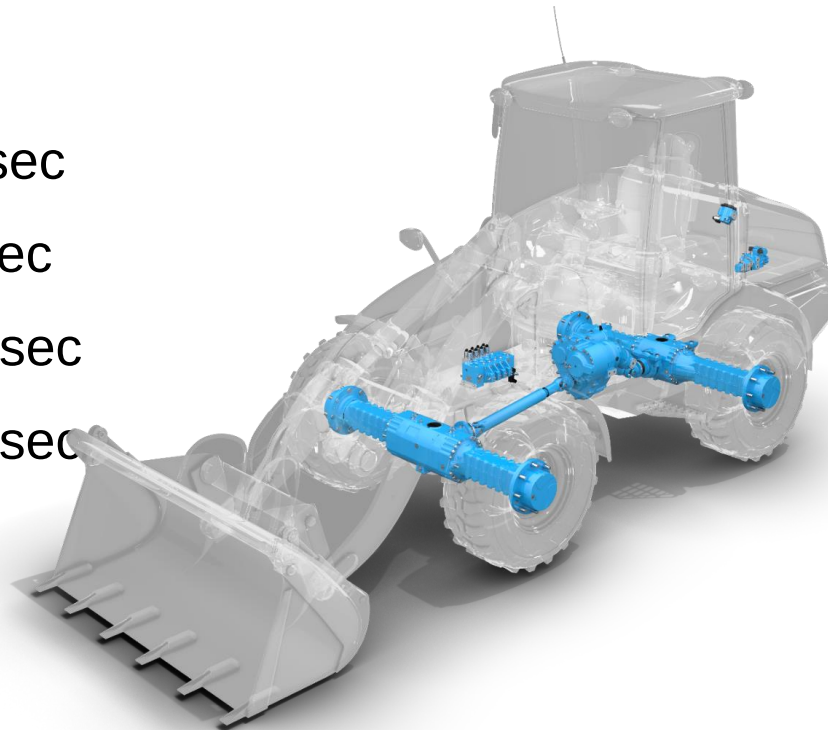


Load at wheels (forces and torques) for 5 vehicle maneuvers

5 vehicle maneuvers with sample duration

- Y-Cycle: 65 sec
- Y-Cycle aggressive: 80 sec
- Paper recycling 1: 48 sec
- Paper recycling 2: 240 sec
- Paper recycling 3: 240 sec

Sampling rate: 200 Hz



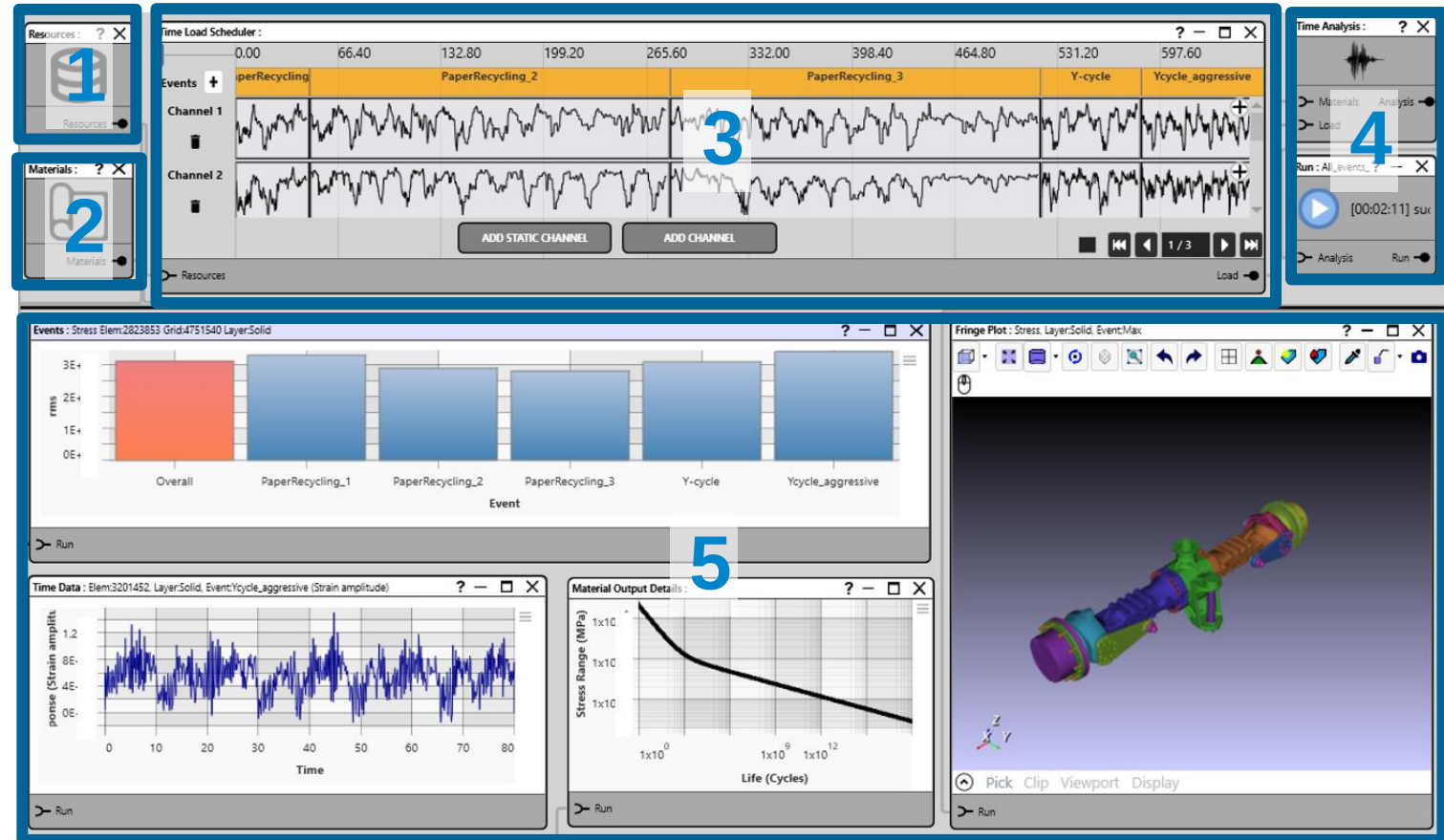
Acquired signals (example)

Fatigue process implementation in CAEfatigue



Process flow driven fatigue duty cycle

1. FE ODB results are the process input “resources”
2. Material library for Stress or Strain life modelling
3. Loading schedule: a 5-events duty cycle is obtained by linear superposition of up to 12 loading channels (6 dof left and right)
4. Process Settings and play button
5. Post processing area



Any process flow template is easily repeated with different versions, loads or new parts

Identification of critical areas and uncertainties

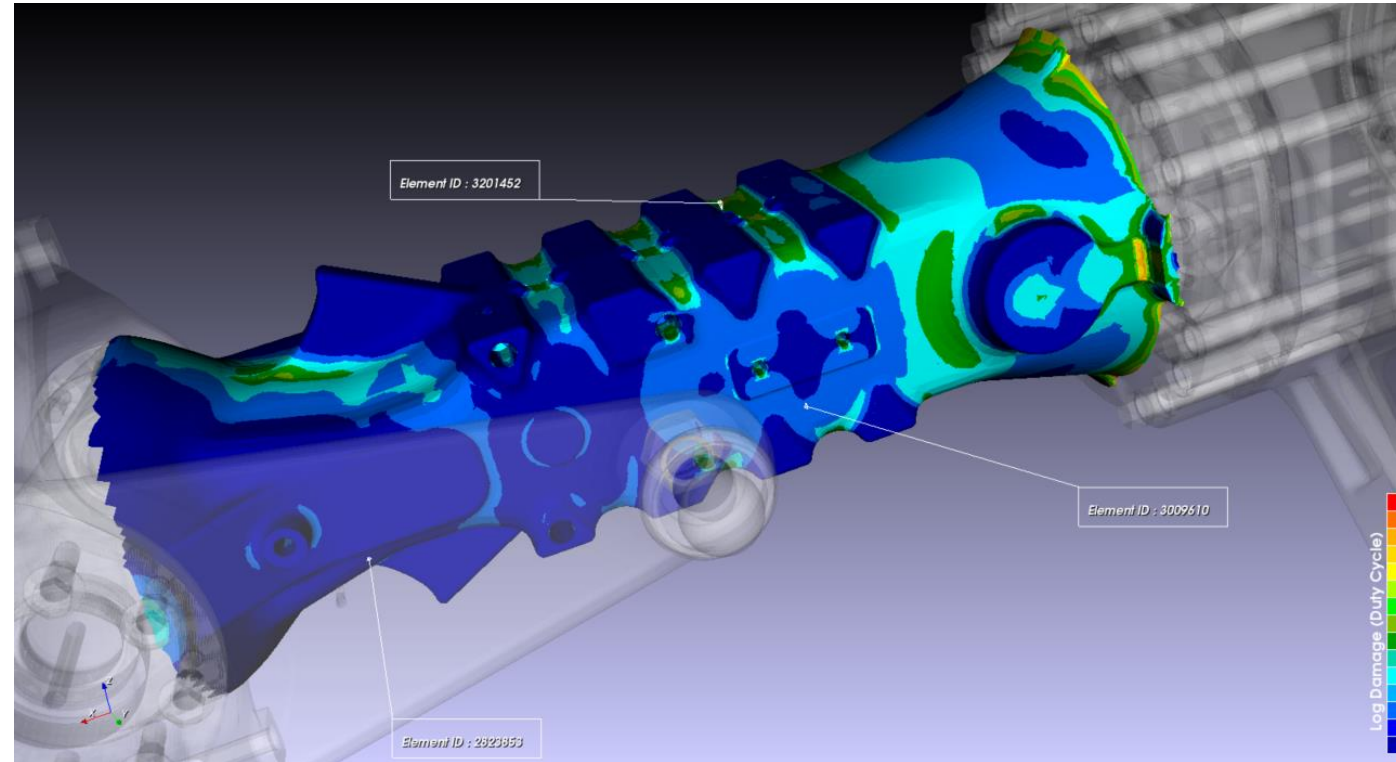


Identification of critical areas for ensuing stochastic analysis

Fatigue results correspond to an assumed reference customer usage, nominal material properties and limited acquired loads

Uncertainty and variability affects primarily:

- Duty cycle agenda (varies with customer)
- Load factors (up to +/- 35%)
- Material properties



A probabilistic approach is undertaken to assess the fatigue robustness of the axle

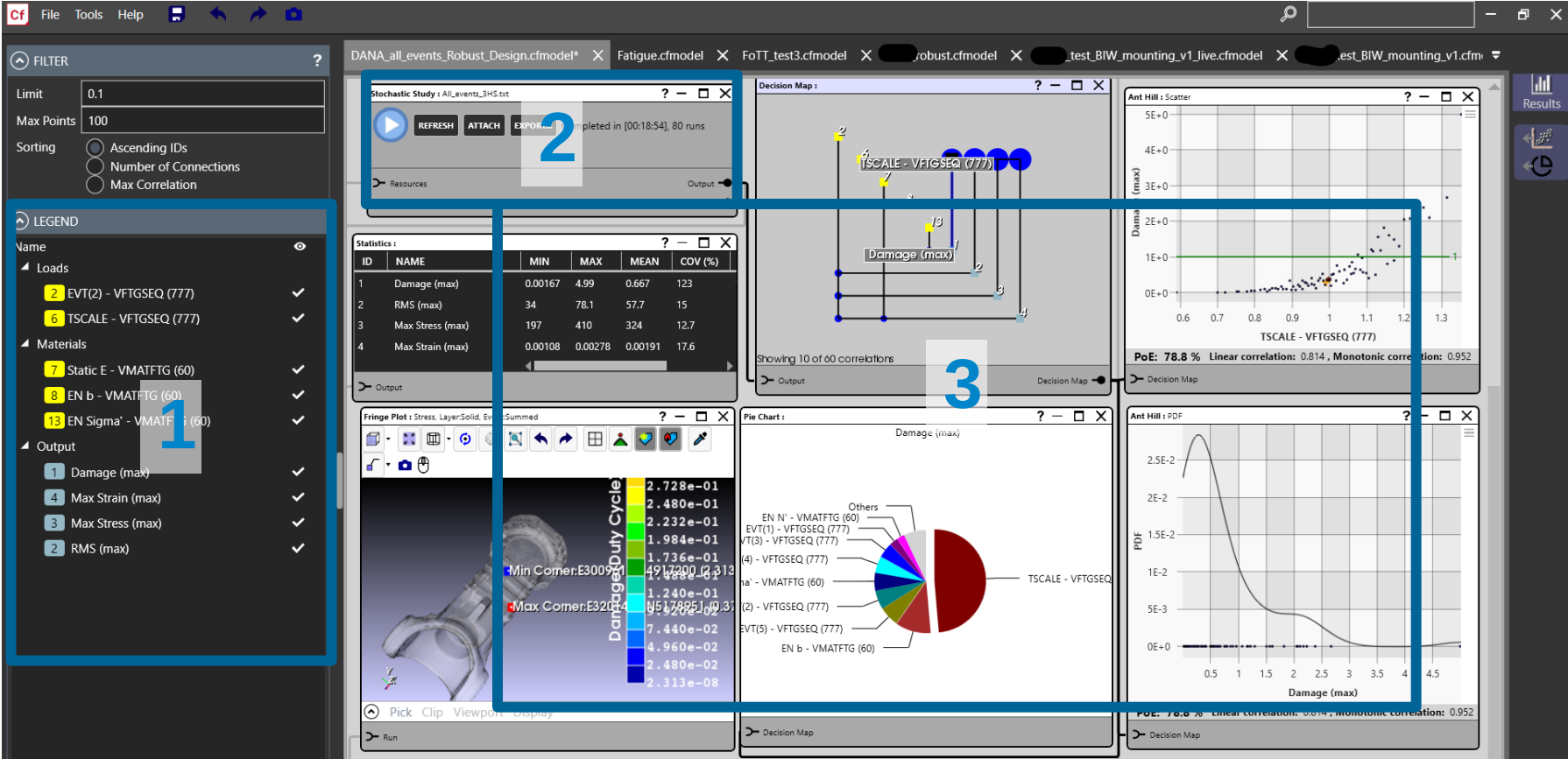
Stochastic design process with Robust Design



Robust Design stochastic function study to build knowledge about model robustness

Process flow driven in CAEfatigue

- 1. FE and Fatigue input randomization and output section
- 2. DOE queue set-up and monitoring
- 3. Output section



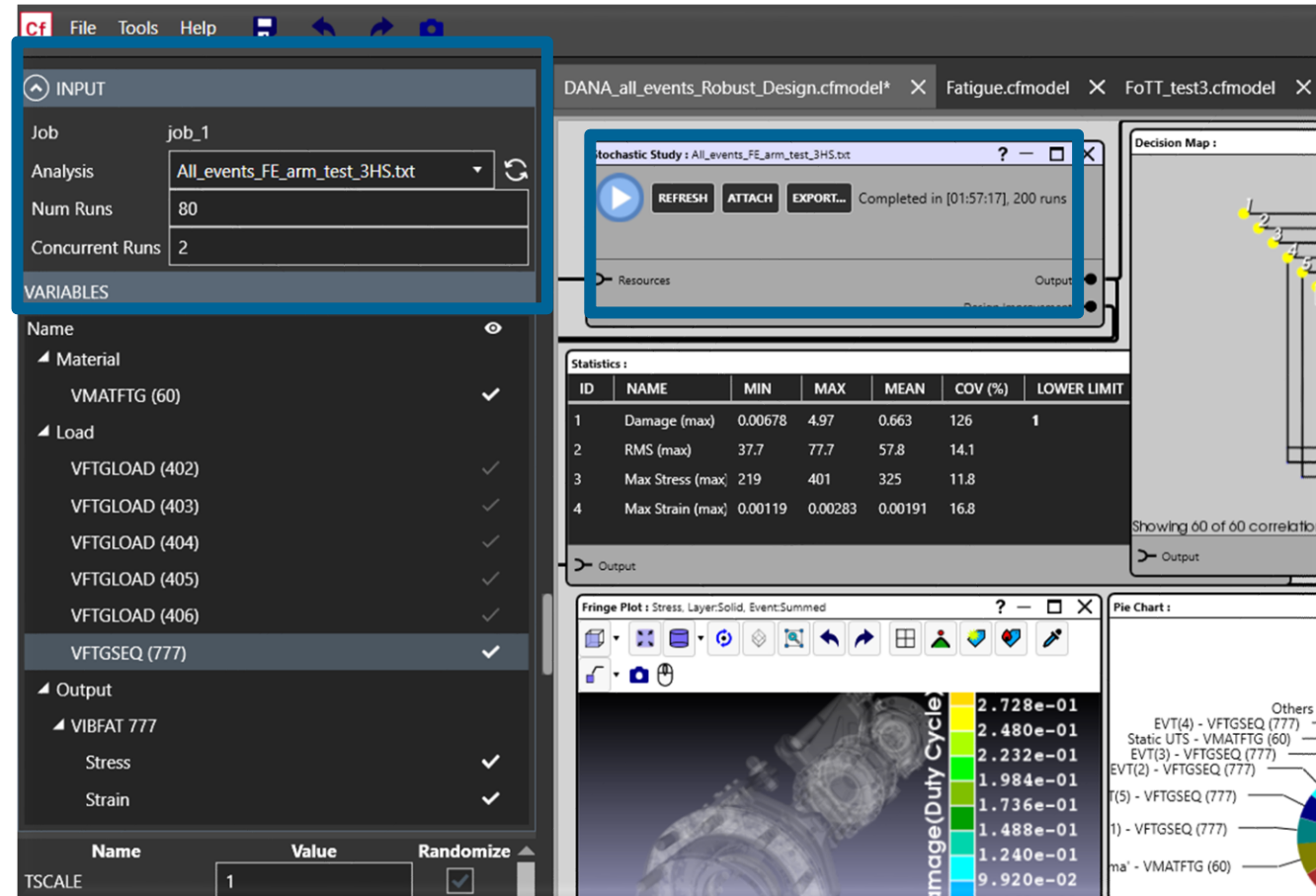
STEP 3: Stochastic simulation and “meta-model”



Robust Design stochastic function study

Process flow driven in CAEfatigue

- DOE set-up and run interactively or in batch
- With a conditioned sampling strategy less than 100 samples are adequate
- Multiple jobs run in parallel, results are monitored as they appear
- Results are collected in portable HDF5 format
- Input/relationship “metamodel” can be exported in CSV file for addition purposes

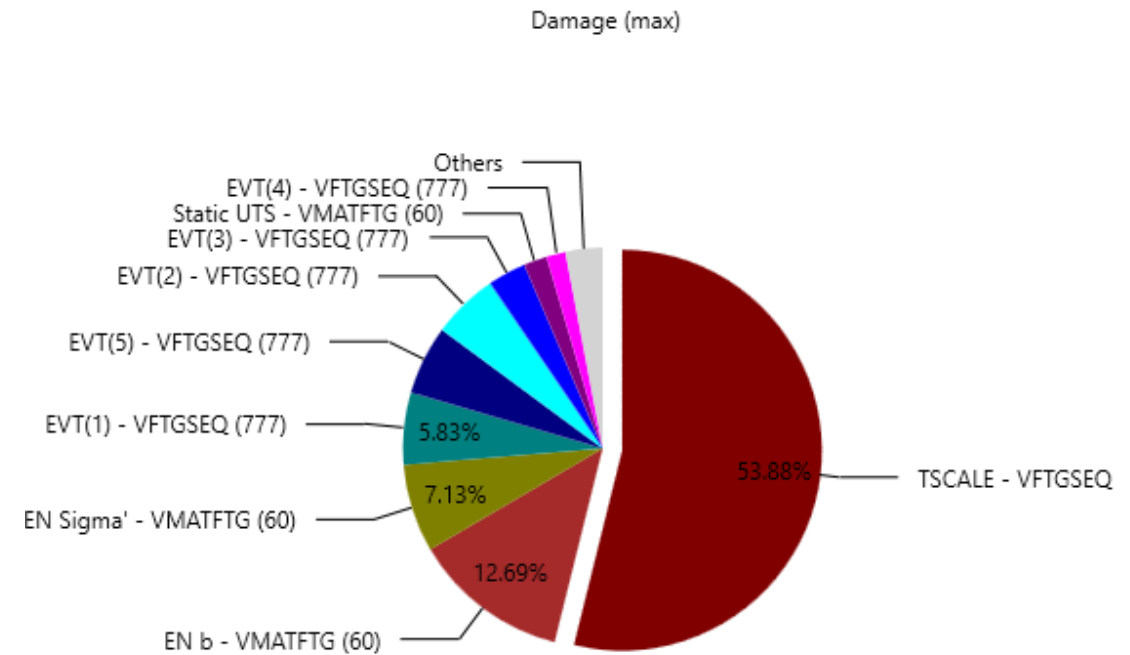
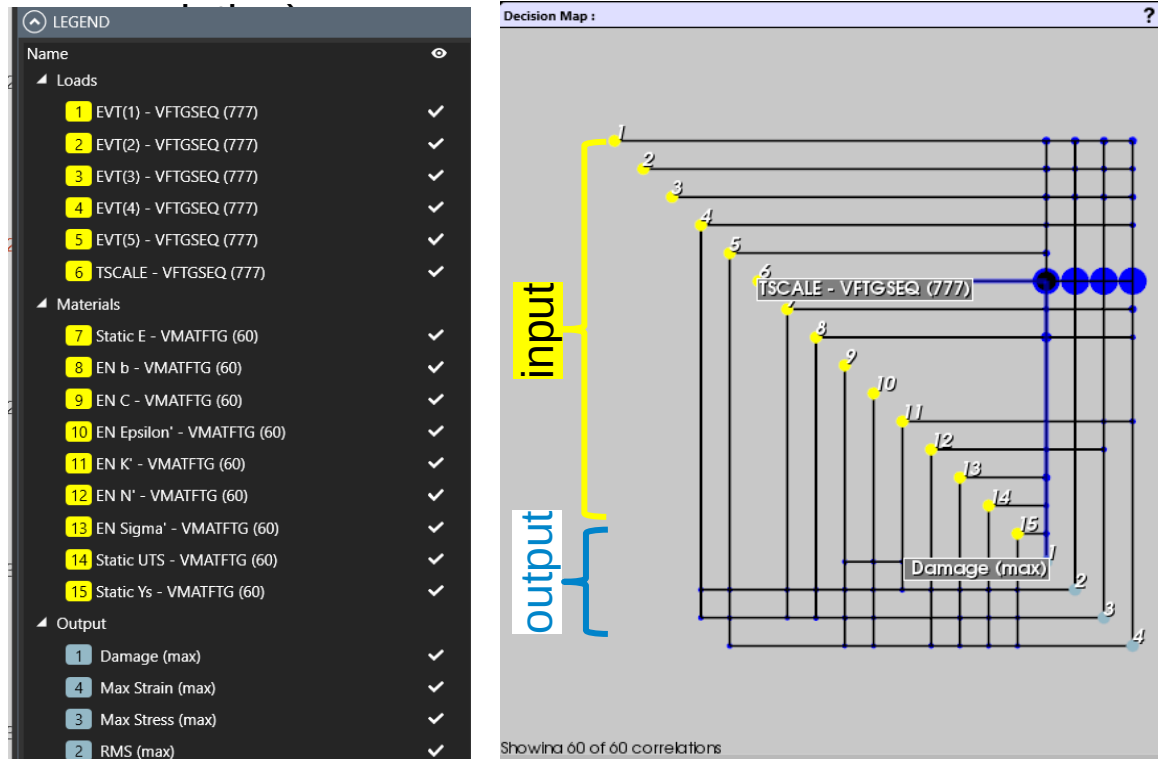


STEP 4: Robust Design postprocessing with Ant hill plots and decision maps



Build actionable knowledge in multiple ways and make decisions

- Decision maps define the relationship between input and output
- Higher correlations are shown on the upper right quadrant of the decision map (dot size reflects)
- Pie charts provide the relative weight and influence of parameter variation on max stress and total cumulative damage

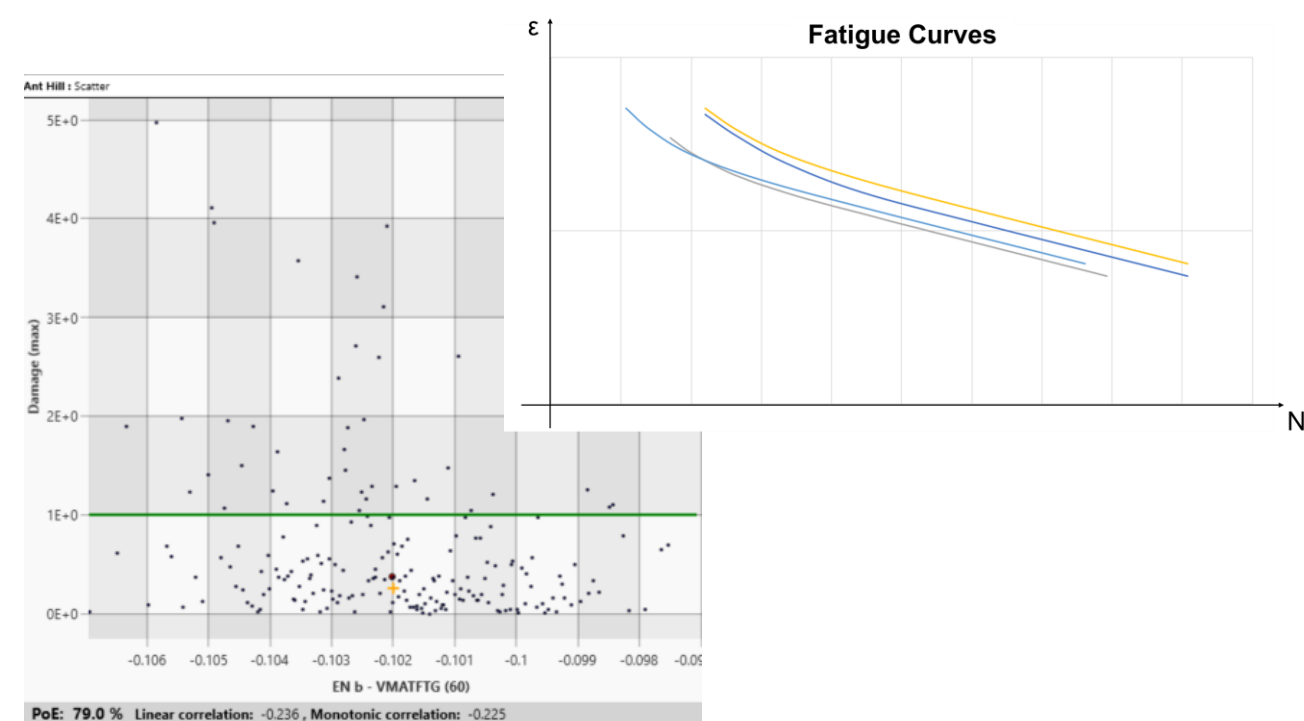
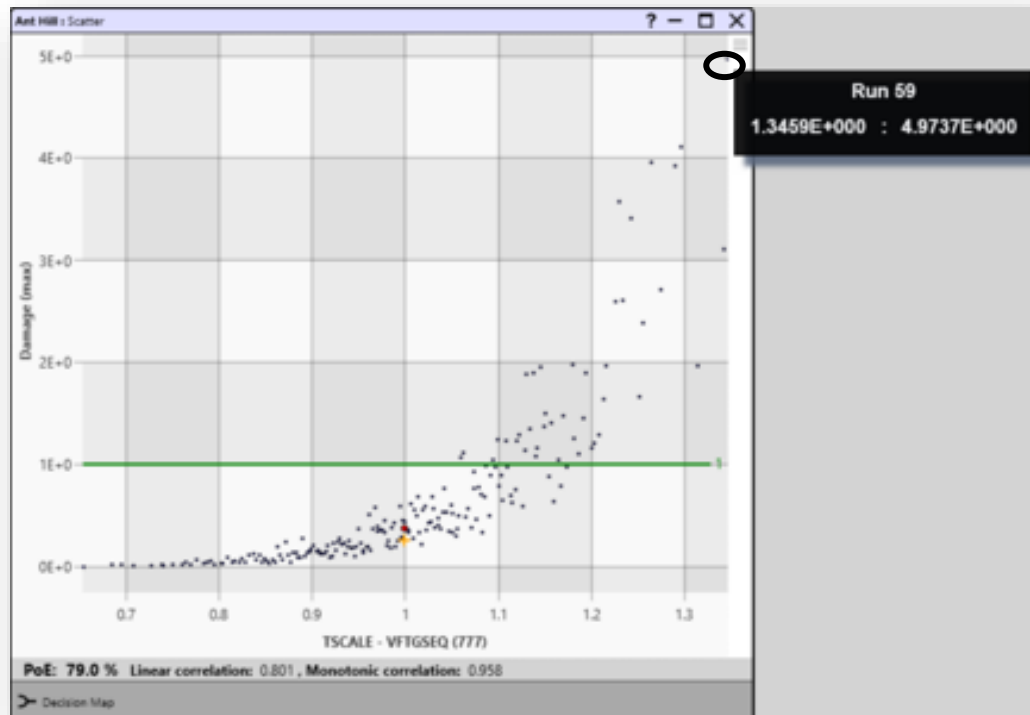


STEP 4: Robust Design postprocessing with Ant hill plots and decision maps



Build actionable knowledge in multiple ways and make decisions

- Identifying parameter combinations and values causing unacceptable damage (thresholds or outliers)
- Material variation leads to a cloud of possible fatigue curves. Material and part supplier (e.g., castings) could be consulted for improved tolerances



STEP 4: Robust Design postprocessing with Ant hill plots and decision maps

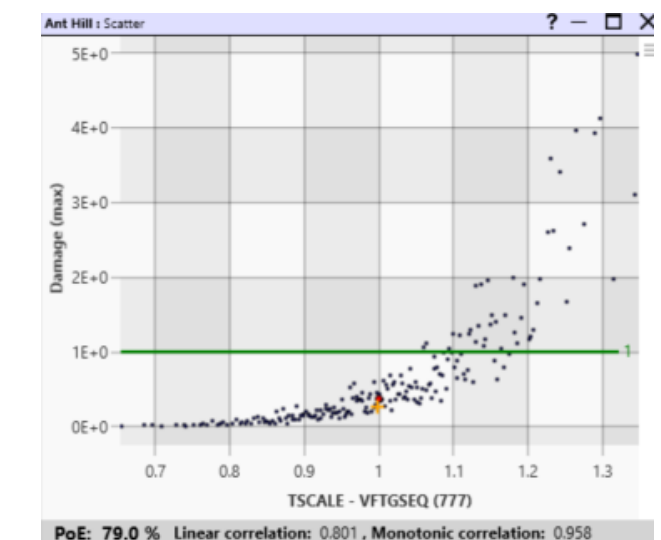
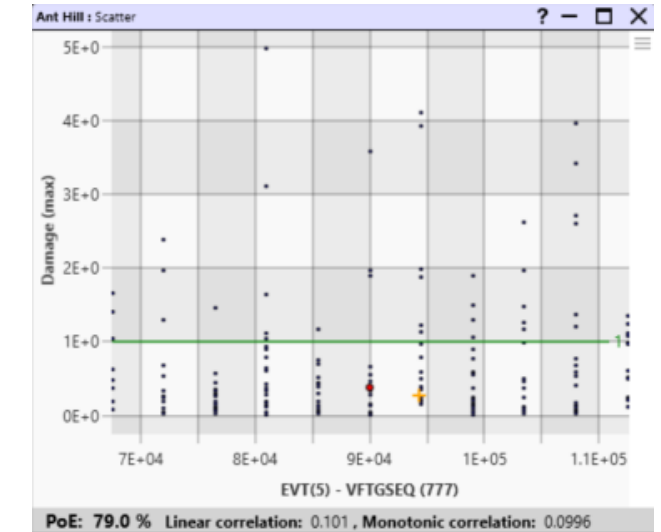
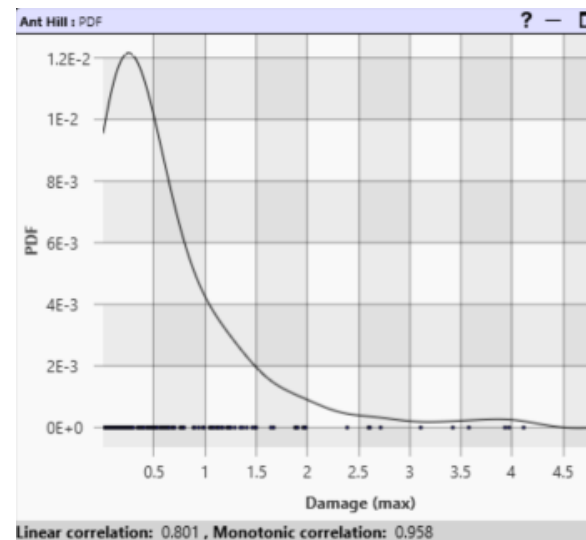


Build actionable knowledge in multiple ways and make decisions

- Probabilistic recommendations for OEM on axle usage limits
- Enforcing/recommending limits on vehicle weight for a specific maneuver
- Restriction on cumulative duty cycle

	A	B	C	D	E	F
1	TSCALE	EVT(1)	EVT(2)	EVT(3)	EVT(4)	EVT(5)
2	1	150000	31035	30000	112500	90000
3	1	180000	32587	31500	95625	76500
4	1	120000	37242	24000	112500	108000
5	1	172500	37242	28500	106875	67500
6	1	157500	24828	34500	118125	85500
7	1	172500	29483	36000	106875	67500
8	1	142500	38794	25500	95625	99000
9	1	127500	29483	30000	123750	99000
10	1	120000	31035	27000	140625	94500

Combinazioni_Condizioni_di_Cari



Conclusions and Next Steps

A Robust Design approach affects the design in multiple ways

- Model Credibility: the inclusion of tolerances and scatter in finite element models significantly increases their level of realism, elevating the credibility of the model
- Reducing physical testing: with the higher levels of model credibility, stochastic simulation may be used to reduce the amount of physical testing with increased confidence
- Assess Risk and liability: stochastic models can be used to assess product robustness, risk, and liability
- Cost Reduction: by identifying components that have little influence on response targets, any associated tolerances may be relaxed
- The tolerances and variability assessed through Robust Design can advice customers about realistic durability expectancy with respect to load factors
- Interaction with supplying foundries and other parts suppliers to monitor material and geometrical variability in interest areas

With the next “synthesis” phase of this project we will be determining a simplified and representative robust surrogate set of loads for test validation

The ultimate aim of the project is the implementation of a fully repeatable process to be consistently and easily applied to multiple parts from the DANA production line



Incorporating uncertainty in durability assessments



Dana improves off-road durability with robust design and stochastic analysis

Dana Incorporated: Federico Bavarese (Driveline Lead Engineer), Tommaso Pirovini (Driveline Engineer), Lorenzo Bellorini (Simulation Engineer), Claudio Maffei (Virtual Verification & Integration Engineer)
Hexagon's Manufacturing Intelligence division: Marco Veltri (Business Developer), Patrik Adolfsson (Product Designer, R&D), Mark Robinson (Technical Consultant)

Challenge

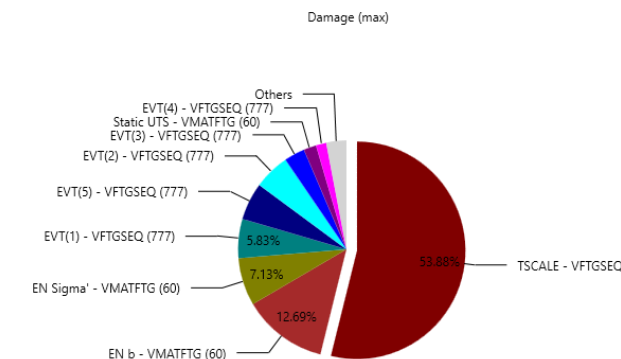
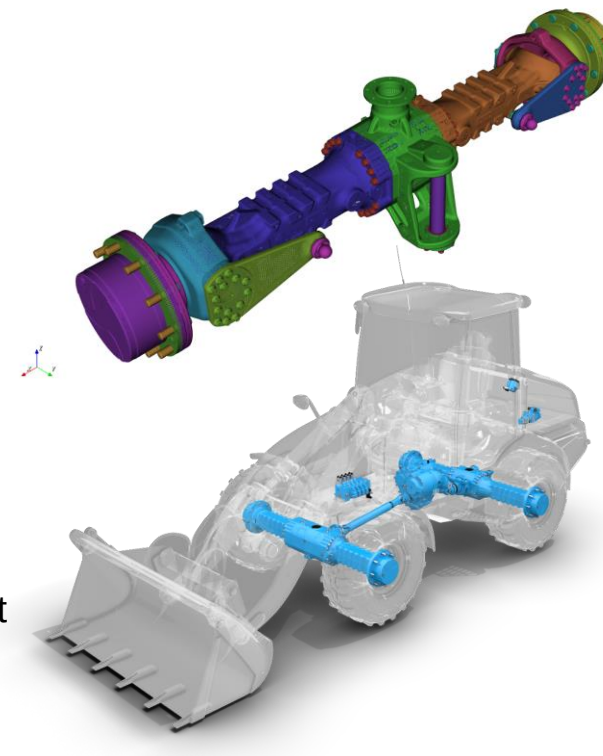
- Incorporate parameters and loads variability in durability design
- Identify loads and locations contributing the most to the damage of the axle structure.
- Determine the best combination of loads and event representative of customer operations

Solution

- Robust Design tool provides actionable outputs that quantify the effect of each variable and loading (e.g., y-cycle loading is the most destructive) while ALL input variables vary.
- Surrogate loads through SDI for a most realistic and robust component certification physical test

Benefits

- Gain insight into the underlying interrelations of variables to understand which are the most influential.
- Reduce and improve physical testing.
- Reduce liability costs by exploring the whole design space of loading conditions upfront.



Dana Improves off-road durability with Robust Design

Article in Engineering REALITY by Hexagon



Dana improves off-road durability with robust design and stochastic analysis

Dana Incorporated: Federico Bavaresco (Driveline Lead Engineer), Tommaso Pirovini (Driveline Engineer), Lorenzo Bellorini (Simulation Engineer), Claudio Maffei (Virtual Verification & Integration Engineer)
Hexagon's Manufacturing Intelligence division: Marco Veltri (Business Developer), Patrik Adolfsson (Product Designer, R&D), Mark Robinson (Technical Consultant)

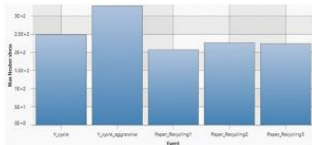


Figure 1: Comparison between Max Neuber Stress results related to every maneuver.

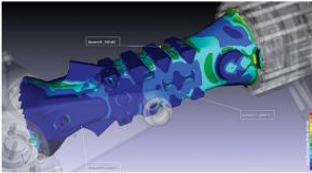


Figure 2: Identification of critical fatigue areas.

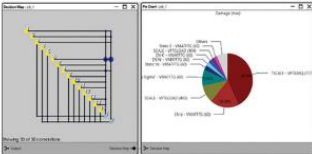


Figure 3: Correlation between parameter variations.

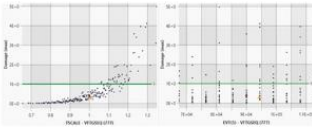


Figure 4: Clouds of input-output pairs obtained with the stochastic study.

Phase 2. Stochastic analysis after randomisation of loads and fatigue parameters to determine the effects of variation

After performing the fatigue simulation in phase 1, the teams deployed CAEfatigue Robust Design to automatically enforce a stochastic variation on all the input variables and perform Monte Carlo simulation. All parameters follow a Gauss Normal distribution based on three pieces of information: the mean value, the coefficient of variation, and the cutoff (the number of standard deviations that determines the accuracy percentage). The only exception is customer usage or the number of repetitions for every different maneuver.

Embracing such probabilistic approach, the teams could evaluate the effects of the uncertainties on the structure's robustness, identifying the input variations that led to the highest potential for damage and other undesirable conditions, thus empowering them to create more robust designs capable of withstanding the uncertainties with in-field conditions.

Hexagon's CAEfatigue Robust Design provides two graphic tools (Figure 3) — the decision map and pie chart — that help engineers understand how different variables influence the axle component performance. The decision map shows the relationship between the input parameters (yellow squares) and requested outputs (light blue squares), with the size of the blue circles showing the relative input-output influence. The pie chart displays all relative parameter influence on a given output—for example, their impact on max damage value.

CAEfatigue Robust Design can also generate a point-cloud visualization of the input-output pairs distribution. This visual representation makes it easy for engineers to identify the combinations of inputs most likely to lead to an unacceptable output (e.g., damage exceedance probability) or undesirable behaviors (e.g., outliers or bifurcations).

Phase 3. Loads synthesis through stochastic design improvement, aimed at designing a representative physical test

Once the engineers reached their robust design objectives, they began the loads synthesis phase, searching for combinations of test loads conditions causing results closest to the target in-field damage. For this endeavor, the teams deployed the stochastic design improvement workflow inside CAEfatigue Robust Design. Through this probabilistic approach, a set number of stochastic simulation steps are undertaken in sequence, each including small variations of key design variable parameters affecting the test bench set-up.

"The third phase is where we identify the procedure that determined the best set of loads in the physical tests that mimicked the real damage accumulated during the simulation," said Bavaresco. "We use a simplification of the

maneuvers, which is represented in different kinds of loads. The whole purpose is to correlate the damage predicted with the damage we actually get in the test setup."

"We'll take the initial results and refine them, taking variability into account to perform the stochastic design improvement. To accomplish that, we start from an initial assumption, then change that initial condition—for example, we may tweak the number of repetitions in the scaling factor, modulating them until we get a damage that is as close as possible to what we'd calculated with the actual maneuvers. It isn't a typical simulation validation exercise; it's that we're trying to get our results to emulate as best as possible what we've calculated in simulation," Bavaresco said.

Dedicated CAEfatigue Robust Design graphic tools allow engineers to quickly interpret a vast number of results, letting them follow design variables trends step-by-step.

All design parameter combinations have the same probability. For every simulation step, CAEfatigue Robust Design identifies the best solution (red cross in the point cloud) as the starting point for the next step (Figure 6). At the steps sequence conclusion, CAEfatigue displays the test parameters combination causing damage closest to the damage calculated with the real in-field load events. Assuming that the conclusion reached meets the target requirements, engineers can perform a final robustness check to validate and confirm their findings, using the improved test parameter combination as input for a new stochastic study.

Results: Randomness and variability

Dana has been pleased with the results that Hexagon and the CAEfatigue and Robust Design solutions have been able to provide. "Knowing the variability of the damage allows us to reduce the margin because we've already taken the variability of the inputs into account. That means we can optimise the weight and close the variability gap," said Bavaresco.

"With Hexagon allowing us to introduce randomness and variability into the equation, we can right-size our components—which helps us ensure reliability in the field and remain competitive. In the end, working with Hexagon creates not just a massive value for us, but also our customers," he shared.

The company plans to continue its stochastic design improvement in the future, further evaluating how various input parameters impact structure damage and determining a simplified, representative robust surrogate load set for test validation. Ultimately, the company will be able to develop a fully-repeatable process that can be easily and consistently applied to multiple parts of its production line.

"Hexagon's simulations allow us to virtually create and predict the same damage we see in real life. Those simulations will serve as the basis for future optimization because if we can optimise in simulation, we protect against failure on the field," Bavaresco concluded.

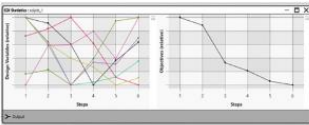


Figure 5: Design parameters variation and damage objectives trend according to the step number.

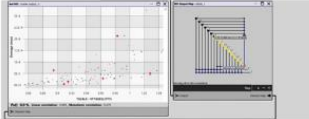
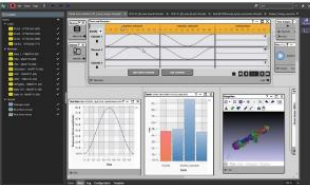


Figure 6: Improved configurations for every step.



Better
Products
Quicker

Time, Budget,
Quality &
Repeatability
important for a
project delivery
organisation

new platforms
– More
complex
engineering,
more levels of
integration

Digital
handshake
between
engineering
toolsets

Digital
handshake
between
engineering and
operations

Product quality
and robustness

Insights into
complex
systems
limited



What differentiates our Durability offering from our competitors?

- We have the **latest and most technologically advanced architecture** combining loads management, fatigue, random response and stochastic simulation software in the market
 - Solver agnostic (Nastran, Marc, Abaqus, Ansys, Optistruct, LS-Dyna)
 - stand-alone GUI, batch control file or FE integrated (Nastran/Marc)
 - multi domain, multi thread, multi tab...
- We deliver these capabilities through **easy-to-use streamlined workflows (templates)** that can be leveraged by durability experts as well as less experienced engineers for **unmatched productivity**
- As a fully Hexagon owned technology, we provide **the most integrated durability and Robust Design solutions**, our **digital handshake** abridging acquired loads, with manufacturing, stress analysis and testing
- Our closely connected teams of **durability simulation experts and solution architects** are super responsive to customer requirements and improvements, always ready to support customers on any of their challenges



HEXAGON

MSC Software

Thank you.



Marco Veltri, PhD
Business Enablement Manager, Multiphysics
Product Management & Business Development

marco.veltri@hexagon.com